

THE SYNTAX OF VERB-OBJECT PHRASES IN CHINESE
CONSTRAINTS AND REANALYSIS

漢語動賓結構的語法研究
限制與重析

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Editions LANGAGES CROISES

CRLAO

Publication Data

PAUL (Waltraud). - The Syntax of Verb-Object Phrases in Chinese: Constraints and Reanalysis / Waltraud Paul. - Paris: Langages croisés, 1988. - VIII-232 p.; 26cm.

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Langages Croisés

Director of Publication: YAU S.C.

EHESS-CRLAO 54, bd. Raspail 75006 Paris

Printed by Feng An Press

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PREFACE

This work is the revised version of my French doctoral dissertation entitled "Syntaxe des syntagmes verbe-objet en chinois: contraintes et restructuration", presented in June 1986 at the Centre de Recherches Linguistiques sur l'Asie Orientale, Ecole des Hautes Etudes en Sciences Sociales in Paris.

A number of people were directly or indirectly involved in the writing of this book and it is a great pleasure for me to acknowledge their contribution here.

First of all, my thanks go to the members of the Centre de Recherches Linguistiques sur l'Asie Orientale for their generous hospitality and friendliness which made my stay in Paris a very pleasant one.

I am particularly indebted to my thesis supervisor, Alexis Rygaloff, who directed my studies for four years and gave me his advice and support.

I would also like to express my gratitude to the following persons whose assistance has been invaluable:

Félicité Maury, who with never-ending patience and kindness shared her linguistic intuitions with me,

Fu Jingqi, Monique Hoa and Xu Dan, who also helped me with the data,

Marie-Claude Paris and Alain Peyraube, who very carefully read the earlier version and whose comments and suggestions considerably improved the form and content of the present book,

Alain Rouveret, who, hopefully, recognizes some of the ideas he was the first to point out to me,

James C.T. Huang, with whom I had several very stimulating discussions during my stay in Taiwan in 1984,

Hélène Gauthier, who corrected the French text and whose encouragement was very precious,

Alain Lucas, who gave me important bibliographical information and also helped me in a lot of other ways,

James Manners, who never lost patience when I lost mine with the computer,

Sue Manners and David Wible, who had the ungrateful task of correcting my English,

Siegfried van Malderen, who cheerfully reminded me that all work and no play makes Jack a dull boy.

It goes without saying that I remain solely responsible for any errors and unclear formulations that occur in the text.

LIST OF ABBREVIATIONS

The following abbreviations are used in glossing examples:

BA	Marker of the object preposed to the verb
BEI	Passive marker
CL	Classifier
DE	Marker of the complement of manner
DIR	Directional (<u>shàng</u> , <u>xiàláí</u> , <u>qǐláí</u> etc.)
DUR	Durative aspect (<u>zài</u> , <u>-zhe</u>)
EXP	Experiential aspect (<u>-guo</u>)
GEI	Marker of the goal and the benefactive (to, for) and other functions
GEN	Conjunction (and, with) and other functions
NEG	Negation (<u>bù</u> , <u>méi</u>)
PART	Sentence-final particle (<u>ba</u> , <u>le</u> , <u>ma</u> , <u>ne</u> etc.)
PERF	Perfective aspect (<u>-le</u>)
PL	Plural (e.g. 3PL = 3rd person plural)
SG	Singular
SUB	Subordinator (<u>-de</u>)

1. INTRODUCTION

The present study examines the syntactic behaviour of transitive verb-object phrases (henceforth VOP) in modern spoken (Mandarin) Chinese (pǔtōnghuà). These VOPs, which consist of a verb and an object, are called transitive because they allow for another object. Thus in (1), bìnggrén 'patient' is the "outer" object of the VOP kāi dāo 'to open knife' = 'to operate on', whereas dāo 'knife' is the "inner" object of the VOP kāi dāo:

- (1) 醫生給病人開刀
 Yisheng gei bingren kai dao
 doctor for/to patient open knife

The doctor operates on the patient.

I have excluded from my investigation intransitive VOPs like bì yè 'to finish studies' = 'to graduate', shāng fēng 'to hurt wind' = 'to catch a cold', chū shì = 'to produce world' = 'to be born', as well as verb-object compounds (like chū-bǎn 'to produce-printing plate' = 'to publish') whose objects can immediately follow and which therefore show the same behaviour as simple verbs, e.g. mǎi 'to buy':

- (2) 他買了/出版了一本書
 Ta mai-le /chu-ban-le yi-ben shu
 3SG buy-PERF/publish-PERF 1 -CL book

He has bought/published a book.

In the case of the VOPs, however, the position to the right of the VOP is unacceptable for the outer object, and the unacceptability of the structure 'Subject verb inner Object outer Object' thus serves as a criterion to distinguish between V-O phrases and V-O compounds:

- (3) *醫生開刀病人
 Yisheng kai dao bingren
 doctor open knife patient

The unacceptability of (3) can be accounted for if we compare it to (2): the outer object bìngrén 'patient' is not allowed to follow the VOP kāi dāo 'to open knife' = 'to operate on' because the object position is already taken by the inner object, dāo.

This explanation is based on the "Phrase Structure Condition" (PSC) proposed by James C.T. Huang(1982). According to the PSC, the internal structure of a phrase XP in Chinese is always head-final, except on the lower level where it may be head-initial. (This does, however, not hold for the NP which is always head-final.) Applied to the VP, the PSC predicts, roughly speaking, that a verb can only be followed by one constituent, for with more than one postverbal constituent we obtain the structure $[_V[_V V NP] NP]$. Such a structure is ruled out by the PSC, because not only is the head of the lowest level $\bar{V}$ on the left, but also the head of the next higher level $\bar{\bar{V}}$. It is the PSC which will serve as our working hypothesis and whose predictions will be compared with the data.

With the aid of the Xiàndài hànyǔ cídiǎn (Peking 1978) and the Chinese-English Dictionary (Peking 1978), a corpus of about 400 VOPs has been established not including dialectal expressions and expressions belonging to the written language.

While concentrating on the syntactic behaviour of VOPs, this study is at the same time an exploration of the Chinese syntax in general. This is also reflected in the organization of this work: chapters devoted to particular problems of the VOPs alternate with chapters of a more general nature which go beyond the domain of the VOPs.

Thus, chapter 2 explains why in most of the sinological literature, the verb-object phrases have been incorrectly analysed as verb-object compounds.

In chapter 3, the PSC and the different phenomena it can account for are discussed at length. We particularly concentrate on the repetition of the verb when followed by two constituents, i.e. the direct object and a complement of duration (or of frequency) or a complement of manner. This process called "verb copying" can save structures otherwise ruled out by the PSC. Another way to make a structure with two postverbal constituents conform to the PSC is restructuring modeled on the double object construction, double object verbs allowing two NPs in the postverbal position. Furthermore, I explain why the second verb in the verb copying construction cannot mark the outer object, although the resulting structure would be well-formed with respect to the PSC.

Chapter 4 gives a complete overview of the forms available for the outer object, marking with gei (cf. (1)) being only one of several possible forms. On the basis of these data, I then discuss whether there is a relation between a given VOP and the form of its outer object and whether the form of the outer object can be predicted for each VOP. I also inquire into the underlying representation for the VOPs and I give arguments in favour of the representation [_V[_V verb inner Object] outer Object].

In chapter 5, I explain why the outer object marked by gei is only allowed in the preverbal position (cf. (1)), and unacceptable in the postverbal position:

- (4) *醫生開刀給病人
 Yisheng kai dao gei bingren
 doctor open knife to/for patient

The unacceptability of (4) is surprising in so far as the post-verbal gěi-phrase is not only allowed for verbs of giving, but also for verbs like qī (chá) 'to brew (tea)' and dǎ (máoyī) 'to knit (a sweater)' which do not belong to the class of verbs of giving: qī chá gěi Zhāngsān 'to make tea for Zhangsan'.

Given the complexity of the problems linked to gěi, I first give a detailed summary of the most exhaustive study on gěi, i.e. that of Zhu Dexi (1979, 1983). I then criticize it point by point and propose my own analysis which differs from Zhu's, especially in the analysis of complex sentences with a purposive clause introduced by gěi: NP V NP gěi NP V (NP)'. Whereas Zhu considers these complex sentences as the extension of a corresponding sentence with a postverbal gěi-phrase: NP V NP gěi NP', I show that for verbs which are not verbs of giving the sentence with a postverbal gěi-phrase is in fact derived from the sentence with a purposive gěi-clause, and not the other way round, as Zhu suggests.

On the basis of this new analysis, it is now possible to explain why the outer object marked by gěi is confined to the preverbal position. VOPs do not allow their outer objects marked by gěi to be in the postverbal position because they are not compatible with a purposive gěi-clause from which the postverbal gěi-phrase would have to be derived.

In chapter 6, I introduce the distinction between "syntactic restructuring" and the "thematic reanalysis". We have an example of restructuring in (5) where the complement of duration, sān-ge zhōngtóu 'three hours', appears as the modifier of the direct object. Sentence (6), on the other hand, illustrates a case of reanalysis: the outer object tā 'he' occupies the modifier position of the inner object, i.e. it takes on the form of the

"possessive object":

- (5) 我看了三個鐘頭的書
 Wo kan -le san-ge zhongtou-de shu
 1SG look-PERF 3 -CL hour -SUB book

I read for three hours.

- (6) 上級革了他的職
 Shangji ge -le ta -de zhi
 boss expel-PERF 3SG-SUB work

The boss fired him.

Though superficially the NP resulting from restructuring, sān-ge zhōngtóu-de shū, and the NP resulting from the reanalysis, tā-de zhí, have the same structure, I show them to behave differently with respect to the deletion of the head noun in coordinated structures, taking the behaviour of a NP with a real modifier (like wǒ-de shū '1SG-SUB book' = 'my book') as the standard of comparison. Since in the deletion of the head noun, the outer object displays the same behaviour as a NP with a real modifier, I examine whether the reanalysed NP 'o.O.de i.O' can be passivised and topicalised to the same extent as a NP with a real modifier. The chapter closes with a comparison between the possessive object and the "possessive subject", i.e. the reanalysis of the subject as the modifier of the object moved to the preverbal position.

In chapter 7, we abandon our working hypothesis, the PSC, because the discussion of the data related to the VOPs has proven the PSC to be inadequate. We propose instead two other conditions, the "Bi-uniqueness Condition" and the "Condition of Predication", which together account for the constraints governing the postverbal position.

2. PRELIMINARIES: V-O PHRASES OR V-O COMPOUNDS ?

Verb-object phrases (VOPs), the subject of my thesis, are often called verb-object compounds in sinological literature, i.e. they are treated as words. In the present chapter, I will first present the arguments which have led to this analysis of V-O phrases as V-O compounds; I then will justify my own analysis according to which they are, in fact, V-O phrases and finally, I will indicate some consequences following from this approach.

The main motivation for talking about V-O compounds instead of V-O phrases is of a semantic nature, i.e. it is based on the observation that in many cases the meaning of a VOP cannot be understood from the meanings of its components and that the VOP must therefore be listed in the lexicon.¹ Accordingly, kāi dāo 'to open knife' = 'to operate on' counts as a compound and is listed in the lexicon, contrary to kāi chuāng 'to open a window', for example, which need not be included in the lexicon. For the followers of this semantic approach (e.g. Lin Handa(1953: 6), Yong Yongquan(1957: 67) and Zhong Benkang (1955: 41-42)), the accessibility of these so-called compounds to different syntactic processes like the separability of the components (e.g. by an aspectual suffix on the verb or by a modifier of the object component) is without any relevance.

Even those Chinese linguists (like Lu Zhiwei(1964: 87), Chao (1968: 425) and Zhao Jinming(1984)) who - among other criteria -

¹ Cf. Richter(1969: 1.1.1.) for a more detailed description of the positions taken by Chinese linguists before 1964. Richter also points out very clearly that the decision for "phrase" or for "compound" depends in fact on adopting a grammatical or a semantic definition of the word.

do use separability as a test to decide upon the question of a V-O compound or a V-O phrase in the end rely on semantic considerations. Lu Zhiwei(1964: 87), for example, hesitates to call xiǎo xīn '(to be) small heart' = 'to be careful' a phrase because of the high degree of semantic cohesion between the components; but at the same time he states the separability of xiǎo xīn in xiǎo diǎnr xīn '(to be) small a little heart' = 'Be a bit careful'. Neither does the separability of shēng qì 'to produce air' = 'to be angry' as illustrated in shēng hái-zi-de qì 'to produce child-SUB air' = 'to be angry with the children' convince Zhao Jinming(1984: 20) to regard shēng qì as a V-O phrase; according to his analysis, shēng qì is a V-O compound, i.e. a word.

Lü Shuxiang(1979: 26, 31) is the only one among the Chinese linguists not to fall into this "trap" of the belief in an isomorphism between semantics and syntax (with semantics being the dominating part). Quite the contrary, he is completely aware of possible conflicts between syntactic structures and the semantic relations associated with them:

We are faced here with a contradiction between the grammar and the lexicon. From the point of view of the lexicon, shuì jiào ['to sleep sleep' = 'to sleep'], dǎ zhàng ['to beat battle' = 'to make war'] etc. can count as words; but from the point of view of the grammar, these combinations can only be considered as phrases.

(Lü Shuxiang 1979: 26; my translation, W.P.)

In the present thesis, we follow Lü Shuxiang's analysis: being separable by aspectual suffixes as well as by modifiers of the object component, the VOPs are phrases, and not compounds. Besides the ungrammaticality of the structure 'Subject verb inner object outer object' (*Yīsheng kāi dāo bìng rén 'doctor open knife patient') which in itself should be enough evidence against the analysis of VOPs as compounds (a real V-O compound

like chū-bǎn 'to produce - printing plate' = 'to publish' allowing its object to follow),² there are more arguments for an analysis of VOPs in terms of phrases, as pointed out convincingly by Huang(1984a).

Huang starts from the concept of "lexical integrity" which says that information concerning the internal structure of words is not accessible to syntactic rules. This idea was originally formulated by Jackendoff(1972) and is known as the "Lexical Integrity Hypothesis"(LIH):

(1) "No phrase-level rule may affect a proper subpart of a word."

(Huang 1984a: 60)

The LIH predicts for example that the conjunction reduction is unable to apply to the components of a word (see (2b)):

(2a) 火車跟汽車
[huo -che] gen [qi -che]
fire-vehicle and air-vehicle
train and car

(2b) *火跟汽車
[huo -gen qi -] che
fire and air vehicle

In the same way, the internal structure of a word is also inaccessible to interpretative rules, which explains why (3) is acceptable:

(3) 一塊綠色的黑板
yi-kuai lüse de hei -ban
1 -CL green SUB black-board
a green blackboard

(Huang 1984a: 61)

² Evidently, this argument is only valid if one accepts the Phrase Structure Condition(PSC) as one of the principles of Chinese syntax. The PSC will be discussed in detail in chapter 3.

With the LIH in mind, let us now examine the criteria which Chao(1968) proposes for the identification of V-O compounds (and which are basically the same as those suggested by Lu Zhiwei(1964)).

A V-O construction is considered to be a V-O compound if it satisfies at least one of the following five conditions:

- (i) one or both of the constituents being bound
- (ii) neutral tone in the object
- (iii) exocentricity of the construction as a whole
- (iv) lexicality (specialization) of meaning
- (v) inseparability of the constituents

(Chao 1968: 415; footnotes omitted)

According to Huang(1984a: 63), these conditions - except that of lexicality (iv) - can in fact be subsumed under the LIH. First, conditions (i) and (ii) can be reduced to condition (v), for in most cases elements carrying a neutral tone are not free, either, which amounts to saying that the components are not separable.³

Besides separability, Huang also adopts Chao's criterion of exocentricity for the identification of compounds. Like inseparability, exocentricity can be incorporated into the LIH: for an exocentric structure is only acceptable if it is not subject to

³ This is somewhat simplified, though, because the neutral tone is not obligatory in V-O compounds and can therefore not be used as a reliable criterion, as pointed out correctly by Chi(1983:3).

Furthermore, there exist VOPs where one or both constituents are bound, but which nevertheless are separable. In fact, Huang himself cites an example of this type, i.e. the VOP yōu mò 'to tease' which is borrowed from English humour and which is separable: yōu tā yī-mò 'yōu 3SG 1-mò' = 'to tease him' (Huang1984a: 65).

But these critical remarks do not refute the validity of the LIH in Chinese.

the well-formedness condition demanding endocentricity for all phrasal structures. In fact, once a structure is marked as a word, it is exempt from this condition because its internal structure is no longer accessible to the endocentricity principle. Bǎng-tuǐ 'tie-leg' = 'legging' for example is a combination of a verb and a noun which results in a noun, thus violating the endocentricity principle. Once it is considered as a (compound) word, however, it is opaque to the endocentricity principle and the separability of its components is excluded. (Huang 1984a:63).

The VOPs examined in the present study obey the endocentricity principle of the $\bar{X}$ -Theory: a phrase consisting of a verbal head and a noun results in a verb phrase. The status which Chao ascribes to the VOPs, however, is not very clear. At first sight, he seems to regard them as exocentric: "A V-O construction functions as a whole as an intransitive verb; when it behaves otherwise, it is an exocentric compound." (Chao 1968: 417). But this point of view can probably be explained by the fact that for Chao only those V-O compounds which admit their object in post-verbal position (and which therefore are also called V-O compounds in our terminology) count as transitive V-O compounds. In the section dealing with "Goals of V-O compounds" (p. 430) where he in fact examines V-O phrases he does not discuss the question of endocentricity or exocentricity, either.

As to the non-compositionality of meaning as identification criterion for compounds, I have already shown at the beginning of this chapter that it is not a valid criterion. But on the other hand, the specialization of meaning explains why VOPs like kāi dāo 'to open knife' = 'to operate on' and kāi wánxiào 'to open joke' = 'to make fun of' are listed in the lexicon, contrary to kāi chuāng 'to open (a) window' for example.⁴

Thus, the VOPs appear in the lexicon for the same reason as idiomatic expressions like guà yángtóu, mài gǒuròu 'to hang sheep-head, to sell dogmeat' = 'to cheat' whose phrasal status is beyond doubt "in spite of" the non-compositionality of its meaning.

In order to avoid the situation where the comparison between VOPs and idiomatic expressions causes more problems than it solves, the fact has to be stressed that Chinese idiomatic expressions (including idiomatic VOPs) have much more freedom than idiomatic expressions in English, but it is the latter which have formed our ideas about the syntactic behaviour of idiomatic expressions in general (cf. Cheng 1983: 185; Herrfurth 1967).

The items examined in the present thesis are phrases whose meaning can be compositional or non-compositional. I hope to have shown convincingly that the choice between V-O phrase and V-O compound is more than a terminological game, for the phrasal status of VOPs helps explaining their behaviour which otherwise would remain mysterious, as for example the link between endocentricity and separability.

That Chao does not pay attention to these regularities is due to his inconsistent use of the terms 'phrase' and 'compound'. The VOP kāi dāo 'to open knife' = 'to operate on' for example is discussed in the section dealing with V-O phrases (p. 319), whereas the VOP zhào xiàng 'to illuminate image' = 'to take a picture' appears among V-O compounds (according to Chao's terminology, p. 430). Despite this terminological inconsistency,

⁴ As we will see in chapter 4.3., in fact all VOPs, including those with a compositional meaning, must be listed in the lexicon because the form the outer object of a given VOP will take is not predictable and must therefore be specified in the lexicon together with the VOP concerned.

Chao's grammar is the best source of information for the problems related to the VOPs.⁵

On the other hand, the lack of terminological precision practised by most of the Chinese linguists explains perhaps why the interpretation of VOPs as compounds had not been contested earlier; for while speaking about compounds, they have nevertheless continued to examine the syntactic behaviour of the components of these so-called compounds. In this context, it is even more surprising that Lü Shuxiang's analysis (cf. the beginning of this chapter) has not attracted due attention.

Though I disagree with the common opinion among Chinese linguists that items like kāi dāo 'to open knife' = 'to operate' are V-O compounds, I do agree with them in calling the components of the VOP 'verb' and 'object', respectively. Although the relation between the verb and the noun in the VOP is not that between a verb and a patient (which seems to be considered as the prototypical relation between a verb and its object), to speak of verb-object phrases reflects the intuition that the postverbal position, i.e. the object position, is not limited to the object role. This "gap" between the object position and the object rôle is not limited to the VOPs, but the case of the VOPs is only a

⁵-----
Chao's inconsistent analysis of kāi dāo and zhào xiàng is even more surprising as they mark their outer object in the same way, namely with gěi:

- (i) 給病人開刀
gei bingren kai dao
to/for sick:person open knife

to operate on a sick person
- (ii) 給大家照相
gei dajia zhao xiang
to/for everybody illuminate image

to take a picture of everybody

reflection of the situation in syntax.⁶ I therefore reject Chi's suggestion to substitute the term verb-noun phrase for verb-object phrase which according to him would be more appropriate for the variety of relations existing between the verb and the noun (Chi 1974, 1984).⁷ While attempting a more adequate description of the data, Chi in fact, obscures an important characteristic of the Chinese language.⁸

⁶ The complement of duration, for example, also occupies the object position (cf. chapter 3.2.).

⁷ According to Chi(1974: 87) zuò lǐbài 'to do worship' = 'to attend worship service in church' and bāng máng 'to help business' = 'to help' are examples of VOPs which do not show a relation verb-object: "... the relationship between the two constituents is not one of action and acted upon."

⁸ For lexical aspects of VOPs (like their productivity) cf. Chi's doctoral thesis (published in 1985) from which the cited articles (Chi 1983, 1984) have been extracted.

3. THE "PHRASE STRUCTURE CONDITION" AND VERB COPYING

3.1. The "Phrase Structure Condition"

In the introduction, I briefly mentioned the "Phrase Structure Condition" (PSC) proposed by James C.T.Huang(1982') in order to explain the ungrammaticality of the structure 'verb inner object outer object' (e.g. *Yīshēng kāi dāo bīnggrén 'doctor open knife patient'). The PSC is defined by Huang as follows:

The PSC (final version)

- a) The internal structure of a phrase XP in Chinese is head-initial for the expansion of complements of X ($X \neq N$), and head-final otherwise.
 - b) At SS [S-structure, W.P.]¹ there is at most one sub-structure X^1 of XP such that X^1 is head-initial and lexically branching.
- (Huang 1982': 2.6)

Applied to the NP, the PSC predicts that the NP will always be head-final, modifiers and complements both preceding the head. (Huang thus explicitly states the well-known fact that within the NP the determiners are always left of the head.)

As to the predictions made by the PSC for the VP, it is important to know that "complement" here refers to the element subcategorised for by the verb, i.e. the inner complement, in contrast to the outer complement which is not subcategorised for by the verb. Consequently, the PSC forbids two (or more) constituents after the verb only if they are on different levels,

¹ The S-structure can be roughly defined as the former surface structure enriched by traces due to transformations. The index i in X^i represents a number which indicates the 2 level of the phrase, i.e. it gives the number of bars. Thus, $V^2 = \bar{\bar{V}}$.

i.e. if they are both inner and outer complements.² Thus, a structure like (1) violates the PSC, whereas the structure in (2) is in harmony with it, the two NPs both being inner complements of the verb:

(1) $*[\bar{V}[\bar{V} \text{ V NP }] \text{ NP }]$

(2) $[\bar{V} \text{ V NP NP }]$

Consequently, double object verbs like sòng 'to give (as a present)', mài 'to sell', wèn 'to ask' and tōu 'to steal' do not violate the PSC, because the indirect and the direct object - being subcategorised - are sisters of the verb. The same holds true for the so-called "pivot" verbs like jiào 'to order, to tell', bī 'to force' and ràng 'to let' which not only subcategorise for a direct object, but also for a sentential complement (cf. Huang 1982': 2.4):

(3) 我逼他去
 Wo bī ta qu
 1SG force 3SG go
 I forced him to go.

A sentence like *Yīsheng kāi dǎo bìngrén can only be excluded by the PSC if we postulate $[\bar{V}[\bar{V} \text{ verb inner object}] \text{ outer object}]$ as the underlying structure for the VOPs; for in such a structure the outer object, not being subcategorised for by the verb, is on a different level from the inner object. This does not mean, however, that the outer object is an outer complement on a par with a circumstantial complement, for example. The outer object does not entertain a relationship with the verb, but with the VOP

² For intransitive verbs, the distinction between inner and outer complements is evidently not important, and the PSC is tantamount to an interdiction of more than one constituent in postverbal position.

as a whole, and I would like to claim that the structure [[verb inner object] outer object] represents the syntactic hierarchies as well as the relations of subcategorisation. Supporting evidence for this point of view comes from the behaviour of the outer object in verb copying (cf. below, 3.2.4.).

3.2. Verb copying

One of the phenomena which in light of the PSC is easily explainable and which loses its "exoticness" (cf. Li 1975: 881) is the so-called "verb copying": the phenomenon that the verb must be repeated when accompanied by its object, the inner complement, and an outer complement which is a complement of duration, frequency or manner:

- (4a) *我看了書三個鐘頭
 Wo kan -le shu san-ge zhongtou
 1SG look-PERF book 3 -CL hour
- (4b) 我看書看了三個鐘頭
 Wo kan shu kan -le san-ge zhongtou
 1SG look book look-PERF 3 -CL hour
 I have read for three hours.
- (5a) *我等信半天
 Wo deng-le xin bantian
 1SG wait-PERF letter long:time
- (5b) 我等信等了半天
 Wo deng xin deng-le bantian
 1SG wait letter wait-PERF long:time
 I have waited for mail a long time.
- (6a) *我騎過馬兩次了
 Wo qi -guo ma liang-ci le
 1SG ride-EXP horse 2 -time PART
- (6b) 我騎馬騎過兩次了
 Wo qi ma qi -guo liang-ci le
 1SG ride horse ride-EXP 2 -time PART
 I have ridden on a horse twice.

- (7a) *他看書得很仔細
Ta kan shu de hen zixi
3SG look book DE very careful
- (7b) 他看書看得很仔細
Ta kan shu kan de hen zixi
3SG look book look DE very careful
He reads very carefully.
- (8a) *他等信得很不耐煩
Ta deng xin de hen bunaifan
3SG wait letter DE very impatient
- (8b) 他等信等得很不耐煩
Ta deng xin deng de hen bunaifan
3SG wait letter wait DE very impatient
He waits for mail very impatiently.³

Sentences (4a) - (8a) are unacceptable because here the verb is not only followed by its inner complement, i.e. the direct object, but also by an outer complement, i.e. a complement of duration, frequency or manner, respectively, thus resulting in the structure $[_{\bar{V}} [_{\bar{V}} V NP] \text{ outer complement }]$ excluded by the PSC. The structure of the sentences (4b) - (8b), $[_{\bar{V}} [_{\bar{V}} V NP] [_{\bar{V}} V \text{ outer complement }]]$, however, is in harmony with the PSC, each verb being followed by one constituent.⁴ We still have to verify the

³ As the status of de which introduces the complement of manner is not clear, I do not analyse it here. Perhaps Huang(1982: 490) is right in considering it as a complementiser.

⁴ If in (4a) - (8a) the direct object is moved to a preverbal position, either by bǎ (see (i)) or by topicalisation (see (ii)), the sentences are acceptable, because now the verb is only followed by one constituent, the outer complement:

- (i) 我把書看了三個鐘頭
Wo ba shu kan -le san-ge zhongtou
1SG BA book look-PERF 3 -CL hour
I have read the book for three hours.
- (ii) 這本書我看了三個鐘頭
Zhei-ben shu, wo kan -le san-ge zhongtou
this-CL book 1SG look-PERF 3 -CL hour
This book, I have read it for three hours.

prediction made by the PSC that it is not the first $\bar{V}$ which is the head of $\bar{V}$ (the head-initial position being allowed only on the lowest level of expansion), but the second $\bar{V}$ containing the copied verb. In fact, there are a number of data confirming this analysis (cf. Li 1975: 880; Li & Thompson 1981: 447-450; Huang 1982': 2.5; Li Linding 1984).

First, only the second verb can be suffixed with an aspect marker:

(9a) 我看書看了三個鐘頭
 Wo kan shu kan -le san-ge zhongtou
 1SG look book look-PERF 3 -CL hour

(9b) *我看了書看三個鐘頭
 Wo kan -le shu kan san-ge zhongtou
 1SG look-PERF book look 3SG-CL hour

I have read for three hours.

(10a) 我等信等了半天
 Wo deng xin deng-le bantian
 1SG wait letter wait-PERF long:time

(10b) *我等了信等半天
 Wo deng-le xin deng bantian
 1SG wait-PERF letter wait long: time

I have waited for mail a long time.

 (footnote 4 continued)

(When the verb is followed by only one constituent as in (i) and (ii), $\bar{v}$ can be substituted for $\bar{V}$, for there is no direct object, or $\bar{v}$ can be declared to be the lowest level here.) If neither preposing by bǎ nor topicalisation is possible, there is another way to "save" the structure [[V NP] outer complement] with regard to the PSC: the complement of duration or of frequency can appear as a modifier of the direct object noun:

(iii) 我看了三個鐘頭的書
 Wo kan -le san-ge zhongtou-de shu
 1SG look-PERF 3 -CL hour -SUB book

I have read for three hours.

This phenomenon will be discussed in chapter 6.1.

Secondly, only the copied verb can be preceded by adverbs like yǐjīng 'already', zhǐ 'only' etc.:⁵

(11a) 他等信已經等了半天
Ta deng xin yijing deng-le bantian
3SG wait letter already wait-PERF long:time

(11b) *他已經等信等了半天
Ta yijing deng xin deng-le bantian
3SG already wait letter wait-PERF long:time

I have already waited a long time for mail.

⁵ The situation is more complicated in the case of negation whose markers bù and méi are commonly considered to be adverbs. Though as predicted by the PSC the negation of the first verb is completely impossible, the negation of the second verb is rarely fully acceptable, either (see (i)). Instead, it is preferable to negate only the complement of duration or of frequency (see (ii)), or to negate the structure where the complement of duration or of frequency appears as a modifier of the object noun (see (iii)):

(i) ?他看書沒看三個鐘頭，只看了一個鐘頭
Ta kan shu mei kan san-ge zhongtou
3SG look book NEG look 3 -CL hour

zhi kan -le yi-ge zhongtou
only look-PERF 1 -CL hour

He has not read for three hours, but only for one hour.

(ii) 他看書看了四個鐘頭，不是三個鐘頭
Ta kan shu kan -le si-ge zhongtou
3SG look book look-PERF 4 -CL hour

bu shi san-ge zhongtou
NEG be 3 -CL hour

He has read for one hour, not for three hours.

(iii) 他沒看三個鐘頭的書，只看了一個鐘頭
Ta mei kan san-ge zhongtou-de shu,
3SG NEG look 3 -CL hour -SUB book

zhi kan -le yi-ge zhongtou
only look-PERF 1 -CL hour (cf. above, footnote 4)

He has not read for three hours, but only for one hour.

A sentence like (i) is problematic because it is probably conceived as contradictory that the affirmation of an event is followed by its negation. It is interesting to observe that in the case of a complement of manner, the negation of the second verb is fully acceptable, thus confirming the prediction made by the PSC:

- (12a) 我看書只看了三個鐘頭
 Wo kan shu zhi kan -le san-ge zhongtou
 1SG look book only look-PERF 3 -CL hour
- (12b) *我只看書看了三個鐘頭
 Wo zhi kan shu kan -le san-ge zhongtou
 1SG only look book look-PERF 3 -CL hour
 I have only read for three hours.
- (13a) 你等車可以等五分鐘，多了不行
 Ni deng che keyi deng wu-fenzhong duo le bu xing
 2SG wait car can wait 5 -minute much PART NEG possible
- (13b) *你可以等車五分鐘，多了不行
 Ni keyi deng che deng wu-fenzhong duo le bu xing
 2SG can wait car wait 5 -minute much PART NEG possible
 You can wait five minutes for the car, but not more.
- (14a) 他看書能連續看四個鐘頭
 Ta kan shu neng lianxu kan si-ge zhongtou
 3SG look book can continuous look 4 -CL hour
- (14b) *他能連續看書看四個鐘頭
 Ta neng lianxu kan shu kan si-ge zhongtou
 3SG can continuous look book look 4 -CL hour
 He can read for four hours without a break.

The suffixation with aspect markers clearly shows that the second verb is the main verb. As to being preceded by adverbs and auxiliaries, this is strictly speaking a property of the verb phrase, and not of the verb. But the compatibility of the second verb with aspect markers together with its compatibility with adverbs and auxiliaries prove that the second $\bar{V}$ is the head of the entire VP, as predicted by the PSC, the first $\bar{V}$ thus being reduced to a kind of degenerate $\bar{V}$ lacking the properties mentioned above.

 (footnote 5 continued)

- (iv) 我今天吃飯並沒吃得很飽
 Wo jintian chi fan bing mei chi de hen bao
 1SG today eat food at:all NEG eat DE very full
 Today I have not eaten to my satisfaction at all.

3.2.1. Restructuring modeled on double object verbs

There exists a set of counter-examples to the PSC where the verbs in question are neither double object verbs nor pivot verbs, but where nevertheless verb copying is not obligatory:

- (15) 我拜訪了張三三個鐘頭
 Wo baifang-le Zhangsan san-ge zhongtou
 1SG visit -PERF Zhangsan 3 -CL hour
 I saw Zhangsan for three hours.
- (16) 我等了三個鐘頭
 Wo deng-le ta san-ge zhongtou
 1SG wait-PERF 3SG 3 -CL hour
 I have waited three hours for him.
- (17) 我看過這個電影三次了
 Wo kan -guo zhei-ge dianying san-ci le
 1SG look-EXP this-CL film 3 -time PART
 I have seen this film three times.
- (18) 他去了北京好幾次了
 Ta qu-le Beijing hao ji -ci le
 3SG go-PERF Peking very several-time PART
 He has been to Peking many times.

The optionality of verb copying is due to the nature of the direct object: in all the examples cited, it is referential and animate ((15), (16)) or referential and definite ((17), (18)), respectively (cf. Li & Thompson 1981: 444).⁶ If the direct object

⁶ Li & Thompson (1981: 126-130) give the following definition of "referential" and "definite": "A noun phrase is referential when it is used to refer to an entity. This entity may be physical or conceptual, real or hypothetical ..." (p. 126). Noun phrases with a predicative function like yīshēng 'doctor' in (i) are examples of non-referential NPs:

- (i) 他是醫生
 Ta shi yisheng
 3SG be doctor
 He is a doctor.

lacks these properties, verb copying is obligatory:⁷

- (19a) *我看了電視三個鐘頭
 Wo kan -le dianshi san-ge zhongtou
 1SG look-PERF television 3 -CL hour
- (19b) 我看電視看了三個鐘頭
 Wo kan dianshi kan -le san-ge zhongtou
 1SG look television look-PERF 3 -CL hour
- I have watched television for three hours.
- (20a) *我等信等了半天
 Wo deng-le xin bantian
 1SG wait-PERF letter long:time
- (20b) 我等信等了半天
 Wo deng xin deng-le bantian
 1SG wait letter wait-PERF long:time
- I have waited for mail a long time.

 (footnote 6 continued)

"Only noun phrases that are referential can be definite or indefinite. They differ in that a definite noun phrase refers to an entity that the speaker believes is known to the hearer, while an indefinite noun phrase refers to an entity about which the speaker believes the hearer does not already know." (Li & Thompson 1981: 129-130; emphasis mine, W.P.)

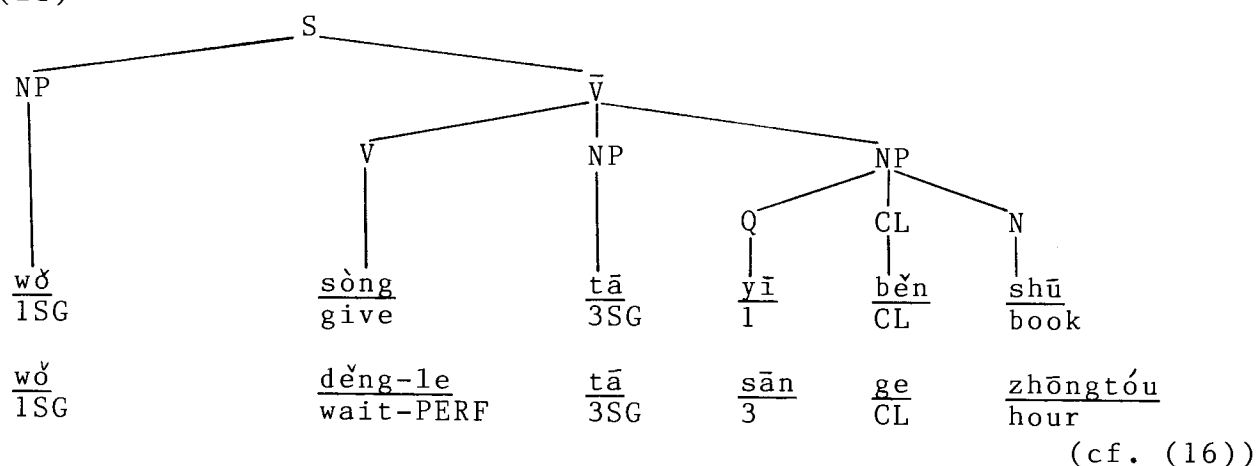
⁷ Verb copying should also be obligatory in the following sentences where the NPs xǐshǒujiān 'restroom' and fànguǎn 'restaurant' are neither definite nor animate;

- (i) 我去洗手間一下
 Wo qu xishoujian yi-xia
 1SG go restroom 1 -moment
- I go to the restroom.
- (ii) 這個禮拜我去了飯館兩次
 Zhei-ge libai, wo qu-le fanguan liang-ci le
 this-CL week 1SG go-PERF restaurant two -time PART
- This week, I have gone to a restaurant twice.

I can offer no explanation for the optionality of verb copying here. Perhaps it has to do with the fact that verbs of motion like qù 'to go' only allow locative nouns as their objects.

Why is verb copying sensitive to the nature of the direct object?⁸ Huang(1982') proposes an analysis which not only explains why verb copying is optional in the case of a referential and definite/animate direct object, but also why such a structure is not excluded by the PSC. According to Huang, sentences like (15) - (18) are the result of restructuring which takes the structure of the double object construction as a model, i.e. they are to be analysed as follows:⁹

(21)



Thus, it is no coincidence that verb copying is optional in the case of a referential and animate/definite direct object because then the typical structure of double object verbs showing an animate indirect object and an inanimate direct object, is imitated: the (animate) direct object of děng 'to wait' occupies

⁸ Only few linguists (e.g. Jia Ying 1985) have examined the cases where verb copying is optional, but none of them has offered any explanation of this phenomenon. This also applies to Li & Thompson(1981) who nevertheless are the first to explicitly state the conditions under which verb copying is optional, i.e. in the case of a referential and definite/animate direct object.

⁹ It is important not to confuse the structure resulting from restructuring with the deep structure of the sentence in question. Thus, (21) [_S NP [_{V̄} V NP NP]] representing the deep structure postulated for double object verbs is not meant to be the deep structure of Wǒ děng-le tā sān-ge zhōngtóu, but only shows the analysis of this sentence at the level of S-Structure.

the indirect object position of the verb sòng 'to give', whereas the complement of duration sān-ge zhōngtóu 'three hours' is in the direct object position of sòng, i.e. in the same position as yī-běn shū 'a book'. By adopting the structural analysis of double object verbs (with both NPs being sisters of the verbs), the PSC is satisfied.¹⁰

The fact that in the case of a complement of manner, verb copying is obligatory (regardless of the nature of the direct object), is another argument for Huang's analysis: a structure like (22a) cannot be analysed in terms of the double object construction, and consequently, verb copying is necessary to satisfy the PSC:

(22a) *我等他得很不耐煩

Wo deng ta de hen bunaifan
1SG wait 3SG DE very impatient

(22b) 我等他等得很不耐煩

Wo deng ta deng de hen bunaifan
1SG wait 3SG wait DE very impatient

I have waited for him very impatiently.

(23a) *他說中文得很流利

Ta shuo zhongwen de hen liuli
3SG speak Chinese DE very fluent

(23b) 他說中文說得很流利

Ta shuo zhongwen shuo de hen liuli
3SG speak Chinese speak DE very fluent

He speaks Chinese very fluently.

¹⁰ In fact, in his discussion of the "mock object" tā, illustrated in

(i) 睡他一天

shui ta yi-tian
sleep Tā 1 -day

to sleep an entire day

Lü Shuxiang (1955: 184) proposes a similar analysis: he compares sentence (i) to double object constructions and calls yī-tiān 'one day' a "quasi-object". (Cf. also Chao 1968: 320.)

(24a) *他看書得很仔細

Ta kan shu de hen zixi
3SG look book DE very careful

(24b) 他看書看得很仔細

Ta kan shu kan de hen zixi
3SG look book look DE very careful

He reads very carefully.

3.2.2. Non-durative verbs and verb copying

So far we have seen that the obligatoriness or optionality of verb copying (in the case of a complement of duration or of frequency) depends on the nature of the direct object. The following sentences (25a) – (27a), however, are all acceptable without verb copying, though in each case the direct object is non-referential, thereby excluding restructuring modeled on the double object construction. In fact, verb copying is even unacceptable here:

(25a) 孩子出世三天了

Haizi chu shi san-tian le
child come:out world 3 -day PART

It has been three days since the baby was born.

(25b) *孩子出世出了三天了

Haizi chu shi chu -le san-tian
child come:out world come:out-PERF 3 -day

(26a) 他畢業一年了

Ta bi ye yi-nian le
3SG finish studies 1 -year PART

It has been one year since he graduated.

(26b) *他畢業畢了一年了

Ta bi ye bi -le yi-nian le
3SG finish studies finish-PERF 1 -year PART

(27a) 汽車出毛病一個禮拜了

Qiche chu maobing yi-ge libai le
car come:out defect 1 -CL week PART

It has been one week since there was something wrong with the car.

(27b) *汽車出毛病出了一個禮拜了

Qiche chu maobing chu -le yi-ge libai le
car come:out defect come:out-PERF 1 -CL week PART

In order to explain the facts which seem to question the validity of the PSC, it is necessary to take a closer look at the verbs involved. As they are all non-durative, the temporal expressions sān-tiān 'three days', yī-nián 'one year' etc. cannot possibly indicate the duration of the event itself. The only interpretation possible is that sān-tiān 'three days', yī-nián 'one year' etc. refer to the lapse of time between the (beginning of the) action and the moment of utterance (or another point of time indicated). This is to say that sān-tiān 'three days' and yī-nián 'one year' do not function as complements of duration here, but that they serve as predicates for the sentential subjects háizi chū shì 'the baby was born' and tā bì yè 'he graduated', respectively. We thus obtain the structure [_S [_S NP V NP] [_V NP]] for (25a) - (27a), and not the structure [_S NP [_V [_V V NP] NP]] excluded by the PSC.

The fact that temporal expressions like sān-tiān 'three days' can be preceded by yǒu 'to have' or its negation méi, as well as by adverbs like yǐjīng 'already', cái 'only, just', hái 'still' and chàbùduō 'approximately' provides supporting evidence for their predicative function in these sentences:

(28) 孩子出世没三天

Haizi chu shi hai mei san-tian
child come:out world still NEG 3 -day

It has not yet been three days since the baby was born.

(29) 他畢業已經一年了

Ta bi ye yijing yi-nian le
3SG finish studies already 1 -year PART

It has already been a year since he graduated.

(30) 汽車出毛病差不多一個禮拜了

Qiche chu maobing chabuduo yi-ge libai le
car come:out defect approximately 1 -CL week PART

It has been about a week now since the car has not been working.

As shown in the following sentences, both yǒu 'to have' and adverbs like yǐjīng 'already' are put at the beginning of the verb phrase:

- (31a) 這本書他已經看了三個鐘頭了
 Zhei-ben shu, ta yijing kan -le san-ge zhongtou le
 this-CL book 3SG already look-PERF 3 -CL hour PART
 This book, he has already read it for three hours.
- (31b) *這本書他已經看了有三個鐘頭了
 Zhei-ben shu, ta yijing kan -le you san-ge zhongtou
 this-CL book 3SG already look-PERF have 3 -CL hour
- (31c) 這本書他看了已經有三個鐘頭了
 Zhei-ben shu, ta kan -le yijing you san-ge zhongtou
 this-CL book 3SG look-PERF already have 3 -CL hour
 This book, it has already been three hours that he has been reading it.

(31b) is ungrammatical because the indications given by yǐjīng 'already' and yǒu 'to have' are contradictory: whereas yǐjīng 'already' marks kàn 'to look' as the head of the verb phrase kàn-le sān-ge zhōngtóu and implies the structure [_S tā [_V yǐjīng kàn-le sān-ge zhōngtóu]] (which is the structure for (28a)), the presence of yǒu 'to have' before sān-ge zhōngtóu points to the predicative function of sān-ge zhōngtóu 'three hours', thus leading to the structure [_S [_S tā kàn-le] [_V yǒu sān-ge zhōngtóu]].

We thus obtain the following situation: if the verb is non-durative, verb copying is excluded, and the temporal expression functions as a predicate.¹¹ But if the verb is durative, verb copying is possible and the temporal expression indicates the

¹¹ As Marie-Claude Paris(p.c.) has pointed out to me, there are, however, cases where a temporal expression appears in the position of the complement of duration, though it co-occurs with a non-durative verb and must therefore denote the amount of time elapsed since the (beginning of the) event, and not the duration of the event itself:

duration of the event. This generalization starting from the durative or non-durative character of the verb can, however, not be inverted.

First, a temporal expression can also be predicated of a sentential subject containing a durative verb:¹²

(footnote 11 continued)

- (i) 他已經了三天
Ta yijing dao -le san-tian
3SG already arrive-PERF 3 -day
He already arrived three days ago.
- (ii) 他已經死了好久了
Ta yijing si -le haojiu le
3SG already die-PERF long:time PART
It is already a long time since he died.
- (iii) 他才畢業一個禮拜了
Ta cai bi ye yi-ge libai le
3SG only finish studies 1 -CL week PART
He graduated just one week ago.
- (iv) 他已經走了五年了
Ta yijing zou-le wu-nian le
3SG already go -PERF 5 -year PART
He already left five years ago.

The position of the adverb yǐjīng 'already' clearly shows that the following verb is the head of the verb phrase and thereby excludes a predicative function of the temporal expression. It seems, however, that only non-durative verbs make use of this structure to indicate the lapse of time since the (beginning of the) event. For with durative verbs, the temporal expression in such a structure is necessarily interpreted as the duration of the event itself; when the lapse of time between the event and the moment of utterance is to be expressed, the structure where the temporal expression is predicated of a sentential subject must be used.

The occurrence of non-durative verbs in the complement structure illustrated in sentences (i) - (iv) remains mysterious to me. Audrey Li (1987: 15) who claims that temporal expressions can appear in both the complement structure and the predicate structure, no matter whether they indicate the duration of the event itself or the amount of time elapsed since the event, considers sentences like (i) - (iv) as complement structures. She does not, however, discuss how to reconcile this analysis with the non-durative character of the verbs in question.

¹² In the case of durative verbs, there is, however, a rather strong tendency to interpret a temporal expression as the

- (32a) 他看書已經有三個鐘頭了
 Ta kan shu yijing you san-ge zhongtou le
 3SG look book already have 3 -CL hour PART
- (32b) 他看書已經看了三個鐘頭了
 Ta kan shu yijing kan -le san-ge zhongtou
 3SG look book already look-PERF 3 -CL hour
- He has read for three hours.

 (footnote 12 continued)

duration of the event itself. This explains why sentences (i) and (ii) lacking yǒu 'to have' or an adverb signaling the predicative function of the temporal expression, are not acceptable:

- (i) *他看書三個鐘頭了
 Ta kan shu san-ge zhongtou le
 3SG look book 3 -CL hour PART
- (ii) *我等信半天了
 Wo deng xin bantian le
 1SG wait letter long:time PART

The corresponding acceptable sentences have yǒu as well as an adverb:

- (iii) 他看書已經有三個鐘頭了
 Ta kan shu yijing you san-ge zhongtou le
 3SG look book already have 3 -CL hour PART
- It has already been three hours that he has been reading.
- (iv) 我等信已經有半天了
 Wo deng xin yijing you bantian le
 1SG wait letter already have long:time PART
- It has already been a long time that I have been waiting for mail.

In fact, it is this structure which is used for the negation of the temporal expression alone. Contrary to the negation of the second verb in the verb copying construction (vi) (cf. footnote 5), it is completely acceptable:

- (v) 他看書沒有三個鐘頭
 Ta kan shu hai mei you san-ge zhongtou
 3SG look book still NEG have 3 -CL hour
- It is not three hours yet that he has been reading.
- (vi) ?他看書沒有三個鐘頭
 Ta kan shu mei kan san-ge zhongtou
 3SG look book NEG look 3 -CL hour
- He has not read for three hours.

As to the differences between the verb copying construction and the structure with the temporal expression in the predicative function, according to Jia Ying (1985: 26), the duration of the event is emphasized more in the verb copying construction.

- (33a) 他學漢語一年了
 Ta xue hanyu yi-nian le
 3SG learn Chinese 1 -year PART
- (33b) 他學漢語學了一年了
 Ta xue hanyu xue -le yi-nian le
 3SG learn Chinese learn-PERF 1 -year PART
- He has been studying Chinese for one year.
- (34a) 張三發燒三天了
 Zhangsan fa shao san-tian le
 Zhangsan produce fever 3 -day PART
- (34b) 張三發燒發了三天了
 Zhangsan fa shao fa -le san-tian
 Zhangsan produce fever produce-PERF 3 -day
- Zhangsan has had a temperature for three days.

There is hardly any difference of meaning between (32a) - (34a), on the one hand, and the corresponding sentences with verb copying (32b) - (34b), on the other hand; from the statement that it is three hours now that he has been reading, it can be deduced that he has been reading for three hours (cf. (32)). The same holds true for (33) and (34): that it is one year now since he started learning Chinese leads to the conclusion that he has been studying Chinese for one year; and that it is three days now that he has had a temperature amounts to saying that he has had a temperature for three days.

Consequently, sentences (32a) - (34a) do not display any structural ambiguity, because the structures themselves are not ambiguous. We can only speak of "ambiguity" in so far as there is a choice between the "original" meaning, i.e. the meaning corresponding to the syntactic structure in question, and the meaning derived from this original meaning by means of a pragmatic deduction. As sentence (35) shows, this choice can nevertheless make a difference:

- (35a) 他起床半小時了，早就洗過臉了

Ta qi chuang ban -xiaoshi le , zaojiu xi -guo
 3SG get:up bed half-hour PART long:ago wash-EXP

lian le
 face PART

He got up half an hour ago and washed his face long since.

- (35b) 他起床半小時了，還沒下床

Ta qi chuang ban -xiaoshi le, hai mei
 3SG get:up bed half-hour PART still NEG

xia chuang
 descend bed

He has made efforts to get up for half an hour, but he has not got out of bed yet.

(Jia Ying 1985: 25-26)

Secondly, 'yǒu + temporal expression' can also be predicated of a sentential subject containing the verb-copying construction, though such a sentence is not felt to be very natural:

- (36) 他等信等了已經有半天

Ta deng xin deng-le yijing you bantian
 3SG wait letter wait-PERF already have long:time

It has already been a long time that he has been waiting for mail.

- (37) 他看書看了已經有三個鐘頭了

Ta kan shu kan -le yijing you san-ge zhongtou le
 3SG look book look-PERF already have 3 -CL hour PART

It has already been three hours that he has been reading.

Thirdly, though verb copying is unacceptable for durative verbs when accompanied by a temporal expression indicating duration, this is not so for a temporal expression indicating frequency. When both the object and the complement of frequency are present, verb copying is obligatory, which is the same situation as with durative verbs (cf. above, (4) - (8):¹³

¹³ Besides the verb copying construction, there is another structure in harmony with the PSC: the complement of frequency

- (38) 汽車出毛病出過五次了
 Qiche chu maobing chu -guo wu-ci le
 car come:out defect come:out-EXP 5 -time PART
 There have been problems with the car five times.
- (39) 他丟錢丟過三次了
 Ta diu qian diu -guo san-ci le
 3SG lose money lose-EXP 3 -time PART
 He has lost money three times.

When the verb denotes an event which can be repeated, it is compatible with a complement of frequency like sān-cì 'three times'; as the complement refers to the frequency of the event itself and not to the temporal relationship between the event and the moment of utterance, verb copying is used.

As to the possibility of functioning as a predicate, a temporal expression of frequency cannot be predicated of a sentential subject as easily as a temporal expression of duration. Thus, (40a) and (41a) where the expression of frequency is preceded by yǒu (indicating its predicative function) are somewhat problematic whereas the corresponding sentences with verb copying (cf. (40b)) or without yǒu (cf. (41b)), respectively, are fully acceptable:

- (40a) ??他放火已經有六次了
 Ta fang huo yijing you liu-ci le
 3SG put fire already have 6 -time PART

 (footnote 13 continued)

appears in the modifier position of the object noun:

- (i) 汽車出過五次毛病了
 Qiche chu -guo wu-ci maobing
 car come:out-EXP 5 -time defect

There have been problems with the car five times.

This phenomenon will be discussed in detail in chapter 6.1.

- (40b) 他放火已經放了六次了

Ta fang huo yijing fang-le liu-ci le
 3SG put fire already put -PERF 6 -time PART

He has already set a fire six times.

- (41a) ??這本書他看過已經有三次了

Zhei-ben shu, ta kan -guo yijing you san-ci le
 this-CL book 3SG look-EXP already have 3 -time PART

- (41b) 這本書他已經看過三次

Zhei-ben shu, ta yijing kan -guo san-ci le
 this CL book 3SG already look-EXP 3 -time PART

This book, he has already read it three times.

This behaviour of the temporal expression of frequency as well as the other facts discussed so far clearly argue against the claim that the complement of frequency or of duration is in fact the main predicate of the sentence and that, in the underlying representation, it is preceded by yǒu (cf. Teng 1975a). If there were really an underlying yǒu before the complement of duration or of frequency, one would expect this yǒu to be able to "save" a structure like [_S NP [_{̄V} [_{̄V} V NP] outer complement]] with respect to the PSC, resulting in the structure [_S [_S NP V NP] [_{̄V} (yǒu) (temporal) NP]]. This, however, is not the case:

- (42) *他等信(有)半天

Ta deng xin (you) bantian
 3SG wait letter have long:time

- (43) *他看書(有)三個鐘頭

Ta kan shu (you) san-ge zhongtou
 3SG look book have 3 -CL hour

- (44) 他等信已經有半天了

Ta deng xin yijing you bantian
 3SG wait letter already have long:time

It has already been a long time that he has been waiting for mail.

- (45) 他看書已經有三個鐘頭了

Ta kan shu yijing you san-ge zhongtou le
 3SG look book already have 3 -CL hour PART

It has already been three hours that he has been reading.

The structure [_S[_S NP V NP] [_{V̄} (temporal) NP]] is much more easily acceptable when the direct object is referential and animate/definite (cf. (46)), which reminds one of the situation where verb copying is optional (cf. (47)):

- (46) 我等他(有)半天了
 Wo deng ta (you) bantian le
 1SG wait 3SG have long:time PART

It is a long time that I have been waiting for him.

- (47) 我等他半天了
 Wo deng-le ta bantian le
 1SG wait-PERF 3SG long:time PART

I have waited for him a long time.

But contrary to (47), the verb in (46) cannot be suffixed with an aspect marker:

- (48) *我等了他有半天了
 Wo deng-le ta you bantian le
 1SG wait-PERF 3SG have long:time PART

This clearly shows that a structure like (47) represents the case where verb copying is optional because the presence of the aspect marker excludes a predicative function of the temporal expression. As examples (49) - (52) further show, the unacceptability of an aspect marker (within the sentential subject) seems to be a general property of the structure [_S[_S NP V NP] [_{V̄} (yǒu) temporal NP]], and thus confirms our analysis of the structure containing a complement of duration and the structure with a temporal expression in predicative function as two distinct types.¹⁴

¹⁴ Marie-Claude Paris (p.c.) has drawn my attention to the fact that aspect markers do not seem to be generally excluded from the sentential subject in the structure [_S[_S NP V NP] [_{V̄} (yǒu) temporal NP]]. Though a new consultation of my informants confirmed the acceptability judgments for sentences of the type illustrated in (49) - (52), the following sentences were equally accepted:

(footnote 14 continued)

- (i) 他走了已經三年了

Ta zou-le yijing san-nian le
3SG go -PERF already 3 -year PART

It has already been three years since he went away.

- (ii) 他死了已經好久了

Ta si -le yijing haojiu le
3SG die-PERF already long:time PART

It has already been a long time since he died.

- (iii) 他來了已經兩年了

Ta lai -le yijing liang-nian le
3SG come-PERF already 2 -year PART

It has already been two years since he came.

- (iv) 他到了台灣已經幾個月了

Ta dao -le Taiwan yijing ji -ge yueh le
3SG arrive-PERF Taiwan already several -CL month PART

It has already been several months since he arrived in Taiwan.

- (v) 他離了歐洲已經有三年了

Ta likai-le Ouzhou yijing you san-nian le
3SG leave-PERF Europe already have 3 -year PART

It has already been three years since he left Europe.

Perhaps the difference in acceptability has to do with the verb class in question; the verbs in sentences (i) - (v) are all either intransitive verbs or verbs of motion.

Furthermore, the unacceptability of aspect markers within a sentential subject, as observed in the structure with a temporal expression in predicative function (cf. (vii)), does not seem to extend to sentential subjects in general (cf. (vi)):

- (vi) 他去了法國真可惜

Ta qu(-le) faguo zhen kexi
3SG go -PERF France really pity

It is really a pity that he left for France.

- (vii) 他去(*了)法國還不到一年

Ta qu(*-le) faguo hai bu dao yi-nian
3SG go -PERF France still NEG arrive 1 -year

It has not been a year yet since he left for France.

I can offer no explanation for these data nor for the acceptability of an aspect marker within a sentential subject when it contains the verb copying construction (cf. (36) and (37) above). A solution to this problem must await further research about embedded sentences, a domain which apparently has not attracted the attention of many Chinese linguists so far.

- (49) 他學(* 了) 漢語沒有一年了

Ta xue (*-le) hanyu hai mei yi-nian
3SG learn -PERF Chinese still NEG 1 -year

It has not yet been a year that he has been studying Chinese.

- (50) 他等(* 了) 信已經有半天了

Ta deng(*-le) xin yijing you bantian le
3SG wait -PERF letter already have long:time PART

It has already been a long time that he has been waiting for mail.

- (51) 他發(* 了) 燒已經有三天了

Ta fa (*-le) shao yijing you san-tian
3SG produce -PERF fever already have 3 -day

It has already been three days that he has had a temperature.

- (52) 他去(* 了) 法國沒有一個月

Ta qu(*-le) faguo hai mei yi-ge yueh
3SG go -PERF France still NEG 1 -CL month

It has not been a month yet since he left for France.

Though I do not agree with the general claim that a complement of duration or of frequency is, in fact, in the underlying representation, the main predicate of the sentence, this does not exclude that a temporal expression of duration or of frequency can, under special circumstances, have a predicative function. This predicative function, however, is not an inherent characteristic of temporal expressions of duration or of frequency in general, but is a property of the structure in question and confirmed by the presence of yǒu as well as by the position of adverbs (cf. above, (28) - (30)).

Though the data cannot be analysed unequivocally in all cases, I nevertheless hope to have shown that there is no equivalence (not to speak of interchangeability) between the structure where the temporal expression is a complement, on the one hand, and the structure where the temporal expression is the main predicate, on the other hand. Besides the nature of the verb

(e.g. durative vs. non-durative), the type of the temporal expression (duration or frequency) as well as the nature of the direct object determine the acceptability of the two structures in question.

3.2.3. Objections to the postulate of restructuring

In light of the preceding discussion showing the difference between the verb copying construction and the structure with a temporal expression as the main predicate, I will now examine possible arguments against the analysis of a structure without verb copying, 'NP V NP Complement (of duration /frequency)' in terms of restructuring modeled on the double object construction.

The main objection is based on the lack of a corresponding structure with yǒu 'to have' for the double object construction (cf. (53b)) which is, however, claimed to exist for the structure resulting from restructuring (cf. (54b)); and on the fact that double object verbs do not allow for verb copying as in (53c), whereas for verbs with only one subcategorised argument, the structure without verb copying (cf. 54a) and that with verb copying (cf. (54c) are both acceptable in the case of a referential and animate/definite direct object:

- (53a) 我送他一本書
 Wo song ta yi-ben shu
 1SG give 3SG 1 -CL book
 I give him a book (as a present).

- (53b) *我送他有一本書
 Wo song ta you yi-ben shu
 1SG give 3SG have 1 -CL book

- (53c) *我送他送一本書
 Wo song ta song yi-ben shu
 1SG give 3SG give 1 -CL book

- (54a) 我等了三個鐘頭了

Wo deng-le ta san-ge zhongtou le
1SG wait-PERF 3SG 3 -CL hour PART

I have waited for him three hours.

- (54b) 我等他已經有三個鐘頭了

Wo deng ta yijing you san-ge zhongtou le
1SG wait 3SG already have 3 -CL hour PART

It has already been three hours that I have been waiting for him.

- (54c) 我等他等了三個鐘頭了

Wo deng ta deng-le san-ge zhongtou le
1SG wait 3SG wait-PERF 3 -CL hour PART

I have waited for him three hours.

As far as the first point is concerned, I have already shown that in a lot of cases, a corresponding structure with verb copying simply does not exist. Even when there is one, these two structures behave differently with respect to the acceptability of aspect markers (cf. (49) – (52)). This shows that they are two distinct types, even if the meaning of one structure can be deduced from the other by means of a pragmatic conclusion. The ungrammaticality of (53c), however, deserves more attention and will be dealt with at the end of this chapter (cf. 3.2.4.).

Two further arguments against explaining the optionality of verb copying in terms of restructuring are as follows. Whereas yī-běn shū 'a book' in (53a) is a subcategorised argument of the verb sòng 'to give', sān-ge zhōngtóu 'three hours' in (54a), being a complement of duration, is not. Moreover, the direct object tā 'he' of děng 'to wait' cannot undergo the same movement as the indirect object tā 'he' of sòng 'to give':

- (55) 我送了一本書給他

Wo song-le yi-ben shu gei ta
1SG give-PERF 1 -CL book GEI 3SG

I have given a book to him (as a present).

(56) *我等了三個鐘頭給他

Wo deng-le san-ge zhongtou gei ta¹⁵
 1SG wait-PERF 3 -CL hour GEI 3SG

Contrary to the first set of objections (i.e. lack of verb copying and of a construction with yǒu for double object verbs), these two latter arguments cannot be countered by referring to some additional data. One can only point out that a structure resulting from restructuring is expected not to allow the same transformations as its model structure (the identity of the two structures being possible only under certain conditions). Thus, the objections discussed here miss, in fact, the very notion of restructuring because they demand such a high degree of identity that, if it really existed, it would be better to speak of one structure instead of two. The different accessibility to syntactic transformations is, on the contrary, a means to identify the "real" structure and thereby to distinguish it from the structure "imitating" it (cf. below, ch. 6.1.).

3.2.4. The VOPs and verb copying

Before closing this chapter, let us go back or rather advance towards the VOPs. If verb copying can "save" the structure $[_V[_V V NP] \text{ Complement}]$ with respect to the PSC, can it do the same for the structure $[_V[_V V \text{ inner Object}] \text{ outer Object}]$? As shown in the following sentences, the second verb in a verb copying construction cannot mark the outer object of a VOP (cf.(57a)); it can only be followed by an outer complement (e.g. a complement of duration):

¹⁵ Given the multiple functions of gěi: full verb 'to give (as a present)', marker of the goal, the benefactive etc. and the fact that its function depends on its position (preverbal vs. postverbal), I only translate gěi if it is the full verb 'to give'. In all the other cases, I simply gloss it as GEI.

- (57a) *醫生開刀開病人
 Yisheng kai dao kai bingren
 doctor open knife open patient
- (57b) 醫生給病人開刀
 Yisheng gei bingren kai dao
 doctor GEI patient open knife
 The doctor operates on the patient.
- (57c) 醫生給病人開刀開了三個鐘頭
 Yisheng gei bingren kai dao kai -le san-ge zhongtou
 doctor GEI patient open knife open-PERF 3 -CL hour
 The doctor has operated on the patient for three hours.

Since the PSC is not violated in (57a), the ungrammaticality of this sentence must be explained otherwise. I would like to claim that the problem involved here is, in fact, the same as in the unacceptability of verb copying for double object verbs (cf. (53c)).

A comparison of the three cases discussed so far (VOPs, double object verbs and other verbs) shows that verb copying is only acceptable if the two NPs occurring with the verb are the inner complement and the outer complement, respectively:

- (58) 他看書看了三個鐘頭
 Ta kan shu kan -le san-ge zhongtou
 3SG look book look-PERF 3 -CL hour

He has read for three hours.

Verb copying is excluded for double object verbs and pivot verbs because it implies incorrect subcategorisation relations. Thus, the structure of (59a) suggests that there is only one argument, tā 'he', which is subcategorised for by the verb sòng 'to give (as a present)' (yī-běn shū 'a book' being treated as an outer complement), though in fact both NPs, tā and yī-běn shū, are subcategorised arguments, i.e. inner complements:

(59a) *我送他送了一本書

Wo song ta song-le yi-ben shu
 1SG give 3SG give-PERF 1 -CL book

(59b) 我送了他一本書

Wo song-le ta yi-ben shu
 1SG give-PERF 3SG 1 -CL book

I have given him a book (as a present).

The explanation of the unacceptability of verb copying for VOPs follows more or less the same reasoning. In a structure like (57a), the outer object is treated as an outer complement of the verb in the VOP. However, the outer object is not related to the verb of the VOP, but to the VOP as a whole. This is to say that the outer object is an argument subcategorised for by the entire VOP; accordingly, it represents the inner complement of the VOP, and not the outer complement of the verb (alone).

Perhaps this in fact rather simple explanation does not seem to be very obvious at first, for the actual relations between the verb, the inner object and the outer object are obscured by the fact that the structure proposed for the VOPs looks the same as the structure for verbs with a direct object and a circumstantial complement. But contrary to the latter case where the outer complement is not related by subcategorisation either to the verb or to the direct object, for the VOPs the structure $[_V[_V V \text{ inner Object}] \text{ outer Object}]$ at the same time shows the syntactic hierarchies as well as the relations of subcategorisation. In the same way that the sister of the verb, i.e. the inner object, functions as the inner complement of the verb, the sister of the verb phrase, i.e. the outer object, functions as the inner complement of the verb phrase, the VOP.

This also explains why the VOPs do not appear in the structure of double object verbs $[_V V \text{ NP NP}]$ with both NPs as sisters of the verb; the outer object is not a second inner

complement of the verb on a par with the inner object, as this structure implies.

It is now possible to account for the structures discussed so far in which VOPs appear.

The structure $[\bar{V}[\bar{V} V \text{ inner Object}] \text{ outer Object}]$ correctly indicates the relations of subcategorisation, but violates the PSC. The verb copying construction, $[\bar{V}[\bar{V} V \text{ inner Obj.}] [\bar{V} V \text{ outer Obj.}]]$ (cf. above, (57a)) is in harmony with the PSC, but does not respect the actual relations of subcategorisation. The structure $[\bar{V}[\text{pp prep outer Object}][\bar{V} V \text{ inner Object}]]$, where the outer object is marked by a preposition and precedes the VOP (cf. (57b)), on the contrary, takes into account both the PSC and the relations of subcategorisation.

In this chapter, I have introduced the PSC and I have shown how it can explain different syntactic phenomena.

First of all, the PSC only excludes two (or more) postverbal constituents if they are not all subcategorised for by the verb. Double object verbs and pivot verbs thus do not violate the PSC, because their two complements are both inner complements.

When a verb occurs with an inner as well as an outer complement, verb copying applies to avoid a violation of the PSC. The first verb takes the inner complement and the second verb the outer complement. Since only the second verb can be suffixed with aspect markers and be preceded by adverbs and auxiliaries, it is the second $\bar{V}$ which is the head of the entire VP, as predicted by the PSC.

Verb copying in the presence of a complement of duration or of frequency is optional when the direct object is referential and animate/definite, for with such an object the typical structure of double object verbs (where the verb is followed by an animate and definite NP, the indirect object, and an inanimate NP, the direct object) can be imitated. Being restructured on the model of double object verbs, the structure now satisfies the PSC.

That the structure resulting from restructuring cannot undergo the same syntactic transformations as the real double object construction does not constitute an argument against the postulate of restructuring. On the contrary, this is the expected behaviour, the identity of the model structure and the restructured one depending on particular conditions.

In the case of non-durative verbs, a temporal expression of duration cannot occur in the verb copying construction, but functions as the predicate of a sentential subject containing the (non-durative) verb in question. The temporal expression here indicates the time elapsed since the event, an interpretation which is compatible with non-durative verbs. Verb copying is acceptable, however, with a temporal expression of frequency.

A NP which is an outer complement has to be identified by a verb in the same way as a NP functioning as an inner complement, which explains the necessity of verb copying in the presence of both an inner and an outer complement. The need for a 1:1 relation between a verb and a NP is not valid if the postverbal constituents are all inner complements, and thus identifiable (by the verb) on the basis of the relation of subcategorisation. Consequently, verb copying is excluded here. It is not appropriate, either, to indicate the outer object of a

VOP, for the outer object depends on the entire VOP, and not only on the verb. The outer object must therefore be related to the VOP as a whole.

4. THE DIFFERENT FORMS FOR THE OUTER OBJECT

So far, I have only dealt with the outer object marked by gěi which is only one of the different forms the outer object can take. In the present chapter, I will introduce the whole array of forms in which the outer object can appear. Starting from these data, I will then discuss the possible links between the outer object and its form as well as the predictability of the form for a given VOP. I will also deal with the problem of which underlying representation is to be postulated for the VOPs and I will give evidence in favour of the representation proposed in the preceding chapter (cf. 3.1.), namely $[_{V}[_{V} \text{ verb inner object }] \text{ outer object }]$.

4.1. Marking with bǎ

That the coverb or preposition bǎ can appear in the structure $[_s \text{ NP } [_{V}[_{pp} \text{ bǎ NP}] [_{V} \text{ V NP}]]$, i.e. that it can mark the outer object of a VOP, is in fact nothing new, for this structure is the same as that in the "retained object construction": besides the object preposed to the verb and marked by bǎ, there is another object following the verb, "the retained object":¹

¹ In the "normal" case of the bǎ-construction, there is no object following the verb, the object being preposed to the verb and marked by bǎ:

- (i) 他打死了張三
 Ta dasi -le Zhangsan
 3SG beat:to:death-PERF Zhangsan
- (ii) 他把張三打死了
 Ta ba Zhangsan dasi -le
 3SG BA Zhangsan beat:to:death-PERF
- He has beaten Zhangsan to death.

(1) 他把橘子剝了皮

Ta ba juzi bo -le pi
3SG BA orange remove-PERF skin

He has peeled an orange.

The retained object construction is said to be possible only if a special kind of relation holds between the postverbal NP and the NP marked by bǎ, e.g. a part - whole relation as illustrated in (1). (For a discussion of the other relations to be observed in the retained object construction, cf. Thompson 1973). But as demonstrated in sentences (2) - (10), the retained object construction is in fact only a special case of the VOPs; for the outer object can be marked by bǎ though none of the relations said to be necessary for that construction holds between the inner and the outer object:

(2) 上級把他罷職了

Shangji ba ta ba zhi le
boss BA 3SG stop work PART

The boss has fired him.

(3) 把這份禮物打一個包

Ba zhei-fen liwu da yi-ge bao
BA this-CL present tie:up 1 -CL parcel

Please wrap up this present.

(4) 你把號碼倒個過兒就對了

Ni ba haoma dao ge guor jiu dui le
2SG BA number invert CL time then correct PART

If you invert the numbers, it will be correct.

(5) 他還沒有把文件歸檔

Ta hai mei ba wenjian gui dang
3SG still NEG BA document return file

He has not filed the documents yet.

(6) 你應該把這個菜回鍋

Ni yinggai ba zhei-ge cai hui guo
2SG must BA this-CL meal return pot

You have to cook this meal again.

- (7) 校長把他降級了
 Xiaozhang ba ta jiang ji le
 headmaster BA 3SG fall rank PART
 The headmaster has sent him to a lower grade.
- (8) 他把這件事情揭蓋子
 Ta ba zhei-jian shiqing jie gaizi
 3SG BA this-CL matter take:off lid
 He has revealed this affair.
- (9) 把你需要的書開一個單子
 Ba ni xuyao de shu kai yi-ge danzi
 BA 2SG need SUB book open 1 -CL list
 Make a list of the books you need.
- (10) 你先把這篇文章起一個稿
 Ni xian ba zhei-pian wenzhang qi yi-ge gao
 2SG first BA this-CL article work:out 1 -CL draft
 Make a draft of this article first.

4.2. Marking with gěi

Another preposition or coverb which can mark the outer object of a VOP is gěi. The cases where it is beyond doubt that gěi really functions as the marker of the outer object are those VOPs which allow their outer object to be marked either by gěi or by bǎ:

- (11) 他給／把門安上鎖
 Ta gei/ba men an -shang suo
 3SG GEI/BA door install-DIR lock
 He has fitted a lock on the door.
- (12) 他給／把這個商品標上價
 Ta gei/ba zhei-ge shangpin biao-shang jia
 3SG GEI/BA this-CL merchandise mark-DIR price
 He has marked the price of this item.
- (13) 你應該給／把乾電池冲點兒電
 Ni yinggai gei/ba gandianchi chong dianr dian
 2SG must GEI/BA battery fill a:little electricity
 You have to load the battery.

- (14) 上級給／把他調職

Shangji gei/ba ta diao zhi
boss GEI/BA 3SG move work

The boss has transferred him to another post.

- (15) 你應該給／把這些材料分分類

Ni yinggai gei/ba zheixie cailiao fen fen lei
2SG must GEI/BA these data divide divide category

You have to classify these data.

- (16) 他給／把門塗塗漆

Ta gei/ba men tu tu qi
3SG GEI/BA door spread: on spread: on paint

He has given the door a coat of paint.

Though there are only a few examples of this type they are important in so far as the alternation of gěi with bǎ shows clearly that gěi here really functions as the marker of the outer object. A possible confusion between this gěi and the gěi indicating the benefactive (cf. (17)) is excluded here because bǎ can never indicate the benefactive:

- (17) 我給他買了一本書

Wo gei ta mai -le shu
1SG GEI 3SG sell-PERF book

I have sold a book for him.²

Thus, the VOPs in the following examples can be divided into two groups. In the first group ((18) - (24)), there is no doubt about the interpretation of the gěi-phrase as the outer object, i.e. as being affected by the action expressed in the VOP. In the second group ((25) - (31)), however, it is not possible to

² Like for him in English, gěi tā can in fact have two interpretations here: (i) 'for his sake' or (ii) 'on his behalf'. But since I want to concentrate on the difference between the gěi marking the outer object of a VOP and the gěi indicating the benefactive, I will put aside these further differences of interpretation within the benefactive reading of gěi tā.

clearly distinguish between the interpretation 'patient' and the interpretation 'benefactive' for the gěi-phrase. In fact, in most cases - especially when the inner object denotes a body part - we obtain a kind of mixed interpretation, i.e. the person profiting from the action is at the same time affected by it:

Group I

- (18) 他們給樹苗編號兒
Tamen gei shumiao bian haor
3PL GEI sapling arrange number
They have numbered the saplings.
- (19) 他在給車胎打氣
Ta zai gei chetai da qi
3SG DUR GEI tyre beat air
He is inflating the tyre.
- (20) 他給朋友打氣
Ta gei pengyou da qi
3SG GEI friend beat air
He cheers up his friends.
- (21) 護士給病人打針
Hushi gei bingren da zhen
nurse GEI patient beat needle
The nurse gives the patient an injection.
- (22) 她給她愛人戴綠帽子
Ta gei ta airen dai lü maozi
3SG GEI 3SG husband wear green hat
She has made her husband a cuckold.
- (23) 醫生給病人開刀
Yisheng gei bingren kai dao
doctor GEI patient open knife
The doctor operates on the patient.
- (24) 朋友給他扣了帽子
Pengyou gei ta kou -le maozi
friend GEI 3SG put:on-PERF hat
His friends have denounced him.

Group II:

- (25) 媽媽去給他報名
 Mama gei ta bao ming
 mother GEI 3SG report name
 His mother went to register for him.
- (26) 他可給我出氣了
 Ta ke gei wo chu qi le
 3SG really GEI 1SG come:out air PART
 He has really given vent to my anger.
- (27) 我們給他打了掩護
 Women gei ta da -le yanhu
 1PL GEI 3SG make-PERF cover
 We have provided cover for him.
- (28) 我給他們講了和
 Wo gei tamen jiang-le he
 1SG GEI 3SG talk -PERF peace
 I have reconciled them.
- (29) 觀眾不停地給演員喝采
 Guanzhong buting -de gei yanyuan he cai
 audience incessant-SUB GEI actor shout applause
 The audience kept on applauding the actors.
- (30) 他給我梳頭
 Ta gei wo shu tou
 3SG GEI 1SG comb head
 She combs my hair.
- (31) 他給我鑲牙
 Ta gei wo xiang ya
 3SG GEI 1SG insert tooth
 He has put in a (new) tooth.

One would like to use passivisation as a criterion to distinguish between the gěi-phrase indicating the benefactive and therefore excluding passivisation, and the gěi-phrase denoting the outer object, i.e. a patient, which in principle can be passivised. Unfortunately, however, passivisation cannot serve as a systematic test here, because for a NP in Chinese to play

the role of patient is a necessary, but not sufficient condition for passivisation.³ Thus, even among those VOPs whose outer object (marked by gěi) clearly plays the role of patient, only a few admit passivisation; additionally, it is impossible to make any predictions on the acceptability or unacceptability of the passive construction for a given VOP. For example, the outer object of dǎ zhēn 'to give an injection', dài lǚ màozi 'to make a cuckold' and kòu màozi 'to denounce' can each be passivised; whereas the VOPs biàn hàor 'to number' and dǎ qì 'to inflate; to cheer up' do not allow passivisation of their outer object:

- (32) 病人被護士打了一針
 Bingren bei hushi da -le yi-zhen
 patient PASS nurse beat-PERF 1 -needle
 The patient has been given an injection by the nurse.
- (33) 他被戴了綠帽子
 Ta bei dai -le lǜ maozi
 3SG PASS wear-PERF green hat
 He has been made a cuckold.
- (34) 病人被醫生開刀
 Bingren bei yisheng kai dao
 patient PASS doctor open knife
 The patient has been operated on by the doctor.
- (35) 他被朋友扣了帽子
 Ta bei pengyou kou -le maozi
 3SG PASS friend put:on-PERF hat
 He has been denounced by his friends.

³ The degree of "Transitivity" (in the sense of Hopper/Thompson 1980) must also be taken into account for the passive. The notion of "Transitivity" will be discussed in more detail in section 4.3. below.

(36) *樹苗被編號兒了

Shumiao bei bian haor le
sapling PASS arrange number PART

(The saplings have been numbered.)⁴

(37) *他的朋友被他打氣了

Ta -de pengyou bei ta da qi le
3SG-SUB friend PASS 3SG beat air PART

(His friends have been cheered up by him.)

This indeterminacy as to the function of the gěi-phrase (patient or benefactive) implies that it cannot be decided with certainty whether a given verb phrase 'V N' is an instance of a transitive VOP or whether it is a verb phrase like zuò fàn 'to prepare a meal' which optionally allows a benefactive. The unacceptability of verb copying which had served to identify the VOPs (cf. above, 3.2.4.) is of no use here because verb copying is in any case not appropriate for indicating the benefactive:

(38a) *我賣一本書賣了他

Wo mai yi-ben shu mai -le ta
1SG sell 1 -CL book sell-PERF 3SG

⁴ There can be no doubt about the role of patient of shùmiáo 'sapling' in (18) Tāmen gěi shùmiáo biàn hào 'They have numbered the saplings', because only animate nouns can possibly play the role of benefactive. Consequently, the unacceptability of (36) cannot be explained by the impossibility to passivise a benefactive NP. However, as the following example (due to Fu Jingqi) shows, passivisation of the outer object of biàn hào 'to number' does not seem to be excluded in general:

(i) 犯人被騙了號兒以後...

Fanren bei bian -le haor yihou
prisoner PASS arrange-PERF number after

After the prisoners had been numbered, ...

Perhaps the difference in acceptability between (36) and (i) has to do with the inanimate vs. animate nature of the outer object.

(38b) 我給他賣了一本書

Wo gei ta mai -le yi-ben shu
1SG GEI 3SG sell-PERF 1 -CL book

I have sold a book for him.

In fact, we can only be certain that a gěi-phrase represents the outer object of a VOP in the case of inanimate NPs, the role of benefactive being limited to animate NPs.

Since passivisation does not work as a test for all VOPs, and since, furthermore, semantic considerations of the type: Is the gěi-phrase affected by the action in the verb phrase or not, are too subjective to be a reliable criterion, I will not refer to these problematic cases of VOPs when trying to decide upon a particular analysis.⁵

Though among the VOPs whose outer object is marked by gěi there are a lot of unclear cases, gěi, nevertheless, is the most frequent marker for the outer object. Thus, even after the (about fifty) dubious cases have been excluded there still remain about a hundred VOPs marking their outer object with gěi (compared to a total of approximately 400 VOPs). Perhaps the comparison of gěi with bǎ can shed some light on the reason for this dominant role of gěi.

4.3. bǎ vs. gěi

Let us again have a look at examples (11) - (16), (partially) repeated here in (39) - (42), where the outer object can be

⁵ Since in a topic-comment structure several relations can hold between the NP in topic position and the comment (sentence), the topicalisation of a NP does not necessarily allow one to conclude its role as the outer object of a VOP. Cf. below, 4.6 and 6.2.

marked by bǎ or by gěi. The choice between bǎ and gěi is, however, not as free as it seems at first sight; thus, for ān suǒ 'to fit a lock', biào jià 'to mark the price', fēn lèi 'to classify' and tú qī 'to give a coat of paint', the marking with bǎ is possible only when the verb is made more complex either by adding the directional -shàng ((39a), (40a)) or by reduplicating the verb ((41a), (42a)):

(39a) 他把門安*(上)鎖

Ta ba men an *(-shang) suo
3SG BA door install -DIR lock

(39b) 他給門安(上)鎖

Ta gei men an (-shang) suo
3SG GEI door install -DIR lock

He has fitted a lock on the door.⁶

(40a) 他把這個商品標*(上)價

Ta ba zhei-ge shangpin biao *(-shang) jia
3SG BA this-CL merchandise mark -DIR price

(40b) 他給這個商品標(上)價

Ta gei zhei-ge shangpin biao (-shang) jia
3SG GEI this-CL merchandise mark -DIR price

He has marked the price of this item.

(41a) 你應該把這些材料分*(分)類

Ni yinggai ba zheixie cailiao fen *(fen) lei
2SG must BA these data divide divide category

(41b) 你應該給這些材料分(分)類

Ni yinggai gei zheixie cailiao fen (fen) lei
2SG must GEI these data divide divide category

You have to classify these data.

⁶ The notation *(X) indicates that the element X must be present and that, consequently, the sentence is unacceptable without X. The notation (*X), however, means that the sentence is acceptable only when X is not present.

(42a) 他把門塗 * (塗) 漆

Ta ba men tu *(tu) qi
3SG BA door spread:on spread:on paint

(42b) 他給門塗 (塗) 漆

Ta gei men tu (tu) qi
3SG GEI door spread:on spread:on paint

He has given the door a coat of paint.

The greater complexity of the verb (phrase) as a condition for the bǎ-construction is well-known and generally interpreted as a means to heighten the effect which the action expressed by the VP has on the bǎ-object. However, no attention has been paid to the fact that gěi can also appear in such a construction and have the same function as bǎ, i.e. to indicate the impact of an action on the object. Thus, there is no fundamental difference between bǎ and gěi, but only a difference in terms of degrees, as the alternation between bǎ and gěi in examples (39) - (42) shows: the outer object marked by bǎ is understood to be more affected by the action than the outer object marked by gěi (which perhaps is the reason for the special conditions the verb has to satisfy in the bǎ-construction.)

The difference between bǎ and gěi resides in the different degree of "Transitivity", Transitivity being used here in the sense of Hopper/Thompson(1980): Transitivity is not reduced to the relation between the verb and its object, but it is regarded as a relationship which obtains throughout the clause and which is graded.⁷ That in the bǎ-construction the relation between the VP and the bǎ object is characterized by a high degree of

⁷ "Transitivity involves a number of components only one of which is the presence of an object of the verb. These components are all concerned with the effectiveness with which an action takes

Transitivity is also a well-known fact, but it has not been taken into account here that gěi is a kind of "weak" variant (with respect to the degree of Transitivity) of bǎ.

This point of view, however, can perhaps explain why for tú yóu 'to spread on grease' = 'to grease' only gěi is allowed as marker of the outer object (cf. (43b)), whereas tú qī 'to spread on paint' = 'to give a coat of paint' allows bǎ as well as gěi (cf. (42)):

(43a) *他把機器部件塗(塗)油
 Ta ba jiqibujian tu (tu) you
 3SG BA machine:part spread:on spread:on grease

(43b) 他給機器部件塗(塗)油
 Ta gei jiqibujian tu (tu) you
 3SG GEI machine:part spread:on spread:on grease

He is greasing the machine parts.

The impossibility for most of the VOPs to substitute gěi for bǎ (for example in the case of bō pí 'to remove skin' = 'to peel' as well as in sentences (2) - (10)) would then be due to the high degree of Transitivity of the given action which cannot be adequately indicated by gěi. On the other hand, that gěi is not replacable by bǎ (cf. (43) and sentences (18) - (24)) would be accounted for by a degree of Transitivity which is not high

 (footnote 7 continued)

place, e.g. the punctuality and telicity of the verb, the conscious activity of the agent, and the referentiality and degree of affectedness of the object. These components co-vary with one another in language after language, which suggests that Transitivity is a central property of language use." (Hopper/Thompson 1980: 251)

Like Hopper/Thompson(1980), I will write "Transitivity" with a capital letter T when referring to the phenomenon of transitivity as defined by them.

enough for the use of bǎ.⁸ One could speculate that it is the non-maximal degree of Transitivity associated with gěi which is the reason for its being the most frequent marker of the outer object. Contrary to bǎ which clearly indicates a maximally high degree of Transitivity, marking with gěi seems to be less "compromising", more vague as to the degree of Transitivity of the action in question, and therefore has a larger distribution. The somewhat imprecise information which the marking with gěi gives about the degree of affectedness of the object is completed by the individual meaning of each VOP.

Given the preceding discussion of bǎ and gěi we are now able to summarize in two points the main conclusions obtained from the existence of different forms for the outer object.

First, the fact that the outer object can be marked by different coverbs (bǎ, gěi, gēn) does not imply that the outer object plays different semantic roles. The role of the outer object is always the same, i.e. patient of the action expressed in the VOP. What can vary, however, is the extent of the impact of the action on the outer object, and it is these variations in the degree of Transitivity of the VOPs which give rise to the different forms.

Secondly, it is impossible to predict the form of the outer object, not to mention the eventuality of a choice between several forms; consequently, the forms of the outer object must

⁸ The only examples in my corpus for a free variation between bǎ and gěi are the VOPs chōng diàn 'to fill electricity' = 'to load (a battery)' and diào zhǐ 'to move work' = 'to transfer sb. (to another post)'; cf. above, (13) and (14).

be included in the lexical entry of each VOP. Probably, the form of the outer object is not predictable because it is impossible to determine the degree of Transitivity for a given VOP in isolation, i.e. without knowing its use in a sentence.

It is this unpredictability which constitutes the major "inelegance" of analyses based on the notion of Transitivity. Once the marking of the outer object is known, it can rather easily be accounted for in terms of Transitivity; the inverse procedure, however (to start from the notion of Transitivity in order to obtain the form of the outer object for a particular VOP), does not work at all.

In fact, we are confronted here with the same problem as in the analysis of the passive. Thus, it is a well-established fact that the verb phrase must show a certain degree of Transitivity for passivisation to be able to apply. But again it is impossible to determine the necessary degree of Transitivity in such a way as to make passivisation a predictable phenomenon.

4.4. Marking with gēn

The difficulty we observed in the case of VOPs marking their outer object with gěi (cf. 4.2.), i.e. to decide whether a given verb phrase represents a transitive VOP or not, is even more complex in the case of the preposition or coverb gēn. As a matter of fact there are only very few examples (cf. (44) - (49)) where the NP marked by gēn can plausibly be said to be affected by the action:

- (44) 別跟我們湊熱鬧
 Bie gen wo cou renao
 NEG GEN 1SG collect bustle
 Don't bother us!⁹
- (45) 他在上課，你別跟他打岔
 Ta zai shang ke , ni bie gen ta da cha
 3SG DUR go class 2SG NEG GEN 3SG beat junction
 He is teaching, don't interrupt him.
- (46) 別跟我打哈哈
 Bie gen wo da haha
 NEG GEN 1SG beat laughter
 Don't make fun of me.
- (47) 他存心跟我搗蛋
 Ta cunxin gen wo dao dan
 3SG intentionally GEN 1SG pound egg
 He intentionally gives me trouble.
- (48) 你對我這樣客氣，別跟我見外
 Ni dui wo zheyang keqi, bie gen wo jian wai
 2SG towards 1SG so polite NEG GEN 1SG see foreign
 You are so polite to me; don't treat me as a stranger.
- (49) 我只是開玩笑，你可別跟我認真
 Wo zhishi kai wanxiao, ni ke bie gen wo
 1SG only open joke 2SG really NEG GEN 1SG
 ren zhen
 recognize true
 I'm only joking; for heaven's sake, don't take me seriously.

In most of the other cases where a verb phrase co-occurs with a gēn-phrase, the NP marked by gēn cannot be considered as being affected, i.e. it cannot be understood as the patient of the

⁹ The coverb or preposition gēn can either function as a coordinating conjunction within the NP meaning 'and' or it can have the meaning '(together) with'. As it is very difficult to find an adequate translation when it marks the outer object of a VOP, I prefer to simply gloss it as: GEN.

action, but represents a kind of comitative. Since it is the meaning of the verb which demands this comitative gēn-phrase, I label it "lexical comitative" following Teng(1975: 11) here who calls the gēn in this quasi-obligatory gēn-phrase a "lexical phrasal conjunction". The lexical comitative is found in combination with the so-called "reciprocal" verbs like dǎ jiāodào 'to make relation' = 'to make contact with' (cf. (50)), dǎ jià 'to beat fight' = 'to fight' (cf. (51)), jiǎng jià 'to talk price' = 'to bargain', jié hūn 'to tie marriage' = 'to marry', tán tiān 'to talk sky' = 'to chat', tǎolùn 'to discuss' and chǎo 'to quarrel' (cf. (52)):

- (50) 我從來沒跟他打過交道
 Wo cong lai mei gen ta da -guo jiaodao
 1SG ever NEG GEN 3SG make-EXP relation
 I have never had any contact with him.

- (51) 張三老跟人打架
 Zhangsan lao gen ren da jia
 Zhangsan always GEN people beat fight
 Zhangsan always fights with somebody.

- (52) 他天天跟我吵
 Ta tiantian gen wo chao
 3SG daily GEN 1SG quarrel
 Every day he quarrels with me.

As shown in examples (50) - (52), the reciprocal verbs comprise VOPs as well as simple verbs. The preverbal position of the gēn-phrase therefore has nothing to do with the presence of a postverbal object in the case of VOPs, but it is the only position acceptable for the gēn-phrase.

Thus, on the one hand, the lexical comitative is different from the gēn-phrase representing the outer object of a VOP, and on the other hand, from the "normal" comitative, the latter playing an additional role and therefore being optional (e.g. Wǒ

(gēn tā) qù mǎi cài '1SG (GEN 3SG) go buy vegetables' = 'I go shopping (with him).'). Though the lexical comitative and the gēn-phrase indicating the outer object both depend on the verb, they must nevertheless be distinguished: whereas the relation between the subject and the lexical comitative is potentially reversible (if e.g. X is in contact with Y, it follows that Y is in contact with X (cf. (50))), the situation is different for VOPs. Here, the action is clearly uni-directional and thus non-reversible, i.e. a patient (the gēn-phrase) is affected by the subject's action (cf. (44) - (49)).

Actually, there are only very few cases where the gēn-phrase can plausibly be said to represent the outer object of a VOP (i.e. the patient). It is mainly for the sake of completeness that I have mentioned gēn in connection with the VOPs; furthermore, gēn will be referred to again in the ensuing discussion of the possessive object (cf. below, 4.5.).

4.5. The possessive object

When the outer object occupies the modifier position of the inner object and is marked by the subordination marker de, the outer object is said to take the form of the "possessive object":

- (53) 張三出你的洋相
 Zhangsan chu ni -de yangxiang
 Zhangsan come:out 2SG-SUB foreign:face
 Zhangsan brings you into derision.

- (54) 他丟我的臉了
 Ta diu wo -de lian le
 3SG lose 1SG-SUB face PART
 He made me lose face.

- (55) 上級革了他的職了
 Shangji ge -le ta -de zhi le
 boss expel-PERF 3SG-SUB work PART
 The boss has fired him.
- (56) 我們開他的玩笑
 Women kai ta -de wanxiao
 1PL open 3SG-SUB joke
 We make fun of him.
- (57) 他揩我的油了
 Ta kai wo -de you le
 3SG wipe 1SG-SUB grease PART
 He scrounged me.
- (58) 法院治了他的罪了
 Fayuan zhi -le ta -de zui le
 law:court punish-PERF 3SG-SUB guilt PART
 The law court sentenced him.
- (59) 你別鑽他的空子
 Ni bie zuan ta -de kongzi
 2SG NEG bore 3SG-SUB opening
 Don't trick him.

One speaks of possessive object here because the outer object looks exactly like a modifier NP indicating possession, as for example wǒ-(de) in wǒ-de shū '1SG-SUB book' = 'my book'. That the term "possessive object" derives from this analogy with a possessive NP like wǒ-de 'my' can be seen very clearly in Lü Shuxiang's description of this phenomenon: "Sometimes, the genitive does not have its common meaning of possession at all, but it represents the person who is directly or indirectly acted upon, thus corresponding to an (accusative, dative, or ablative etc.) object." (Lü Shuxiang 1955: 186; my translation, W.P.). The same point of view is expressed by Chao:

The grammatical modifier of an object sometimes does not modify the object semantically, but represents something else. When an apparent possessive modifier ... represents the goal, I call it "possessive object". For example, in Bié kāi tā-de wánxiào 'Don't make his fun - don't make fun of him'

[cf. (56), W.P.], it is not his fun, but your own fun that you are told not to make of him, so that the real goal is the tā of tā-de.

(Chao 1968: 321)

Lü Shuxiang further stresses the fact that the possessive object plays the role of patient with respect to the VOP as a whole:

This kind of genitive [i.e. the possessive object, W.P.] most often occurs with verbs which have an idiomatic object; the two components play, in fact, the role of one simple verb. This kind of verb-object construction can on its part have a (logical) object (the person acted upon), but since there is already an object on the surface and since there is no appropriate preposition available, either, this logical object often takes on the form of a genitive. ... This phenomenon reminds one of the idiomatic expressions in English like take care of, get hold of etc. This kind of idiomatic expressions allows passivisation of the form it is well taken care of ...

(Lü Shuxiang 1955: 186 - 187; my translation, W.P.)

The syntactic proof for Lü Shuxiang's analysis is given by the possibility - at least in some cases - to passivise the possessive object (cf. (60) - (62)); a real modifier, however, can never be passivised (cf. (63b)):

(60) 他被上級革了職

Ta bei shangji ge -le zhi
3SG PASS boss expel-PERF work

He was fired by his boss.

(cf. (55))

(61) 他被法院治了罪

Ta bei fayuan zhi -le zui
3SG PASS law:court punish-PERF guilt

He was sentenced by the law court.

(cf. (58))

(62) 他被人家開了個大玩笑

Ta bei renjia kai -le ge da wanxiao
3SG PASS people open-PERF CL big joke

He was terribly made fun of.

(cf. (56))

(63a) 他打破了我的杯子

Ta dapo -le wo -de beizi
3SG break-PERF 1SG-SUB cup

He broke my cup.

(63b) *我被他打破了杯子

Wo bei ta dapo -le beizi
1SG PASS 3SG break-PERF cup

(63c) 我的杯子被他打破了

Wo -de beizi bei ta dapo -le
1SG-SUB cup PASS 3SG break-PERF

My cup was broken by him.

Though passivisation of the outer object is not predictable for a given VOP, it nevertheless can serve as a criterion for distinguishing between the possessive object and a real modifier, for a true modifier NP can never be passivised, even if the verb shows the necessary high degree of Transitivity. (As illustrated in (63c), dǎpò 'to break' can appear in a passive sentence, provided the entire NP wǒ-de bēizi 'my cup' is the subject.) (Cf. chapter 6 for a comparison between the syntactic behaviour of a true modifier NP and that of the possessive object.)

Besides the VOPs given in examples (53) - (59) whose outer object can only take on the form of the possessive object, the following VOPs allow their outer object either to appear as the possessive object or to be marked by a coverb (also cf. Chao 1968: 321):

(64a) 上級罷了他的職了

Shangji ba -le ta -de zhi le
boss stop-PERF 3SG-SUB work PART

(64b) 上級把他罷了職了

Shangji ba ta ba -le zhi le
boss BA 3SG stop-PERF work PART

The boss has fired him.

- (65a) 我幫他的忙
Wo bang ta -de mang
1SG assist 3SG-SUB activity
- (65b) 我給他幫忙
Wo gei ta bang mang
1SG GEI 3SG assist activity

I have given him my assistance.
- (66a) 你別打他的岔
Ni bie da ta -de cha
2SG NEG beat 3SG-SUB junction
- (66b) 你別跟他打岔
Ni bie gen ta da cha
2SG NEG GEN 3SG beat junction

Don't interrupt him.
- (67a) 他揭了這件事的蓋子
Ta jie -le zhei-jian shi de gaizi
3SG take:off-PERF this-CL matter SUB lid
- (67b) 他把這件事揭了蓋子
Ta ba zhei-jian shi jie -le gaizi
3SG BA this-CL matter take:off-PERF lid

He has revealed this affair.
- (68a) 他拍張三的馬屁
Ta pai Zhangsan-de mapi
3SG pat Zhangsan-SUB horse:buttocks
- (68b) 他給張三拍馬屁
Ta gei Zhangsan pai mapi
3SG GEI Zhangsan pat horse:buttocks

He flatters Zhangsan.
- (69a) 我不捧他的場
Wo bu peng ta -de chang
1SG NEG hold 3SG-SUB stage
- (69b) 我不給他捧場
Wo bu gei ta peng chang
1SG NEG GEI 3SG hold stage

I don't boost him.

But there are also cases where only the possessive object indicates the outer object, whereas the coverb phrase plays a

different role (cf. Chao 1968: 322):

- (70a) 張三說你的壞話
 Zhangsan shuo ni -de huaihua
 Zhangsan talk 2SG-SUB vicious:talk
 Zhangsan gossips about you.
- (70b) 張三跟你說壞話
 Zhangsan gen ni shuo huaihua
 Zhangsan GEN 2SG talk vicious:talk
 Zhangsan gossips with you (about sb. else).
- (71a) 他告我的狀
 Ta gao wo -de zhuang
 3SG accuse 1SG-SUB complaint
 He has told on me.
- (71b) 他跟我告狀
 Ta gen wo gao zhuang
 3SG GEN 1SG accuse complaint
 He informed me (about sb. else).

But it is certainly no coincidence that the meaning of these VOPs implies a third person to whom the agent directs its action. Since the coverb phrase and the possessive object play different roles, they can co-occur in the same sentence:

- (72) 我跟張三說李四的壞話
 Wo gen Zhangsan shuo Lisi-de huaihua
 1SG GEN Zhangsan talk Lisi-SUB vicious:talk
 I gossip with Zhangsan about Lisi.
- (73) 我跟你父母告你的狀
 Wo gen ni fumu gao ni -de zhuang
 1SG GEN 2SG parents accuse 2SG-SUB complaint
 I inform your parents about you.

After the presentation of the data concerning the possessive object, let us now see what can be said about the possessive object in terms of the notion of Transitivity, paying special attention to those cases where the outer object can either appear as the possessive object or be marked by a coverb.

That the outer object of both bà zhí 'to stop work' = 'to fire' and jiē gàizi 'to take:off lid' = 'to reveal' can be passivised (cf. (74) and (75)) is not very surprising because besides taking on the form of the possessive object, the outer object can also be marked by bǎ which implies a degree of Transitivity (in principle) high enough for passivisation. (An outer object marked by gěi, though, can also be passivised, cf. (32) - (35).)

- (74) 他被上級罷了職了
 Ta bei shangji ba -le zhi le (cf. (64))
 3SG PASS boss stop-PERF work PART
 He has been fired by his boss.

- (75) 這件事被他揭蓋子了
 Zhei-jian shi bei ta jie gaizi le (cf. (67))
 this-CL matter PASS 3SG take:off lid PART
 This affair has been revealed by him.

It is much more difficult to determine the degree of affectedness of the outer object when it exclusively appears as the possessive object and cannot be alternatively marked with a coverb. Thus, gé zhí 'to expel work' = 'to fire' (cf. (55)), kāi wánxiào 'to open joke' = 'to make fun of sb.' (cf. (56)), zhì zuì 'to punish guilt' = 'to sentence/condemn' (cf. (58)) and zuān kòngzi 'to bore opening' = 'to trick, to deceive' (cf. (59)) all allow their outer object to be passivised, in contrast to the remaining VOPs in the group of examples (53) - (59). From the VOPs where the possessive object and the coverb phrase play different roles (cf. (70) and (71)), only the VOP gào zhuàng 'to tell on sb., to sue sb.' allows passivisation (Wǒ bèi wǒ dìdì gào-le zhuàng '1SG PASS 1SG brother accuse-PERF complaint' = 'I was given away by my brother').

If it is difficult to set up a direct correlation between the possessive object and a certain degree of Transitivity, the

degree of Transitivity associated with the form of the possessive object can perhaps be determined indirectly by means of a correlation between the possessive object and a particular coverb. But as shown in examples (64) - (69), this procedure fails as well because the possessive object can alternate with the coverbs bǎ ((64), (67)), gěi ((65), (68), (69)) and gēn (66), respectively.

This lack of consistency displayed by the forms available for the outer object becomes especially obvious in the case of the VOPs with zhí 'work' as inner object which all have the meaning 'to fire': bà zhí 'to stop work', chè zhí 'to remove work', gé zhí 'to expel work', jiě zhí 'to separate work', miǎn zhí 'to exempt work' and tíng zhí 'to stop work'.¹⁰ Though they all allow passivisation of their outer objects and can therefore be supposed to have more or less the same degree of Transitivity, they nevertheless differ in the form of their outer objects. Whereas they all accept the form of the possessive object, only the outer objects of bà zhí, chè zhí, jiě zhí and miǎn zhí, but not that of gé zhí and tíng zhí can be marked by bǎ. Marking with gēi, on the other hand, seems possible in all cases, though only in a very colloquial style.

That even in the case of VOPs with the same inner object and the same meaning, the form of the outer object varies and is not predictable, either, considerably weakens the explanatory value of the concept of Transitivity.

¹⁰ I have not included diào zhí 'to move work = 'to transfer sb. to another post' here because it belongs to the written language (as was pointed out to me by Fu Jingqi).

4.6. The underlying representation for the VOPs

If one gives up trying to account for the distribution of the different forms the outer object can take and instead concentrates on the given forms themselves, one observes that all the VOPs with zhí 'work' as the inner object allow the possessive object. This seems to suggest the underlying representation $[_V^V [_{NP} \text{ outer } 0. \text{ inner } 0.]]$ for the VOPs (which is different from the one postulated so far, i.e. $[_V^V [_V^V \text{ V inner } 0.] \text{ outer } 0.]$ (cf. above, 3.2.4.)).

In this representation, the outer object is part of a NP whose head is the inner object which seems to do more justice to the apparently very tight relationship between the inner object and the form of the outer object (as observed in the case of the VOPs 'V zhí' 'to fire'). Accordingly, the inner object, being the head of the NP, should play an important role in determining the form of the outer object (as is suggested by the VOPs 'V zhí'). In order to obtain an acceptable surface structure from this underlying representation, either the subordinator -de must be introduced between the outer object and the inner object, or the outer object must be marked by a coverb and appear in the preverbal position.

The underlying representation where the outer object occupies the modifier position of the inner object had first been proposed for the retained object construction (cf. (Cheung 1973: 377 - 380) because of pairs like (76a) and (76b):

- (76a) 他把橘子剥了皮
 Ta ba juzi bo -le pi
 3SG BA orange remove-PERF skin

He has peeled the orange.

- (76b) 他剝了橘子皮
 Ta bo -le juzi -de pi
 3SG remove-PERF orange-SUB skin

He has peeled the orange.

The alternation between a (preverbal) coverb phrase and a NP in a modifier position in the case of a possessive relation was seen as a further argument for the underlying representation in question:

- (77a) 他給孩子換衣服
 Ta gei haizi huan yifu
 3SG GEI child change clothes

- (77b) 他換孩子的衣服
 Ta huan haizi-de yifu
 3SG change child-SUB clothes

He changes the child's clothes.

In my opinion, however, the fact that a part-whole relation or a possessive relation must hold between the two NPs for the alternation to be possible seems too special a condition to be counted as an argument in favour of [_V V [_{NP} inner 0. outer 0.]] as the underlying representation for the VOPs. Besides, there are cases where no alternation is observed though the supposedly necessary relation (in this case a part-whole relation) exists:

- (78a) 我給他梳頭
 Wo gei ta shu tou
 1SG GEI 3SG comb head

I comb her hair.

- (78b) *我梳他的頭
 Wo shu ta -de tou
 1SG comb 3SG-SUB head

- (79a) 我給他擦背
 Wo gei ta ca bei
 1SG GEI 3SG rub back

I rub her back.

- (79b) *我擦他的背
 Wo ca ta -de bei
 1SG rub 3SG-SUB back

Furthermore, the alternation depends not only on the kind of relation between the two NPs, but also on the type of verb. If for example huàn 'to change' in huàn yīfu 'to change clothes' (cf. (77)) is replaced by chuān 'to wear, to put on', the resulting sentence with háizi 'child' as part of the coverb phrase (cf. 80a)) has a meaning completely different from the sentence where háizi is in the modifier position of yīfu 'clothes' (cf. (80b)); accordingly, the former sentence cannot be considered as a variant derived from the latter:

- (80a) 他給孩子穿衣服
 Ta gei haizi chuan yifu
 3SG GEI child wear clothes

He dresses the child.

- (80b) 他穿孩子的衣服
 Ta chuan haizi-de yifu
 3SG wear child-SUB clothes

He wears the child's clothes.

All these data demonstrate that there is no systematic alternation between the structure with the NP in a coverb phrase and the structure with the same NP in a modifier position. Consequently, there is no relation of derivation between these two structures; this hypothesis is probably due to the confusion between the identity of meaning and the identity of structure.

There are also considerations of a more general nature which speak in favour of the underlying representation [_V [_V V inner 0.] outer 0.] and against [_V V [_{NP} outer 0. inner 0.]].

First of all, the movement of the outer object out of the modifier position into the preverbal position (the coverb phrase) is a movement prohibited by Ross' "Left Branch Condition" (LBC).

This constraint is formulated as follows:

No NP which is the leftmost constituent of a larger NP can be moved out of this NP.

(Ross 1967: 4.181)

Thus, the derivation of e.g. Tā bǎ júzi bō-le pí (cf. (81a)) from Tā bō-le júzi-de pí (cf. (81b)) violates the LBC because in such a derivation júzi 'orange' being the leftmost constituent of júzi-de pí moves out of its original position (cf. Huang 1982': footnote 21):

(81a) 他把橘子剥了皮
Ta ba juzi bo -le pi
3SG BA orange remove-PERF skin

(81b) 他剥了橘子的皮
Ta bo -le juzi -de pi
3SG remove-PERF orange-SUB skin

He has peeled the orange.

Before pursuing this line of reasoning, I will first show that the LBC is also valid in Chinese.

At first sight, nothing speaks against the hypothesis that (82a) results from the movement of júzi 'orange' out of the modifier position of the NP into the topic position. Under this assumption, the part-whole relation between pí 'skin' and júzi 'orange' would be accounted for as well (cf. (82b)):

(82a) 橘子，他剥了皮
Juzi, ta bo -le pi
orange 3SG remove-PERF skin

The orange, he has peeled it.

(82b)

Juzi, ta bo -le [NP pi]

↑

But if (82b) represents the correct analysis for (82a), then how can the ungrammaticality of (83) and (84) which seem to have the

same structure as (82a) be explained ?

- (83) *張三，我看見了爸爸
 Zhangsan, wo kanjian-le baba
 Zhangsan 1SG see -PERF father

- (84) *張三，我看見了狗
 Zhangsan, wo kanjian-le gou
 Zhangsan 1SG see -PERF dog

Let us try to account for the three sentences (82a), (83) and (84) assuming that the LBC is valid in Chinese.

In this case, the ungrammaticality of (83) and (84) is expected because they both violate the LBC. The only link possible between the topic and the comment is that provided by a possessive relation between Zhāngsān and the NPs bàba 'father' and gǒu 'dog', respectively. Consequently, Zhāngsān must have moved to the topic position from its position within the NP [Zhāngsān-de bàba] and [Zhāngsān-de gǒu], respectively, thus violating the LBC and giving rise to ungrammatical sentences.

We still have to explain why (82a) is not unacceptable though it looks like (83) and (84). This entails having to inquire again about its underlying representation because the one proposed in (82b), violating the LBC, cannot be correct. According to Huang(1982': footnote 21), júzi 'orange' in (82a) is not the (topicalised) modifier of pí 'skin' (the assumption of which would imply a movement violating the LBC), but it represents the outer object of the VOP bō pí 'to peel' and accordingly can be topicalised. The part-whole relation observed between pí and júzi is not obtained from the syntactic configuration, but results from a pragmatic deduction: if the act of removing the skin is meant to affect a certain orange, the skin to be removed must evidently be the skin of the orange in question, and not that of a different one.

Thus, (82a) is acceptable because the noun in the topic position, júzi 'orange' is the outer object of the VOP bō pí 'to peel' contained in the comment sentence. Such an interpretation is impossible for (83) and (84), for it is not clear how Zhangsan could be affected by my having seen his father or his dog. This is clearly not to say that every noun in the topic position must be interpreted as an outer object. There is also the case of the base-generated topic which gives rise to the same structure as that in (82a), but where the link between the topic and the comment is established by the situational context, and not by a syntactic relationship:

- (85) 這個車禍，幸虧沒有人死了
 Zhei-ge chehuo, xingkuì mei you ren si -le
 this-CL accident fortunately NEG have people die-PERF

As for this accident, fortunately nobody died.

(83) and (84) are unacceptable because there is no link between the topic and the comment. First, the situational context does not give any clue as to the possible relation between the topic and the comment which excludes the analysis of (83) and (84) as structures with a base-generated topic. Secondly, a link between the topic and the comment via a possessive relation between the topic, Zhāngsān, on the one hand, and bàba 'father' gǒu 'dog', on the other hand, is excluded by the LBC. Thirdly, the last possibility of establishing a link between the topic and the comment, (i.e. to consider the noun in the topic position as the outer object of the verb phrase in the comment sentence) is not feasible, either, because it is semantically implausible.

To summarize, the relation between the topic and the comment can be of a syntactic, semantic or pragmatic nature, but it can also involve more than only one of these three components:

- (86) 張三，爸爸很有錢
 Zhangsan, baba hen youqian
 Zhangsan father very rich

As for Zhangsan, his father is very rich.

The acceptability of (86) seems to contradict our claim that the LBC is valid in Chinese, but Huang(1982: 516, footnote 4) shows that it can be accounted for by the special status of bàba. Being an expression of inalienable possession, bàba 'father' implies a possessor for which - by pragmatic deduction - the nearest NP in the topic position, Zhāngsān, is chosen. That (86) really involves a pragmatic procedure which interprets the nearest NP as possessor (and not a structural relation) is shown by the unacceptability of (83), repeated here as (87):

- (87) *張三，我看見爸爸了
 Zhangsan, wo kanjian baba le
 Zhangsan 1SG see father PART

Under the pragmatic procedure, the nearest NP, wǒ 'I', is understood to be the possessor implied by bàba; consequently, bàba is interpreted as wǒ bàba 'my father'. (87) is unacceptable because wǒ kànjiàn bàba le (which is perfectly acceptable as an independent sentence) does not say anything about Zhangsan and therefore is inappropriate as a comment on Zhangsan.

The situation is clearer when the subject in the comment sentence is not an expression of inalienable possession and where, accordingly, the NP in the comment sentence cannot be related to the NP in the topic position by a pragmatic deduction (cf. Huang 1982:516, footnote 4):

- (88) *張三，狗死了
 Zhangsan, gou si -le
 Zhangsan dog die-PERF

(88) is unacceptable for the same reason as (84): a possessive relation between Zhāngsān and gǒu which would provide the

necessary link between the topic and the comment is excluded by the LBC.

The difference between the structure with a base-generated topic and the structure where the outer object of a VOP is topicalised is, admittedly, not always very clear-cut. Sentence (82a) (Júzi, ta bō-le pǐ 'The orange, he has peeled it'), for example, could also be regarded as involving a base-generated topic; in that case, the general knowledge that oranges have to be peeled would provide the link between the topic and the comment. It is the existence of a corresponding sentence with bǎ which makes us analyse the topic noun in (82a) as the outer object of the VOP in the comment sentence (cf. (81a): Tā bǎ júzi bō-le pǐ 'He has peeled the orange').

What has been demonstrated clearly, however, is that a noun in the topic position cannot have moved there from the modifier position of a NP in the comment. The impossibility of such a derivation (regardless of the kind of relation holding between the two NPs in question) clearly argues against [_V V [_{NP} outer O. inner O.]] as the underlying representation for VOPs.

Besides the LBC, the impossibility of passivising a (real) modifier NP (cf. (63b)) constitutes additional evidence against the underlying representation in question and contrasts with the passivisability of the outer object. This latter argument is particularly strong because passivisation of the outer object also applies in those cases where the possessive relation or the part-whole relation between the inner object and the outer object seems so obvious as to speak in favour of the underlying representation [_V V [_{NP} outer O. inner O.]]:

- (89) 橘子被剥了皮
 Juzi bei bo -le pi (cf. (81))
 orange PASS remove-PERF skin

The orange has been peeled.

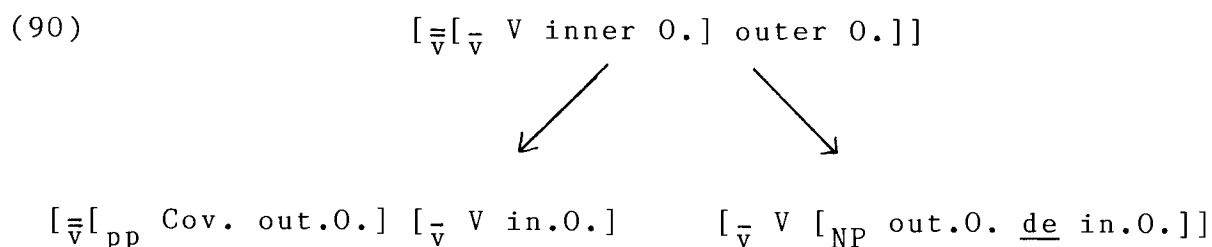
Even if the generalization that a modifier NP cannot be passivised were slightly modified so as to allow for passivisation of the possessive object, the passivisability of the outer object still remains a problem for the underlying representation under discussion, only removed to another level; for now there would exist two classes of modifier NPs (one allowing passivisation and the other excluding it) which would have to be distinguished in order to prevent ungrammatical sentences.

The third argument against [$\bar{V}$ V [$_{NP}$ outer O. inner O.]] as the underlying representation for the VOPs is closely linked to the passivisability of the outer object just discussed. Though an underlying representation should, as much as possible, reflect the semantic relations involved, [$\bar{V}$ V [$_{NP}$ outer O. inner O.]] does not take into account the passivisability of the outer object at all. On the contrary, it implies that it is the inner object, being the head noun of the NP, which is affected by the verb. In fact, however, it is the outer object which is the patient with respect to the VOP as a whole (Huang, p.c.).¹¹

The analysis I propose is as follows. The "real" possessive object with VOPs where no part-whole relation or possessive

¹¹ With [$\bar{V}$ V [$_{NP}$ outer O. inner O.]] as the underlying representation it would probably also be difficult to account for the alternation between bǎ and gěi as markers of the outer object. As we have shown above (cf. 4.3.), the choice of bǎ or gěi depends on the degree of Transitivity of the verb phrase in question. In the underlying representation under discussion, however, the outer object only entertains a relation with the inner object, but not with the verb, and it is therefore totally unexpected that a modification of the verb phrase can have an influence on the form of the outer object.

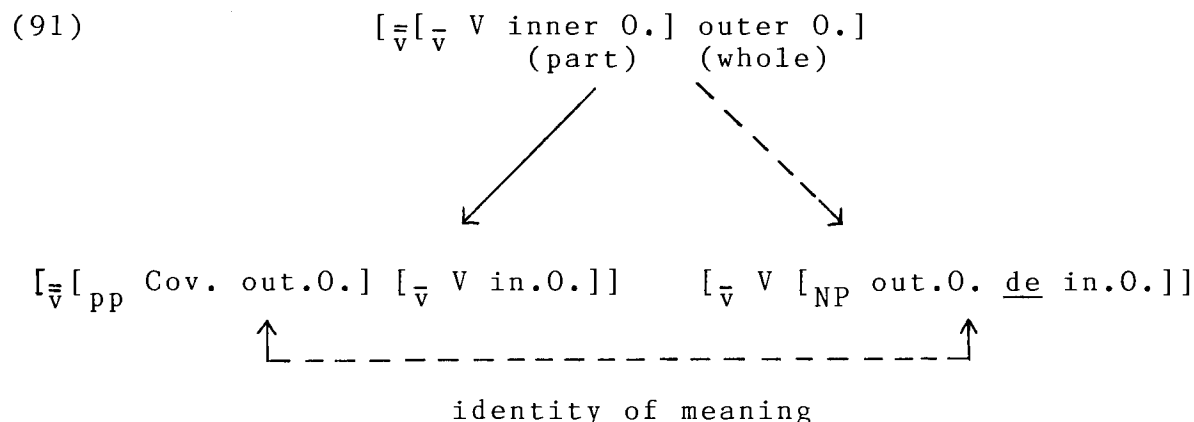
relation exists between the inner object and the outer object is the result of a reanalysis¹² operating on the underlying representation $[\bar{V}[\bar{V} V \text{ inner } 0.] \text{ outer } 0.]$. Accordingly, the underlying representation is the same for the VOPs whose outer object appears as the possessive object and the VOPs whose outer object is marked by a coverb (which makes plausible the availability of both forms for the outer object of some VOPs). We thus obtain the following relations between the underlying representation and the two surface structures where the outer object appears as the possessive object or is marked by a coverb, respectively:



The situation is more complicated for VOPs like bō pí 'to remove skin' = 'to peel' where a special relation (a part-whole relation) exists between the inner object and the outer object. The structure where the outer object is marked by a coverb, bǎ júzi bō pí (cf. (81a)) is clearly derived from the underlying representation $[\bar{V}[\bar{V} \text{ bō pí}] \text{ júzi}]$ (cf. (90)). Bō júzi-de pí (cf. (81b)), however, allows for two analyses: either júzi-de pí represents the possessive object and accordingly, bō júzi-de pí is a variant of bǎ júzi bō pí, or júzi-de pí is a "normal" NP with júzi as a real modifier and does not result from a reanalysis (as does the possessive object). In this latter case, it is the (quasi-) identity of meaning between the two sentences (resulting from the part-whole relation between pí 'skin' and júzi

¹² The reason why I speak of "reanalysis" here and not of "restructuring" will be discussed in chapter 6.

'orange'), and not the derivation from the same underlying representation which establishes a relation between bǎ júzi bō pí and bō júzi-de pí. Figure (91) depicts the two kinds of relations between bǎ júzi bō pí and bō júzi-de pí:



To summarize, there are several reasons why the underlying representation $[\bar{v} \text{ V } [\text{NP out.0. inner 0.}]]$ based on the existence of the possessive object is not appropriate. First, the movement of the outer object from its original position to the preverbal position, where it is marked by a coverb, constitutes a movement prohibited by the LBC. Secondly, with this underlying representation, the passivisability of the outer object is completely unexpected; furthermore, the so far valid generalization that modifier NPs cannot be passivised must be weakened to allow for those cases where the modifier is the possessive object (and thus passivisable). Thirdly, the relation of 'action - affected object' existing between the VOP and its outer object is not reflected at all in an underlying representation where the outer object appears as the modifier of the inner object.

4.7. The outer object in the indirect object position

In this final section, I will briefly discuss those cases where the outer object, instead of appearing as the possessive object or being marked by a coverb, simply follows the verb:

- (92) 我可以幫你一個忙?
 Wo keyi bang ni yi-ge mang? (cf. 65))
 1SG can assist 2SG 1 -CL activity
 Can I help you?
- (93) 你跟他說我還有別的女朋友，這是拆我台嘛!
 Ni gen ta shuo wo hai you biede nüpengyou;
 2SG GEN 3SG talk 1SG still have another girlfriend
 zhei shi chai wo tai ma!¹³
 this be take:apart 1SG platform PART
 You told her that I have another girlfriend; there you really let me down!
- (94) 他們要我唱歌，這可將了我一軍
 Tamen yao wo chang ge ,
 3PL want 1SG sing song
 zhei ke jiang -le wo yi-jun¹⁴
 this really embarrass-PERF 1SG 1 -general
 They wanted me to sing which really embarrassed me.
- (95) 他開了我一個玩笑
 Ta kai -le wo yi-ge wanxiao (cf. (56))
 3SG open-PERF 1SG 1 -CL joke
 He made fun of me.

¹³ This sentence is only acceptable in a very colloquial style. Otherwise, the possessive object is preferred:

- (i) 他拆我的台
 Ta chai wo -de tai
 3SG take:apart 1SG-SUB platform
 He lets me down.

¹⁴ jiāng X yī-jūn represents an idiomatic expression because there is no other form available for the outer object (which is always preceded by yī 'one'). But there can be no doubt that wǒ is the outer object, for it can be passivised:

- (i) 我被他將了一軍
 Wo bei ta jiang -le yi-jun
 1SG PASS 3SG embarrass-PERF 1 -general
 I was brought into an embarrassing situation by him.

- (96) 小心，他佔你便宜
 Xiaoxin, ta zhan ni pianyi
 attention 3SG seize 2SG cheap

Be careful, he wants to profit at your expense.

The comparison between the structure of sentences (92) - (96) and the double object construction illustrated in (97):

- (97) 我送他一本書
 Wo song ta yi-ben shu
 1SG give 3SG 1 -CL book

I give him a book (as a present).

shows that in sentences (92) - (96), the outer object occupies the same position as the indirect object in (97). The phenomenon involved here apparently is the same as that observed in connection with verb copying, namely restructuring modeled on the double object construction. As shown in chapter 3.2.1., verb copying (in the presence of a direct object and a complement of duration or of frequency) is not obligatory if the direct object is referential and animate/definite, for the resulting structure looks like the typical structure of double object verbs, where the verb is followed by an animate indirect object and an inanimate direct object. In fact, in all the sentences (92) - (96), the outer object is referential, animate and definite and thereby allows restructuring modeled on the double object construction: the (animate) outer object occupies the indirect object position and the (inanimate) inner object occupies the direct object position.

But contrary to verb copying whose optionality is predictable on the basis of the referential and definite/animate nature of the direct object, the cases where the outer object appears in the indirect object position cannot be predicted. On the contrary, this structure is only acceptable for some VOPs and represents the exception rather than the rule for referential and

animate/definite outer objects. I have no explanation to offer at this time but this phenomenon seems to deserve more attention because it can perhaps give us some indication as to the constraints governing restructuring.

In this chapter, we have given a complete overview of the forms available for the outer object. The outer object can be marked by bǎ, gěi, gēn, it can appear as the possessive object or be restructured as the indirect object in a double object construction.

The existence of different forms for the outer object does, however, not imply that the outer object plays different roles. The role of the outer object is always the same, i.e. patient of the VOP. What can vary, though, is the extent of the impact of the action on the outer object, and it is these variations in the degree of Transitivity of the VOPs which can give rise to the different forms.

Marking with the coverb gěi is by far the most frequent form, probably because gěi is associated with a non-maximal degree of Transitivity.

It is impossible to predict which form the outer object of a particular VOP will take, not to mention the eventuality of a choice between different forms. Consequently, the form(s) of the outer object must be included in the lexical entry of each VOP. Apparently, the form of the outer object is not predictable because it is impossible to determine the degree of Transitivity for a given VOP in isolation, i.e. without knowing its use in a sentence.

5. COMPLEX SENTENCES WITH GEI

In this chapter, I will explain why the outer object marked by gěi is only allowed in the preverbal position, and unacceptable in the postverbal position:

- (1a) 醫生給病人開刀
 Yisheng gei bingren kai dao
 doctor GEI patient open knife
 The doctor operates on the patient.

- (1b) *醫生開刀給病人
 Yisheng kai dao gei bingren
 doctor open knife GEI patient

In this respect, the VOPs are different from both verbs of giving (e.g. sòng 'to give (as a present)' and verbs like qī (chá) 'to brew (tea)', dǎ (máoyī) 'to knit (a sweater)' which all allow for a postverbal gěi-phrase:

- (2a) 我送一本書給他
 Wo song yi-ben shu gei ta
 1SG give 1 -CL book GEI 3SG
 I give him a book (as a present).

- (2b) 我沏茶給他
 Wo qi cha gei ta
 1SG brew tea GEI 3SG
 I make him some tea.

For verbs of giving, the acceptability of a postverbal gěi-phrase can be considered as a defining property of this verb class. Verbs like qī (chá) 'to brew (tea)' and dǎ (máoyī) 'to knit (a sweater)', however, do not belong to a clearly defined verb class; they only have in common the ability to express the production of a concrete entity and to optionally allow for a benefactive.

Given the complexity of the problems linked to gěi, I will first give a detailed summary of the most exhaustive study on gěi, i.e. Zhu Dexi(1979, 1983). I will then criticize it point by point and propose my own analysis which is different from Zhu's. This new analysis of the functioning of gěi will then be shown to account for the acceptability of the outer object marked by gěi in the preverbal position only.

5.1. The analysis of Zhu Dexi(1983, 1979)

In this section, I will present Zhu Dexi's article on "Bāohán dòngcí 'gěi' de fùzá jùshì" [Complex constructions with the verb gěi] published in 1983. His article dating from 1979 "Yǔ dòngcí 'gěi' xiāngguān de jùfǎ wèntí" ["Sentences containing the verb gěi] and dealing with more or less the same subject matter will only be taken into consideration here if it contains some information which is relevant to the problem but which has not been mentioned in the later article.

In order to facilitate reference to Zhu's article, I have followed his terminology as far as the summary is concerned. In my critical comment as well as in the presentation of my own analysis, however, I have sometimes, for the sake of convenience, preferred to use terms different from Zhu's.

Zhu Dexi(1983) distinguishes the following six construction types containing gěi:

S'1: Ns V1 gěi N' N V2P

我送給他一本書看

Wo song gei ta yi-ben shu kan
1SG give GEI 3SG 1 -CL book look
I give him a book to read.

S'2: Ns V1 N gěi N' V2P

我送一本書給他看

Wo song yi-ben shu gei ta kan
1SG give 1 -CL book GEI 3SG look
I give him a book to read.

S'3: Ns gěi N' V1 N V2P

我給他買輛車騎

Wo gei ta mai liang che qi
1SG GEI 3SG buy Cl bicycle ride
I buy a bicycle for him to ride on.

(p. 161)¹

S'4: Ns V1 N' N V2P

我送他一本書看

Wo song ta yi-ben shu kan
1SG give 3SG 1 -CL book look
I give him a book to read.

S'5: 我講個故事給你聽

Wo jiang ge gushi gei ni ting
1SG tell CL story GEI 2SG listen
I tell a story for you to listen to.

S'6: (你以為我不會騎馬)，我騎給你看

(Ni yiwei wo bu hui qi ma) wo qi gei ni kan
2SG think 1SG NEG know ride horse 1SG ride GEI 2SG look
(You think I don't know how to ride a horse), I'll ride one
to show you. (p. 162)

(Ns stands for the subject NP, V1 for the first verb, and V2P for the second VP; N' and N are the indirect and the direct object, respectively.)

The verbs occurring in these six constructions S'1 - S'6 can be divided into four different classes:

¹ If not indicated otherwise, the page numbers refer to Zhu(1983).

- Va: verbs of giving like sòng 'to give (as a present)', to send off'; mài 'to sell'; huán 'to give back'; dì 'to pass on';
- Vb: verbs of taking like tōu 'to steal'; zhuàn 'to win'; qiǎng 'to seize'; mǎi 'to buy';
- Vc: verbs denoting the production of a concrete entity like chǎo (cài) 'to fry (a meal)'; qī (chá) 'to brew (tea); dǎ (máoyī) 'to knit (a sweater)';
- Vd: verbs denoting the creation of an abstract entity like chàng (gē) 'to sing (a song)'; jiǎng (gùshi) 'to tell (a story)'; zuò (guǎiliǎn) 'to make (faces);

As the first step in the analysis of the six structures S'1 - S'6, Zhu divides them into two groups, depending on whether a corresponding structure without V2P exists or not. Thus, for S'1 - S'4, we obtain matching constructions without V2P, i.e. S1 - S4, whereas the deletion of V2P in S'5 and S'6 does not yield acceptable structures.

S1: Ns V gěi N' N

我送給他一本書

Wo song gei ta yi-ben shu
1SG give GEI 3SG 1 -CL book
I give him a book (as a present).

S2: Ns V N gěi N'

我送一本書給他

Wo song yi-ben shu gei ta
1SG give 1 -CL book GEI 3SG
I give him a book (as a present).

S3: Ns gěi N' V N

我給他買輛車

Wo gei ta mai liang che
1SG GEI 3SG buy CL bicycle
I buy a bicycle for him.

S4: Ns V N' N

我送他一本書

Wo song ta yi-ben shu
1SG give 3SG 1 -CL book
I give him a book (as a present).

*我講個故事給你

Wo jiang ge gushi gei ni
1SG tell CL story GEI 2SG (S'5 without V2P)

*(你以爲我不會騎馬)我騎給你

(Ni yiwei wo bu hui qi ma) wo qi gei ni
2SG think 1SG NEG know ride horse 1SG ride Gei 2SG

(S'6 without V2P)

According to Zhu, the existence of a pair of sentences $S_n:S'_n$ constitutes the evidence for the derivability of S'_n from S_n . Consequently, he makes his structural analysis of the S'_n sentences depend on the acceptability or the unacceptability of the supposedly basic structure S_n . If the corresponding structure without V2P, i.e. S_n , is acceptable, then the first VP (the VP of the sentence S_n) and the added VP, V2P, make up a new (complex) VP in S'_n . If, however, there is no corresponding S_n , then the gei-phrase and the V2P form a group separate from the first VP:

S'1, S'4: Ns [[V1 (gěi) N' N] V2P]

S'3: Ns [[gěi N' V1 N] V2P]

S'2: Ns [[V1 N gěi N'] V2P]

S'5, S'6: Ns [V1 N][gěi N' V2P]²

5.1.1. S'1 and S'4: Ns V1 (gěi) N' N V2P

As the distinction between S'1 and S'4 is not necessary for their analysis, I will, contrary to Zhu, consider them as one

² Zhu does not indicate whether [V1 N] and [gěi N' V2P] belong to the same complex VP or not.

type, S'1,4.³

The verbs allowed in the V1 position of S'1,4 are limited to Va verbs, i.e. verbs of giving for which the immediately following gěi is optional. As S1 and S4 equally admit only verbs of giving Va with an optionally adjacent gěi, Zhu postulates that S'1 and S'4 are derived from S1 and S4, respectively. There is, however, one additional condition to be satisfied in S'1 and S'4,

³ There are, however, two cases where the distinction between S'1 and S'4 has to be maintained, namely a) with a special subset of verbs of giving called Va/c by Zhu and b) with verbs of taking like tōu 'to steal'.

Va/c verbs like xiě 'to write', jì 'to send' differ from "normal" verbs of giving (like sòng 'to give') in two respects: 1) For Va/c verbs, the gěi immediately following the verb is obligatory, i.e. Va/c verbs are only acceptable in S1, but not in S4:

- (i) 我寫給他一封信
 Wo xie gei ta yi-feng xin (S1)
 1SG write GEI 3SG 1 -CL letter

I have written him a letter.

- (ii) *我寫他一封信
 Wo xie ta yi-feng xin
 1SG write 3SG 1 -CL letter (S4)

2) Va/c verbs allow their recipient to appear in a preverbal gěi-phrase, i.e. Va/c can appear in S3:

- (iii) 我給他寫了一封信
 Wo gei ta xie -le yi-feng xin
 1SG GEI 3SG write-PERF 1 -CL letter

I have written him a letter.

With "normal" verbs of giving, however, the preverbal gěi-phrase can never indicate the recipient, but only the person on whose behalf or for whose benefit the action was done, i.e. the benefactive:

- (iv) 我給他賣了一本書
 Wo gei ta mai-le yi-ben shu
 1SG GEI 3SG buy-PERF 1 -CL book

I have sold a book on his behalf.

But as Zhu(1983: 161) points out himself, the Va/c verbs are not important for the problem under investigation and their particular behaviour need not to be taken into account.

As to the second case where the distinction between S'1 and S'4

i.e. the intentionality of the verb V1 (p. 163). Thus, non-intentional verbs of giving like chuánrǎn 'to infect' and shū 'to convey, to lose' are totally acceptable in S1 and S4 (cf. (3a) and (4a)), but they are excluded from S'1 and S'4 (cf. (3b) and (4b)):

- (3a) 他傳染(給)我天花
 Ta chuanran (gei) wo tianhua
 3SG infect GEI 1SG smallpox
 He infected me with smallpox.

- (3b) *他傳染(給)我天花死
 Ta chuanran (gei) wo tianhua si
 3SG infect GEI 1SG smallpox die

- (4a) 我輸(給)他很多錢
 Wo shu (gei) ta hen duo qian
 1SG lose GEI 3SG very much money

I lost a lot of money to him.

- (4b) *我輸(給)他很多錢花
 Wo shu (gei) ta hen duo qian hua
 1SG lose GEI 3SG very much money spend

 (footnote 3 continued)

has to be maintained, i.e. the verbs of taking, they are only admitted in S'4, and not in S'1:

- (v) 我偷了他一點酒喝
 Wo tou -le ta yi-dian jiu he (S'4)
 1S steal-PERF 3SG 1 -little wine drink

I have stolen him a little wine to drink.

- (vi) *我偷給他一點酒喝
 Wo tou gei ta yi-dian jiu he (S'1)
 1SG steal GEI 3SG 1 -little wine drink

But in fact Zhu does not give any example where a Vb verb is followed by both its arguments as in (v).

My presentation in which S'1 and S'4 are treated as one sentence type is therefore only of limited validity for Va/c verbs and Vb verbs.

There are two kinds of relations between the direct object of the first verb V1 and V2P: either the direct object of V1 is understood to be the direct object of V2 as well (cf. (5)), or the direct object of V1 indicates the instrument of the action expressed by V2P (cf. (6) and (7)). (If V2P comprises a verb and a NP, only the latter interpretation is possible.)

(5) 我送給你一瓶酒喝

Wo song gei ni yi-ping jiu he
1SG give GEI 2SG 1 -bottle wine drink

I have given you a bottle of wine to drink.

(6) 我借給你一隻筆寫

Wo jie gei ni yi-zhi bi xie
1SG lend GEI 2SG 1 -CL pen write

I have lent you a pen to write with.

(7) 我送給你一把刀切肉

Wo song gei ni yi-ba dao qie rou
1SG give GEI 2SG 1 -CL knife cut meat

I have given you a knife to cut meat with.

As the verbs of giving Va do not only appear in S'1, but also in S'2 (Ns V1 N gei N' V2P), and as in S'2 we observe the same relations between V2P and the direct object of V1 as in S'1 (i.e. the relation verb - object and the relation verb - instrument), Zhu postulates a transformational link between S'1 and S'2:⁴

(8) 我借給你一隻筆寫

Wo jie gei ni yi-zhi bi xie
1SG lend GEI 2SG 1 -CL pen write

(S'1)

I have lent you a pen to write with.

⁴ The possibility to be transformed into S'2 only exists for S'1 (Ns V gei N' N V2P), but not for S'4 (Ns V N' N V2P) because it depends on the presence of gei.

- (9) 我借一隻筆給你寫
 Wo jie yi-zhi bi gei ni xie (S'2)
 1SG lend 1 -CL pen GEI 2SG write
 I have lent you a pen to write with.

This transformation of S'1 into S'2 is, however, irreversible for verbs other than verbs of giving, i.e. in general we only have S'1 $\rightarrow$ S'2, and not S'2 $\rightarrow$ S'1. For besides verbs of giving Va, S'2 also allows for Vb and Vc verbs. These two latter verb classes not being admitted in S'1, a transformation from S'2 into S'1 is therefore excluded in that case. This point will be discussed in more detail in my own analysis of S'2 (cf. below, 5.2.4.1.).

5.1.2. S'3: Ns gěi N' V1 N V2P

This structure allows verbs of taking Vb (cf. (10)) as well as verbs Vc denoting the production of a concrete entity (cf. (11)) in its V1 position:

- (10) 我給你買了輛車騎
 Wo gei ni mai-le liang che qi
 1SG GEI 2SG buy-PERF CL bicycle ride
 I have bought you a bicycle to ride on.
- (11) 我給你打了件毛衣穿
 Wo gei ni da -le jian maoyi chuan
 1SG GEI 2SG knit-PERF CL sweater wear
 I have knitted a sweater for you to wear.

Verbs of giving Va are excluded from this structure, because the preverbal gěi-phrase does not indicate the recipient, but the benefactive, i.e. the person for whose benefit or on whose behalf something has been done:⁵

⁵ This observation is not valid for Va/c verbs like xiě (xìn) 'to write (a letter)', where the preverbal gěi-phrase can indicate the recipient. Cf. above, footnote 3.

- (12) 我給他送了一本書
 Wo gei ta song-le yi-ben shu
 1SG GEI 3SG give-PERF 1 -CL book

I have given a book (to a third person) on his behalf.

It is not very clear to me why Zhu fails to include the verbs Vd denoting the creation of an abstract entity among the verbs admitted in S'3, for sentences (13) - (15) are obviously well-formed:

- (13) 你給我唱個歌聽
 Ni gei wo chang ge ge ting
 2SG GEI 1SG sing CL song listen

Sing a song for me to listen to.

- (14) 你給我講個故事聽
 Ni gei wo jiang ge gushi ting
 2SG GEI 1SG tell CL story listen

Tell a story for me to listen to.

- (15) 你給我做個鬼臉看
 Ni gei wo zuo ge guilian kan
 2SG GEI 1SG make CL grimace look

Make faces for me to see.

This omission of Vd verbs is the more striking because Vd verbs are not only allowed in S'3, but also in S3 (Ns gěi N' V N) (cf. (18)) and thus show the same distribution as Vb and Vc verbs whose acceptability in S'3 Zhu acknowledges:⁶

⁶ Probably Zhu has excluded Vd verbs because only those structures S3 (Ns gěi N' V N) which can be transformed into S2 (Ns V N gěi N') count as true S3 structures. As shown in (i) - (iii), this transformation cannot take place when S3 contains a Vd verb:

- (i) 你給我唱個歌聽
 Ni gei wo chang ge ge ting (S'3)
 1SG GEI 1SG sing CL song listen

Sing a song for me to listen to.

- (16) 我給你買了輛車
 Wo gei ni mai-le liang che (S3, Vb)
 1SG GEI 2SG buy-PERF CL bicycle
 I have bought a bicycle for you.
- (17) 我給你打了件毛衣
 Wo gei ni da -le jian maoyi (S3, Vc)
 1SG GEI 2SG knit-PERF CL sweater
 I have knitted a sweater for you.
- (18) 我給你唱了一首歌
 Wo gei ni chang-le yi-shou ge (S3, Vd)
 1SG GEI 2SG sing -PERF 1 -CL song
 I have sung a song for you.

Vd verbs will be discussed in more detail in the section dealing with S'5 (cf. below, 5.1.4.).

5.1.3. S'2: Ns V1 N gei N' V2P

This structure allows for Va, Vb and Vc verbs, provided they are intentional (p. 164):

- (19) 我借了一隻筆給你寫
 Wo jie -le yi-zhi bi gei ni xie (Va)
 1SG lend-PERF 1 -CL pen GEI 2SG write
 I have lent you a pen to write with.

 (footnote 6 continued)

- (ii) 你給我唱個歌
 Ni gei wo chang ge ge (S3)
 2SG GEI 1SG sing CL song
 Sing a song for me.
- (iii) *你唱個歌給我
 Ni chang ge ge gei wo (S2)
 2SG sing CL song GEI 1SG

By analogy, a structure S'3 which does not come from a true S3 structure is probably not regarded a true S'3 either and is therefore discarded by Zhu.

- (20) 他偷了點酒給我喝
 Ta tou -le dian jiu gei wo he (Vb)
 3SG steal-PERF a:little wine GEI 1SG drink
 He has stolen a little bit of wine for me to drink.

- (21) 我打了件毛衣給你穿
 Wo da -le jian maoyi gei ni chuan (Vc)
 1SG knit-PERF CL sweater GEI 2SG wear
 I have knitted a sweater for you to wear.

Zhu excludes Vd verbs from S'2 because contrary to Va, Vb and Vc verbs, there is no corresponding sentence S2 without V2P for Vd verbs. An abstract entity not being transferable, the NP introduced by gei cannot be interpreted as recipient, an interpretation which is, however, possible in the case of Va, Vb and Vc verbs:

- (22) 我借了一隻筆給你
 Wo jie -le yi-zhi bi gei ni (Va)
 1SG lend-PERF 1 -CL pen GEI 2SG
 I have lent you a pen

- (23) 他偷了點酒給我
 Ta tou -le dian jiu gei wo (Vb)
 3SG steal-PERF a:little wine GEI 1SG
 He has stolen some wine for me.

- (24) 我打了件毛衣給你
 Wo da -le jian maoyi gei ni (Vc)
 1SG knit-PERF CL sweater GEI 2SG
 I have knitted a sweater for you.

- (25) *我唱了一首歌給你
 Wo chang-le yi-shou ge gei ni (Vd)
 1SG sing -PERF 1 -CL song GEI 2SG

The acceptability of S2 as well as the identity of the verb classes admitted in both S2 and S'2 further substantiate Zhu's claim that S'n has to be derived from Sn, a hypothesis which has already been confirmed for S'1, S'4 and S'3.

Though I do not want to expose my own analysis of the problem yet, I nevertheless think it necessary to make a first comment on Zhu's verb classification, in particular on his distinction between verbs of taking Vb and verbs denoting the production of a concrete entity Vc. It is certainly correct to point out that verbs like tōu 'to steal' and qiǎng 'to snatch away' are double object verbs. But this characteristic of Vb verbs does not play any role in the generalizations Zhu draws from his data. A sentence like (23) is not an example of a double object construction, because the gěi-phrase does not indicate the second argument of tōu (i.e. the source), but an optionally permitted benefactive. Gěi tā in (23) plays the same role as gěi tā in (24), where we have a Vc verb which subcategorises for only one argument (i.e. for its direct object) and which admits a benefactive as an additional role. The structure which really corresponds to the double object construction: Ns Va N gěi N' (cf. (27)) is therefore not the one illustrated in (23), but the structure: Ns Vb N' N as in (26):

- (26) 我偷他點酒
 Wo tou ta dian jiu (Vb)
 1SG steal 3SG a:little wine
 I steal a little bit of wine from him.

- (27) 我送一本書給他
 Wo song yi-ben shu gei ta (Va)
 1SG give 1 -CL book GEI 3SG
 I give him a book (as a present).

As shown in (28), a verb of taking Vb can naturally be accompanied by its two arguments and a benefactive. Such a sentence, though, is totally different from those given by Zhu where the source argument never appears (cf. (29)):

(28) 我偷他點酒給你(喝)

Wo tou ta dian jiu gei ni (he)
 1SG steal 3SG a:little wine GEI 2SG drink

I steal a little bit of wine from him for you (to drink).

(29) 我偷點酒給你(喝)

Wo tou dian jiu gei ni (he)
 1SG steal a:little wine GEI 2SG drink

I steal a little bit of wine for you (to drink).

Thus, the fact that verbs Vb like tōu 'to steal', mǎi 'to buy' etc. belong to the class of double object verbs has no influence whatsoever on the structures examined by Zhu, and consequently, the distinction between Vb verbs and Vc verbs can be ignored here. This point of view is confirmed by the observation that in all of the four structures S'1 - S'4, Vb and Vc verbs are simultaneously present or absent, i.e. we do not find any structure where Vb is admitted, but where Vc is excluded and vice versa. In the data examined by Zhu(1983), Vb and Vc verbs thus behave as a single class which can be described as the class of verbs Vb+c with a concrete object, and which has to be distinguished from verbs with an abstract object Vd, on the one hand, and from verbs of giving Va, on the other hand.⁷

⁷ In his 1979 article, Zhu notices that Vb and Vc verbs behave identically in structures S1, S2 and S3 and thus form a group which is different from that of verbs of giving Va (Zhu 1979: 84, 3.5.). He also emphasizes the fact that in a sentence like (i), the source argument of the Vb verb is not mentioned, whereas in (ii) the source argument of the Vb verb is present and (ii), accordingly, illustrates a double object construction:

(i) 我買一點水果給病人

Wo mai yi-dian shuiguo gei bingren
 1SG buy 1 -little fruit GEI patient

I buy some fruit for the patient.

(ii) 我買他家一所房子

Wo mai ta jia yi-suo fangzi
 1SG buy 3SG home 1 -CL house

I buy a house from his family.

(Zhu 1979: 82)

5.1.4. S'5: Ns V1 N gěi N' V2

This structure only allows Vd verbs with an abstract object in its V1 position. Though being superficially identical, S'5 and S'2 have to be distinguished, because S'2 has a corresponding structure S2 without V2P (cf. above, (22) - (24)), whereas S'5 lacks such a corresponding structure (cf. above (25)). To say that there is no S5 is in fact another way to say that Vd verbs do not admit a postverbal gěi-phrase; Zhu explains this unacceptability of a postverbal gěi-phrase with the non-transferability of an abstract object.⁸ Another reason for the necessity to distinguish between S'5 and S'2 are the restrictions on V2P in S'5.

First of all, only verbs of perception like kàn 'to look', tīng 'to listen', xīnshǎng 'to appreciate' are admitted:

- (30) 你唱個歌給我聽
 Ni chang ge ge gei wo ting
 2SG sing CL song GEI 1SG listen

Sing a song for me to listen to.

- (31) 你變個戲法兒給我
 Ni bian ge xifar gei wo kan
 2SG perform CL trick GEI 1SG look

Make a trick for me to see.

 (footnote 7 continued)

He does not, however, include this observation in his 1983 article where, on the contrary, he keeps the distinction between Vb verbs and Vc verbs.

⁸ This means that there is no structure S5 from which S'5 would have been derived, and in that respect it does not make much sense to speak of S'5, "5 prime" presupposing the existence of a corresponding structure noted as "5". The same observation holds true for S'6 which also lacks a corresponding structure without V2P, S6.

- (32) 你講個笑話給我聽
 Ni jiang ge xiaohua gei wo ting
 2SG tell CL joke GEI 1SG listen
 Tell a joke for me to listen to.

- (33) 你做個鬼臉給我看
 Ni zuo ge guilian gei wo kan
 2SG make CL grimace GEI 1SG look
 Make faces for me to look at.

Secondly, V2P cannot consist of a verb and its object, probably because such a V2P would imply the role of instrument (with respect to V2P) for the object of V1, a role which is difficult to conceive of for an abstract object. Consequently, the object of V1 (an abstract entity) is understood to be the object of V2 as well, which is perhaps the reason why the verbs in V2 position are limited to verbs of perception. Zhu states the existence of this restriction but does not propose any kind of explanation for it.

5.1.5. S'6: Ns V1 (N) gěi N' kàn

This structure is exceptional among the structures discussed so far in several respects.

First, V1 is not subjected to any constraint (except intentionality), so that all kinds of verbs are admitted here, including those which do not belong to one of the four classes Va - Vd.

Secondly, there is no choice for V2, only kàn 'to look' being allowed here.

Thirdly, the implicit object of kàn is not the object of V1, but the entire action expressed by V1P. This can easily be demonstrated in a sentence with an intransitive verb where there is simply no object kàn could be in relation with:

- (34) (你以爲我不敢死)我死給你看
 (Ni yiwei wo bu gan si) wo si gei ni kan
 2SG think 1SG NEG dare die 1SG die GEI 2SG look
 (You think I don't dare to die), I'll die to show you.
 (p. 164)

Sentence (35) provides another clear example, because qí (mǎ) 'to ride (a horse)' does not belong to any of the four verb classes Va - Vd and hence, (35) cannot be confused with S'2 or S'5:

- (35) (你說我不會騎馬)我騎給你看
 (Ni shuo wo bu hui qi ma) wo qi gei ni kan
 2SG say 1SG NEG know ride horse 1SG ride GEI 2SG look
 (You say I don't know how to ride a horse) I'll ride to show you.

5.2. Critical review of Zhu's analysis

In this section, I will give my own analysis of the data presented in Zhu(1983). I will first of all discuss structure S'6 because it obviously represents a structure apart from the other structures S'1 - S'5.

5.2.1. S'6: Ns V1 (N) gěi N' kàn

The first indication of the difference between S'6 and the other structures comes from Zhu's observation that while in the structures S'1 - S'5 the object of V1 is also understood to be the object of V2, in S'6 it is the entire action expressed by V1P which functions as the implicit object of kàn. This explains why it is possible to repeat the sentence containing the V1P after kàn:

- (36) (你說我不會騎馬)我騎給你看我會(騎馬)
 (Ni shuo wo bu hui qi ma)
 2SG say 1SG NEG know ride horse

 wo qi gei ni kan wo hui (qi ma)
 1SG ride GEI 2SG look 1SG know ride horse

 (You say I don't know how to ride a horse)
 I'll ride one to show you (I do know).

The acceptability of (36), though, is not only due to the relationship between kàn and the entire action, but also to the fact that wǒ huì qí mǎ depends on gěi nǐ kàn 'to show you', and not on kàn 'to look' alone. This latter point is more visible in the corresponding structure with gěi in preverbal position for S'6 (cf. (38b)) because here gěi nǐ kàn as a whole occupies the preverbal position, a possibility which is excluded in the corresponding structure with a preverbal gěi for S'2 and S'5, i.e. in S'3 (cf. (39b) and (40b)):

- (37) 我給你看我的新衣服
 Wo gei ni kan wo -de xin yifu
 1SG GEI 2SG look 1SG-SUB new dress

 I show you my new dress.
- (38) 你說我不會騎馬
 Ni shuo wo bu hui qi ma
 2SG say 1SG NEG know ride horse
- a) 我騎給你看
 wo qi gei ni kan (S'6)
 1SG ride GEI 2SG look
- b) 我給你看我會(騎馬)
 wo gei ni kan wo hui (qi ma)
 1SG GEI 2SG look 1SG know ride horse

 You say I don't know how to ride a horse,
 a) I'll ride one to show you.
 b) I show you I do know (how to ride a horse).
- (39a) 我買了一張畫給你看
 Wo mai-le yi-zhang huar gei ni kan (S'2)
 1SG buy-PERF 1 -CL painting GEI 2SG look
- (39b) 我給你買了一張畫看
 Wo gei ni mai-le yi-zhang huar kan (S'3)
 1SG GEI 2SG buy-PERF 1 -CL painting look

 I have bought a painting for you to look at.

(40a) 我做個鬼臉給你看

Wo zuo ge guilian gei ni kan
1SG make CL grimace GEI 2SG look

(S'5)

(40b) 我給你做個鬼臉看

Wo gei ni zuo ge guilian kan
1SG GEI 2SG make CL grimace look

I make faces for you to look at.

(38b) is thus exactly parallel to (37), except that we have a sentential object in (38b), while in (37) the object of gěi nǐ kàn 'to show you' is a NP (wǒ-de xīn yīfu 'my new dress'). The corresponding structure with gěi in preverbal position for S'6 also shows that V1 and V2 have the same subject, whereas in S'2 and S'5, the subject of V2P is the NP preceded by gěi.

I have shown enough evidence now for the analysis of S'6 as a complex sentence with 'Ns V1P' as the matrix sentence and 'gěi NP kàn' as a purposive clause with an empty subject which is coreferential with the subject of the matrix sentence. The most important point here is that 'gěi NP' is not in a postverbal position (with respect to V1P), but it is related to V2 which it precedes. This is accounted for in the structural analysis I propose for S'6:⁹

(41) [_S NP[_V [_V V (NP)] [_S Ø [_V [_{pp} gěi NP] kàn]]]]

This analysis is confirmed by the fact that the purposive clause marker lái 'in order to' can precede gěi NP kàn (though the sentence sounds more natural without lái):¹⁰

⁹ I will not examine the exact nature of the empty subject in the purposive clause. (According to the theory of government and binding (Chomsky 1981), this empty subject is an instance of PRO, the empty category which is both pronominal and anaphoric and which only appears in non-finite clauses.)

¹⁰ There are several restrictions on the use of lái 'in order to'.

First, lái can only introduce a purposive clause if the

- (42) 你說我不會騎馬，我騎來給你看
 Ni shuo wo bu hui qi ma
 2SG say 1SG NEG know ride horse

wo qi lai gei ni kan
 1SG ride in:order:to GEI 2SG look

You say I don't know how to ride a horse, I'll ride one to show you.

- (43) 你說我不會做鬼臉，我做來給你看
 Ni shuo wo bu hui zuo guilian,
 2SG say 1SG NEG know make grimace

wo zuo lai gei ni kan
 1SG make in:order:to GEI 2SG look

You say I don't know how to make faces, I'll make faces to show you.

The only requirement for V1P in S'6, i.e. intentionality, can now be explained by the incompatibility between a purposive clause and a matrix clause with a non-intentional verb. The lack of any further conditions for V1P (besides that of intentionality) is then due to the relationship of gěi NP kàn 'to show sb.' with the entire action expressed by V1P, and not only with the object of V1.

 (footnote 10 continued)

latter follows the matrix clause: 'NP V (NP), lái V (NP)', but not *lái V (NP), NP V (NP).

Secondly, a purposive clause introduced by lai only allows an empty subject which, in general, is coreferential with the subject of the matrix clause. (An overt subject is, however, admitted in a purposive clause when it is introduced by gěi, as is the case in structure S'2.)

Thirdly, lái is not obligatory, for a sentence like (i) can have a purposive reading even without lái:

- (i) 他搬出去來照顧他媽媽

Ta banchuqu lai zhaogu ta mama
 3SG move in:order:to take:care:of 3SG mother

- (ii) 他搬出去照顧他媽媽

Ta banchuqu zhaogu ta mama
 3SG move take:care:of 3SG mother

He has moved in order to take care of his mother.

Unfortunately, there are only very few studies on lai as a purposive clause marker or on the purposive clause in general (among them cf. Rygaloff 1975: 123 - 125.)

That in the examples illustrating sentence type S'6, gěi NP kàn 'to show sb.' has the effect of refuting an erroneous presupposition ('to show sb. that he was wrong') is due to the context (e.g. Nǐ shuō wǒ bù huì ... 'You think I don't dare...') and does not follow from a particularity in the structure of gěi NP kàn in S'6 which is the same as for gěi NP kàn in its literal meaning 'to show sb.'.¹¹

I have thus found enough evidence for the fundamental difference between S'6 on the one hand, and S'2 and S'5 on the other hand. Although Zhu has noticed the exceptional behaviour of S'6 (with respect to S'2 and S'5), he has not seen that S'6 is an

¹¹ kàn means 'to look; to read' (with objects like shū 'book', bàozhǐ 'newspaper'). Now, when kàn is preceded by a gěi-phrase, the whole sequence gěi NP kàn has the meaning 'to show sb.'. Accordingly, sentence (i) can have two different interpretations which are made explicit in (ii) and (iii). Furthermore, (iiia) and (iiib) on their parts illustrate the difference between the literal meaning of gěi NP kàn 'to show sb.' (with shū 'book' as the implicit object) and the extended meaning of 'showing sb. that he was wrong' which can be induced by the context.

(i) 我買一本書給你看
 Wo mai yi-ben shu gei ni kan
 1SG buy 1 -CL book GEI 2SG look

(ii) 我買一本書給你讀
 Wo mai yi-ben shu gei ni du
 1SG buy 1 -CL book GEI 2SG read

I bought a book for you to read.

(iiia) 我買一本書給你看
 Wo mai yi-ben shu gei ni kan
 1SG buy 1 -CL book GEI 2SG look

I buy a book to show to you (e.g. its printing, its binding etc.)

(iiib) 你說我不敢買，我買給你看
 Ni shuo wo bu gan mai yi-ben shu ,
 2SG say 1SG NEG dare buy 1 -CL book

wo mai gei ni kan
 1SG buy GEI 2SG look

You say I don't dare to buy a book; I'll buy one to show you.

entirely different structure from S'2 and S'5, and that the latter two just happen to be identical with S'6 on the surface in the case of kàn 'to look' being V2. Consequently, all the sentences S'2 and S'5 with kàn in V2 position (e.g. Wǒ sòng yī-běn shū gěi nǐ kàn 'I give you a book to read'; Wǒ zuò guǐliǎn gěi nǐ kàn 'I make faces for you to look at') can as well be interpreted as S'6 ('I give you a book to show you (that I do give presents)' and 'I make faces to show you (that I dare do things like that)', respectively.) (p. 164, 2.2). This, however, is not due to a structural ambiguity of S'2 and S'5, but can be explained by the availability of two different interpretations for gěi NP kàn (cf. footnote 11). In the first case, gěi NP kàn is a purposive clause with an overt subject introduced by gěi, i.e. a purposive gěi-clause, the whole clause meaning 'so that sb. can look at (it)/ read (it)'. In the second case, gěi NP kàn means 'to show sb.' and the matrix subject is understood to be the subject of gěi NP kàn as well, i.e. gěi NP kàn represents a purposive clause with an empty subject.

5.2.2. S'3: Ns gěi N' V1 N V2P

Having shown that gěi in S'6, being related to kàn 'to look' and not to V1P, is in fact in a preverbal position and that, consequently, S'6 cannot give us any clue to the problem of the postverbal gěi-phrase, I will now demonstrate that S'3 is not relevant to our discussion, either. For the structure of S'3 does not change when the gěi-phrase is dropped, because this gěi-phrase indicates an optional role, namely the benefactive of the action:¹²

¹² This does not hold for the subset of verbs labeled Va/c verbs like xiě (xìn) 'to write (a letter)'. For these verbs allow the

(44a) 我給他偷了一杯酒喝

Wo gei ta tou -le yi-bei jiu he
 1SG GEI 3SG steal-PERF 1 -cup wine drink

I have stolen a cup of wine for him to drink.

(44b) 我偷了一杯酒喝

Wo tou -le yi-bei jiu he
 1SG steal-PERF 1 -CL wine drink

I have stolen a cup of wine to drink (myself).

(45a) 我給他打了件毛衣穿

Wo gei ta da -le jian maoyi chuan
 1SG GEI 3SG knit-PERF CL sweater wear

I have knitted a sweater for him to wear.

(45b) 我打了件毛衣穿

Wo da -le jian maoyi chuan
 1SG knit-PERF CL sweater wear

I have knitted a sweater to wear (myself).

To be more precise, the gěi-phrase in sentences (44a) and (45a) discussed by Zhu simultaneously indicates the benefactive of the action as well as the recipient of the direct object (i.e. the wine and the sweater, respectively). The interpretation of tā as the recipient and its interpretation as the subject of V2 (i.e. of hē 'to drink' and chuān 'to wear') result from the same semantic-pragmatic deduction: in order for tā 'he' to benefit from my action of stealing some wine and knitting a sweater, he must necessarily get the wine and the sweater. This further implies that tā 'he' must be the implicit subject of hē 'to drink' and chuān 'to wear'.

V2P in S'3 thus represents a purposive clause whose empty subject is coreferential with the subject of the matrix clause,

 (footnote 12 continued)

recipient (marked by gěi) in the postverbal as well as in the preverbal position. Cf. above, footnote 3.

except when the latter contains a preverbal gěi-phrase. In this case, it is the NP marked by gěi which is understood to be the subject of the purposive clause. This interpretation, however, is based on semantic-pragmatic inferences, the general structure of S'3 being the same whether the gěi-phrase is present or not. I thus propose the following structural analysis for S'3:

[_S NP [_V [_{pp} gěi NP] [_V V NP] [_S ∅ [_V V (NP)]]]]

That V2P functions as a purposive clause independently from the presence of the gěi-phrase is confirmed by the acceptability of the purposive clause marker lái 'in order to' in both (46a) and (46b) (though the sentences sound more natural without lái, cf. (42) and (43) above):

(46a) 我給他打了件毛衣來穿

Wo gei ta da -le jian maoyi lai chuan
1SG GEI 3SG knit-PERF CL sweater in:order:to wear

I have knitted a sweater for him to wear.

(46b) 我打了件毛衣來穿

Wo da -le jian maoyi lai chuan
1SG knit-PERF CL sweater in:order:to wear

I have knitted a sweater to wear (myself).

To summarize, S'3 illustrates a complex sentence containing a purposive clause with an empty subject. The preverbal gěi-phrase is optional and its presence does not change the general structure of S'3.

5.2.3. S'1 and S'4: Ns V1 (gěi) N' N V2P

After having demonstrated that neither S'6 nor S'3 are relevant for the problem of why the outer object marked by gěi is unacceptable in postverbal position, I will now examine S'1 and S'4 in more detail.

As shown in sentences (47) - (49), V2P is not dependent on the presence of gěi. This is not a surprising finding in so far as verbs of giving Va admitted in S'1 and S'4 are precisely characterized by the optionality of the gěi adjacent to the verb (with the exception of Va/c verbs like xiě (xìn) 'to write (a letter)'; cf. above, footnote 3):

- (47) 我送(給)你一把刀切肉

Wo song (gei) ni yi-ba dao qie rou
1SG give GEI 2SG 1 -CL knife cut meat

I have given you a knife to cut meat with.

- (48) 我借(給)他一百塊錢還賬

Wo jie (gei) ta yibai-kuai qian huan zhang
1SG lend GEI 3SG 100 -CL money return debt

I have lent you 100 dollars to pay back your debts with.

- (49) 我遞(給)他一瓶酒喝

Wo di (gei) ta yi-ping jiu he
1SG pass:on GEI 3SG 1 -bottle wine drink

I have passed on a bottle of wine to him for drinking.

As in S'3 and S'6, V2P in S'1 and S'4 represents a purposive clause with an empty subject which can optionally be introduced by the purposive clause marker lai (whose presence is preferred when the relationship between V2 and the object of V1 is not one of verb - object (cf. (50) and (51)):

- (50) 我送(給)你一把刀來切肉

Wo song (gei) ni yi-ba dao lai qie rou
1SG give GEI 2SG 1 -CL knife in:order:to cut meat

I have given you a knife to cut meat with.

- (51) 我借(給)他一百塊錢來還賬

Wo jie (gei) ta yibai-kuai qian
1SG lend GEI 3SG 100 -CL money

lai huan zhang
in:order:to return debt

I have lent him 100 dollars to pay his debts back.

(52) 我遞(給)他一瓶酒來喝

Wo di (gei) ta yi-ping jiu lai he
 1SG pass:on GEI 3SG 1 -bottle wine in:order:to drink

I have passed on a bottle of wine to him for drinking.

The interpretation of the indirect object NP as the subject of the purposive clause (V2P) is obtained by a semantic-pragmatic inference similar to that already observed for the preverbal gěi-phrase in S'3: since the relationship between V2P and the direct object of V1 is one of verb - object or of verb - instrument, and since the indirect object is the recipient of the direct object, it follows that the recipient is interpreted as the subject of V2P.

The acceptability of a purposive clause with an empty subject does not depend on the presence of gěi; the only condition for V1 to satisfy is that of intentionality, a purposive clause being incompatible with a non-intentional verb in the matrix clause. The non-intentional verb shū 'to lose', for example, excludes a purposive clause:

(53a) 我輸(給)他一百塊錢

Wo shu (gei) ta yibai-kuai qian
 1SG lose GEI 3SG 100 -CL money

I have lost 100 dollars to him.

(53b) *我輸(給)他一百塊錢還賬

Wo shu (gei) ta yibai-kuai qian huan zhang
 1SG lose GEI 3SG 100 -CL money return debt

(cf. (48))

Although the presence or absence of gěi is not decisive in S'1 and S'4, Zhu nevertheless postulates a transformation relation between S'1 and S'2 (Ns V1 N gěi N' V2P) only because gěi is present in both structures. According to my analysis, however, S'1 and S'2 represent totally different structures, especially in the function of gěi. In S'1, gěi marks the indirect

object and thus belongs to the matrix clause, V2P being a purposive clause with an empty subject. Gěi in S'2, however, introduces the subject of the purposive gěi-clause and therefore is not part of the matrix clause (cf. below, 5.2.4.1.).

It is certainly correct to say that the meanings of S'1 and S'2 are approximately equivalent:

- (54) 我送給你件毛衣穿
 Wo song (gei) ni jian maoyi chuan (S'1)
 1SG give GEI 2SG CL sweater wear
 I have given you a sweater to wear.

- (55) 我送件毛衣給你穿
 Wo song jian maoyi gei ni chuan (S'2)
 1SG give CL sweater GEI 2SG wear
 I have given a sweater so that you wear it.

(Though so far I have given identical translations for S'1,4 and S'2 in order to facilitate the discussion, in what follows I will translate S'1,4 differently from S'2. In the case of S'2, this procedure may lead to some awkward sentences, but they have the advantage of rendering the actual structure of the Chinese sentence more faithfully.)

But the relationships between the verb V1 and the NPs in S'2 are not the same as those in S'1. This in fact trivial observation is obtained from the simple fact that S'2, contrary to S'1, also admits other verbs than verbs of giving in the V1 position, e.g. Vc verbs like dǎ (máoyī) 'to knit (a sweater)':

- (56) 我打了件毛衣給你穿
 Wo da -le jian maoyi gei ni chuan (S'2)
 1SG knit-PERF CL sweater GEI 2SG wear
 I have knitted a sweater so that you wear it.

S'1 only allows verbs of giving Va and consequently, the NPs máoyī 'sweater' and nǐ 'you' in (54) are both arguments

subcategorised for by the verb sòng. Verbs admitted in S'2, however, also include Vb and Vc verbs which only subcategorise for one argument.¹³ Therefore, máoyī 'sweater' and nǐ 'you' cannot both be arguments of the verb dǎ 'to knit' in (56). It is this difference between S'1 and S'2 which explains the unidirectionality of the transformation relation claimed by Zhu for S'1 and S'2: S'1 can be transformed into S'2, but S'2 cannot be transformed into S'1, because S'2 also allows Vb and Vc verbs which are unacceptable in S'1 (cf. above, 5.1.1.).

As shown in the following structural analysis, S'1,4 does not involve a purposive clause with an overt subject introduced by gěi, i.e. a purposive gěi-clause, but a purposive clause with an empty subject:

[_S NP [_V [_V V (gěi) NP NP] [_S Ø V2P]]]

For gěi belongs to the matrix clause where it marks one of the two arguments subcategorised for by the verb, i.e. the indirect object. Consequently, the presence or absence of gěi has no bearing whatsoever on V2P.

5.2.4. S'2: Ns V N gěi N' V2P

In order to substantiate my claim that S'1 and S'2 are different structures, I will now discuss S'2 in more detail. After the comparison between S'1 and S'2 with verbs of giving (S'2(Va)), I will examine S'2 showing Vb+c verbs in the

 13 This description is in fact incorrect for Vb verbs because like Va verbs they belong to the class of double object verbs. But we have already seen (cf. above, 5.1.3.) that in the sentences discussed by Zhu, the second argument of Vb verbs, the source, is never present and that consequently, Vb verbs here show the same behaviour as Vc verbs.

V1 position and contrast this second subtype of S'2 (S'2(Vb+c)) with S'5.

5.2.4.1. S'2 vs. S'1

Zhu's postulate that S'2 results from a transformation of S'1 is based on the fact that verbs of giving appear likewise in two structures, labeled S1 and S2 by Zhu(1979). In these two structures, gěi has the same function, i.e. it marks the indirect object:

(57a) 我送給他一件毛衣
 Wo song (gei) ta jian maoyi (S1)
 1SG give GEI 3SG CL sweater

(57b) 我送一件毛衣給他
 Wo song jian maoyi gei ta (S2)
 1SG give CL sweater GEI 3SG

I have given him a sweater (as a present).

When comparing S1 with the corresponding S'1 (cf. (58)), we can see that in S'1 as well as in S1 all the NPs are arguments of the verb, and that S1 and S'1 are thus identical in that respect. As in S1, gěi is not obligatory in S'1, either:

(58) 我送(給)他一件毛衣穿
 Wo song (gei) ta jian maoyi chuan (S'1)
 1SG give GEI 3SG CL sweater wear

I have given him a sweater to wear.

S'2 (cf. (59a)), however, shows a structure totally different from S2, especially in that gěi does not have the same function. If we want to place the purposive clause marker lái 'in order to' in S'2, it has to precede gěi, and not the V2P chuān:

(59a) 我送件毛衣給他穿
 Wo song jian maoyi gei ta chuan (S'2)
 1SG give CL sweater GEI 3SG wear

I have given a sweater so that he wears it.

(59b) 我送件毛衣來給他穿

Wo song jian maoyi lai gei ta chuan
1SG give CL sweater in:order:to GEI 3SG wear

I have given a sweater so that he wears it.

(59c) *我送件毛衣給他來穿

Wo song jian maoyi gei ta lai chuan
1SG give CL sweater GEI 3SG in:order:to wear

Gěi in S'2 (59a) does not mark the indirect object of sòng (as the gěi in S2 (57b) does), but it is part of the purposive gěi-clause (gěi tā chuān) whose subject tā 'he' it introduces.

Thus, the two gěi in S2 and S'2 have clearly different functions; consequently, S'2 cannot be analysed as S2 plus a VP added, and S2 and S'2 must represent different structures. They can roughly be analysed as follows:

S2: [_S NP [_V V NP [_{pp} gěi NP]]]¹⁴

S'2: [_S NP [_V [_V V NP] [_S gěi NP V (NP)]]]

In S2, all the NPs are actually arguments of the verb, whereas in S'2, only one of the two arguments of the verb of giving is present in the matrix clause; it is the subject of the purposive clause which is understood to play the role of recipient with respect to the verb in the matrix clause.

The phenomenon that tā 'he' in (59a) is syntactically not depending on sòng 'to give', but is interpreted as its second argument, does not invalidate our analysis, because it can be explained by a semantic-pragmatic inference: if I give a sweater

14 I have chosen the label "prepositional phrase" for [gěi NP] merely to indicate that it is the entire phrase [gěi NP] which functions as the second argument of the verb. This labeling, however, is not intended to be an answer to the much debated issue whether gěi in such a structure is a verb or a preposition.

as a present having the intention that X wears it, X must receive it, i.e. he must be the recipient of the sweater.

Thus, in S'2, a verb of giving can put but one of its arguments (the patient) in the appropriate position, i.e. in an argument position. The interpretation of a NP elsewhere in the sentence as recipient is the result of a pragmatic-semantic inference.

This fact can perhaps explain why for verbs of giving, S'2 is not felt to be more natural than S2, whereas for Vb+c verbs, S'2 is in fact preferred to S2 (cf. below, 5.2.4.2.). In fact, I would like to claim that for verbs of giving, the structure S'2 is rather peripheral, and indeed there are certain (intentional) verbs of giving for which S'2 is unacceptable:

(60a) 我打電話給他

Wo da dianhua gei ta
1SG dispatch phone:call GEI 3SG

I ring him up.

(60b) *我打電話給他接

Wo da dianhua gei ta jie
1SG dispatch phone:call GEI 3SG receive

(61a) 我寫一封信給他

Wo xie yi-feng xin gei ta
1SG write 1 -CL letter GEI 3SG

I write him a letter.

(61b) *我寫一封信給他看

Wo xie yi-feng xin gei ta kan
1SG write 1 -CL letter GEI 3SG look

In the case of xiě 'to write', we have very clear evidence to show that it is the fact of subcategorising for two arguments which makes S'2 unacceptable for some verbs of giving. For as pointed out by Zhu(1979: 85), xiě 'to write' behaves like a verb of giving only when in combination with xìn 'letter'. With an

object like shī 'poem', however, xiě does not behave any longer like a verb of giving and can now very well figure in S'2:

(62) 我寫了好幾首詩給他看

Wo xie -le haoji-shou shi gei ta kan
1SG write-PERF many -CL poem GEI 3SG look

I have written many poems so that you read them.¹⁵

The semantic or pragmatic plausibility is another factor determining the acceptability of a purposive gěi-clause. In (63a), for example, the purposive gěi-clause (gěi wǒ yòng) was either refused or considered to have a very limited acceptability, for the reason that the person selling the knife will not be interested in what I want to do with it:

(63a) ??他賣了一把刀給我用

Ta mai -le yi-ba dao gei wo yong (S'2)
3SG sell-PERF 1 -CL knife GEI 1SG use

(He has sold a knife so that I use it.)

(63b) 他賣了一把刀給我

Ta mai -le yi-ba dao gei wo (S2)
3SG sell-PERF 1 -CL knife GEI 1SG

He has sold me a knife.

The sentence corresponding to (63a) with mǎi 'to buy' instead of mài 'to sell', however, is fully acceptable and thereby confirms our distinction between S'2 with verbs of giving and S'2 with the remaining verb classes:

¹⁵ In fact, sentence (62) is also acceptable with xìn 'letter' as a direct object (cf. (i)). In this case, however, tā 'he' is not interpreted as the recipient of the letter, but as a third person (besides the letter writer and the recipient) who has a look at the letters, e.g. in order to check whether they are in order:

(i) 我寫了好幾封信給他看

Wo xie -le haoji-feng xin gei ta kan
1SG write-PERF many -CL letter GEI 3SG look

I have written many letters so that he checks them.

- (64) 他買了一把刀給我用
 Ta mai-le yi-ba dao gei wo yong
 3SG buy-PERF 1 -CL knife GEI 1SG use

He has bought a knife so that I use it.

The contrast of acceptability between (63a) and (64) confirms that the quasi-unacceptability of (63a) is due to the semantic-pragmatic improbability of a purposive gěi-clause (gěi wǒ yòng) as well as to the limited acceptability of S'2 for verbs of giving in general. (With verbs like jiè 'to lend' and sòng 'to give (as a present)' a purposive gěi-clause seems to make more sense, which is probably the reason why they figure in the examples for S'2 given by Zhu. We do not, however, find an example of S'2 with mài 'to sell', though mài 'to sell' belongs to the verbs of giving as well as jiè 'to lend' and sòng 'to give (as a present)').

It is important to point out that the limited compatibility between verbs of giving and a purposive clause (observed in S'2) is not a general characteristic of this verb class, but has to do with the particularity of S'2. For in S'2 there is a conflict between the subcategorisation frame of the verb *Va* containing two arguments and the availability of only one argument position; the identification of a NP able to play the role of the second "missing" argument is arrived at by a semantic-pragmatic inference. In a structure like S'1, however, all the arguments of the verb occupy argument positions and thus are immediately identifiable. A purposive clause (with an empty subject) is acceptable here provided that it is semantically or pragmatically plausible:

- (65) 他借給我一隻筆寫
 Ta jie gei wo yi-zhi bi xie zi
 3SG lend GEI 1SG 1 -CL pen write character

He has lent me a pen to write with.

Thus, S'1 and S'2 are totally different structures and their relationship is just not parallel to that between S1 and S2. S1 and S2 can be considered equivalent structures in so far as the two postverbal NPs in both structures are arguments of the verb (the recipient in each structure being marked by gěi).

In S'1, the arguments subcategorised for by the verb are equally present. As in S1, gěi marks the recipient and therefore belongs to the matrix clause which contains the verb of giving. In the matrix clause of S'2, however, there is only one of the two arguments of the Va verb. Gěi here introduces the subject of the purposive gěi-clause - which by inference is interpreted as the second "missing" argument of the matrix verb - and therefore does not belong to the matrix verb.

The purposive clause marker lái 'in order to' can serve to delimit the matrix clause from the purposive clause: in S'1, lái precedes V2P, whereas in S'2, lái is placed before gěi NP V2P.

5.2.4.2. S'2 with Vb+c verbs

S'2 with Vb+c verbs has to be distinguished from S'2 with verbs of giving Va. Whereas S'2(Va) and S2(Va) are not related to each other, I claim that for verbs other than verbs of giving, S2 is in fact to be derived from S'2.

This hypothesis is based on the observation that sentences of type S2(Vb+c), though acceptable as such, are quasi-automatically completed by a verb (i.e. changed into S'2(Vb+c)), and it is S'2 which is felt to be the more natural of the two:

- (66) 我做了飯給他(吃)
 Wo zuo -le fan gei ta (chi)
 1SG make-PERF food GEI 3SG eat

I have made a meal for him (to eat).

- (67) 我沏了杯茶給他(喝)

Wo qi -le bei cha gei ta (he)
1SG brew-PERF cup tea GEI 3SG drink

I have made a cup of tea for him (to drink).

- (68) 我買了件毛衣給他(穿)

Wo mai-le jian maoyi gei ta (chuan)
1SG buy-PERF CL sweater GEI 3SG wear

I have bought a sweater for him (to wear).

The situation we have in (66) - (68) is thus totally different from that with verbs of giving, where S'2 is less natural than S2, and even unacceptable for some verbs of giving (cf. above, (60b) and (61b)).

Since according to Zhu the verbs in the V1 position of S'2 must be intentional, a hypothesis which considers S2 as resulting from the deletion of V2P in S'2 predicts that the V1 position in S2 is also subject to the condition of intentionality. This prediction, in which S2 with a non-intentional verb (which is not a verb of giving) is as unacceptable as S'2 with a non-intentional verb, is borne out: sentences (69a) - (71a) containing non-intentional verbs are all ungrammatical and thus contrast clearly with their acceptable counterparts (69c) - (71c) where the verbs are intentional:

- (69a) *我在圖書館發現了一本很有趣的書給你(看)

Wo zai tushuguan faxian -le yi-ben hen
1SG in library discover-PERF 1 -CL very

youqu -de shu gei ni (kan)
interesting-SUB book GEI 2SG look

(I have discovered a very interesting book in the library for you (to read).)

- (69b) 我在圖書館發現了一本很有趣的書

Wo zai tushuguan faxian -le yi-ben hen
1SG in library discover-PERF 1 -CL very

youqu -de shu
interesting-SUB book

I have discovered a very interesting book in the library.

(69c) 我在圖書館借了一本很有趣的書給你(看)

Wo zai tushuguan jie -le yi-ben hen
1SG in library borrow-PERF 1 -CL very

youqu -de shu gei ni (kan)¹⁶
interesting-SUB book GEI 2SG look

I have borrowed a very interesting book from the library
for you (to read).

(70a) 我見到了一件很漂亮的毛衣給你(穿)

Wo jiandao-le yi-jian hen piaoliang-de maoyi
1SG see -PERF 1 -CL very nice -SUB sweater

gei ni (chuan)
GEI 2SG wear

(I have seen a very nice sweater for you (to wear).)

(70b) 我見到了一件很漂亮的毛衣

Wo jiandao-le yi-jian hen piaoliang-de maoyi
1SG see -PERF 1 -CL very nice -SUB sweater

I have seen a very nice sweater.

(70c) 我打了一件很漂亮的毛衣給你(穿)

Wo da -le yi-jian hen piaoliang-de maoyi
1SG knit-PERF 1 -CL very nice -SUB sweater

gei ni (chuan)
GEI 2SG wear

I have knitted a very nice sweater for you (to wear).

(71a) 我找到了一隻筆給你(使)

Wo zhaodao-le yi-zhi bi gei ni (shi)
1SG find -PERF 1 -CL pen GEI 2SG use

(I have found a pen for you (to use).)

(71b) 我找到了一隻筆

Wo zhaodao-le yi-zhi bi
1SG find -PERF 1 -CL pen

I have found a pen.

¹⁶ Jiè can either be a verb of giving: 'to lend', or a verb of taking: 'to borrow'.

- (71c) 我買了一隻筆給你(使)
 Wo mai-le yi-zhi bi gei ni (shi)
 1SG buy-PERF 1 -CL pen GEI 2SG use
 I have bought a pen for you (to use).

For verbs of giving, however, the intentionality requirement is only valid in S'2, and not in S2, where non-intentional verbs like shū 'to lose' and chuánrǎn 'to infect' are allowed:

- (72) 我輸了很多錢給他
 Wo shu hen duo qian gei ta (S2)
 1SG lose very much money GEI 3SG
 I have lost a lot of money to him.

- (73) 他傳染天花給我
 Ta chuanran tianhua gei wo (S2)
 3SG infect smallpox GEI 1SG
 He has infected me with smallpox.

There is thus enough evidence to analyse the postverbal gěi-phrase when occurring with verbs different from verbs of giving as being derived from a purposive gěi-clause.

As already observed for S'2 with verbs of giving, in S'2(Vb+c) the purposive clause marker lái (optionally) introduces the whole purposive gěi-clause (cf. (74a)), but cannot be put in front of V2P (cf. (74b)). In general, then, gěi in S'2 is part of the purposive gěi-clause 'gěi NP V2P', no matter to what verb class the verb in the matrix clause belongs:

- (74a) 我打了件毛衣來給他穿
 Wo da -le jian maoyi lai gei ta chuan
 1SG knit-PERF CL sweater in:order:to GEI 3SG wear
 I have knitted a sweater so that he wears it.

- (74b) *我打了件毛衣給他來穿
 Wo da -le jian maoyi gei ta lai chuan
 1SG knit-PERF CL sweater GEI 3SG in:order:to wear

Further evidence for deriving S2(Vb+c) from S'2(Vb+c) comes from the unacceptability of a postverbal gěi-phrase in case there

is no appropriate verb available. The VP in the purposive gěi-clause is "appropriate" when its relationship with the object of the matrix clause can be immediately established. The kind of relationship which best answers this demand of an unambiguous relationship is the relationship verb - object as in (66) - (68), where the object of the matrix clause is also understood to be the object of the purposive gěi-clause: zuò fàn ... chī 'to prepare a meal ... to eat; mǎi máoyī ... chuān 'to buy a sweater to wear'. The second best relationship, so to speak, is the instrumental relationship (in a large sense), i.e. the object of the matrix clause is interpreted as the instrument for the action expressed in V2P:

- (75) 我買了一把刀(來)給他切肉
 Wo mai-le yi-ba dao (lai) gei ta qie rou
 1SG buy-PERF 1 -CL knife in:order:to GEI 3SG cut meat
 I have bought a knife so that he cuts meat with it.

- (76) 他租一間房間(來)給我堆東西
 Ta zu -le yi-jian fangjian (lai)
 3SG rent-PERF 1 -CL room in:order:to
 gei wo dui dongxi
 GEI 1SG pile:up thing
 He has rented a room so that I can put my things there.

Besides using lai to make explicit the link between the matrix clause and the purposive gěi-clause (especially when the object of the matrix clause plays the role of instrument with respect to V2P), yòng 'to use' can also be used:

- (77) 我買了一把刀給他切肉用
 Wo mai-le yi-ba dao gei ta qie rou yong
 1SG buy-PERF 1 -CL knife GEI 3SG cut meat use
 I have bought a knife so that he cuts meat with it.

Yòng 'to use' takes yī-bǎ dāo 'a knife' as its object and we thereby arrive at a verb - object relationship between V2P and the object of the matrix clause.

However, lái 'in order to' and yòng 'to use' can only emphasize the relationship between the matrix clause and the purposive gěi-clause; they are not capable of establishing such a relationship. If the link between V2P and the object of the matrix clause functioning as the circumstantial complement of V2P is not so tight that V2P becomes virtually predictable on the basis of the matrix object, then a purposive gěi-clause is not acceptable (whether lái is present or not):

(78) *我打了件毛衣給你有漂亮的樣子

Wo da -le yi-jian maoyi (lai)
1SG knit-PERF 1 -CL sweater in:order:to

gei ni you piaoliang-de yangzi
GEI 2SG have nice -SUB look

(I have knitted a sweater so that you look nice.)

(79) *我買了一本書(來)給你學語言學

Wo mai-le yi-ben shu (lai)
1SG buy-PERF 1 -CL book in:order:to

gei ni xue yuyanxue
GEI 2SG learn linguistics

(I have bought a book so that you learn linguistics.)

That also implies that Wǒ dǎ-le yī-jàn máoyī gěi nǐ 'I have knitted a sweater for you' and Wǒ mǎi-le yī-běn shū gěi nǐ 'I have bought a book for you' will not be interpreted as resulting from the deletion of the V2P (yǒu piàoliang-de yàngzi 'to look nice' and xué yǔyánxué 'to learn linguistics') in (78) and (79), but as being derived from Wǒ dǎ-le yī-jàn máoyī gěi nǐ chuān and Wǒ mǎi-le yī-běn shū gěi nǐ kàn, respectively, where the V2P (chuān 'to wear' and kàn 'to look, read') shows the required tight relationship with the object of the matrix clause.

On the basis of this observation, one would expect that a purposive gěi-clause is unacceptable when there is a choice of several verbs for V2P in the purposive gěi-clause, and none of these verbs has a sufficiently tight relationship with the object

of the matrix clause to qualify as an appropriate verb. In such a case, then, a postverbal gěi-phrase is not acceptable, either, because it is derived from the purposive gěi-clause and presupposes its acceptability. This is in fact the explanation I propose for the ungrammaticality of sentences (80a) and (81a):

(80a) *我買了一張票給你(看電影/去博物館)

Wo mai-le yi-zhang piao gei ni
1SG buy-PERF 1 -CL ticket GEI 2SG

(kan dianying/ qu bowuguan/ ...)
look film go museum

(I have bought a ticket for you (to watch a movie/ to go to the museum/ ...))

(80b) 我給你買了一張票

Wo gei ni mai-le yi-zhang piao
1SG GEI 2SG buy-PERF 1 -CL ticket

I have bought a ticket for you.

(81a) *我做了這件事給你(早一點回家/讀書)

Wo zuo -le zhei-jian shi gei ni
1SG make-PERF this-CL matter GEI 2SG

(zao yi-dian hui jia / du shu / ...)
early 1 -little return home read book

(I have taken care of this matter for you (to go home earlier/ to study/ ...))

(81b) 我給你做了這件事

Wo gei ni zuo -le zhei-jian shi
1SG GEI 2SG make-PERF this-CL matter

I have taken care of this matter for you.

As shown in (80b) and (81b), instead of the postverbal gěi-phrase the preverbal gěi-phrase must be used which indicates the benefactive and which is not related at all to the purposive gěi-clause.

We have seen so far that S2, with verbs different from verbs of giving, is in fact to be derived from S'2. Consequently, in

S2(Vb+c) only intentional verbs are allowed, exactly as in the matrix clause of S'2. That the purposive clause marker lái must precede 'gěi NP V2P' in both S'2(Va) and S'2(Vb+c) confirms the general analysis of S'2 as consisting of a matrix clause and a purposive gěi-clause. Furthermore, S'2 is only acceptable when the V2P (in the purposive gěi-clause) can be predicted on the basis of the matrix object.

The requirement of a tight relationship between the VP of the purposive gěi-clause and the object of the matrix clause does not represent a condition for the purposive clause in general, but only holds for the purposive clause with an overt subject marked by gěi. For a purposive clause with an empty subject, the relationship between its VP and the matrix clause can be less tight, provided it is semantically or pragmatically plausible. Consequently, for structures S'2 and S'5 with a purposive gěi-clause, the VP must be appropriate, i.e. predictable on the basis of the object in the matrix clause; whereas for structures S'1+4 and S'3 containing a purposive clause with an empty subject, the range of possible verbs in V2P position is larger. Xué yǔyánxué 'to learn linguistics', for example, does not qualify as an appropriate verb for shū 'book' in S'2 (cf. (82)), but it is admitted as the VP of the purposive clause with an empty subject in S'3 (cf. (83)):

- (82) *我買了一本書(來)給你學語言學
 Wo mai-le yi-ben shu (lai) (= (79))
 1SG buy-PERF 1 -CL book in:order:to
 gei ni xue yuyanxue
 GEI 2SG learn linguistics
 (I have bought a book so that you learn linguistics.)

- (83) 我給你買了一本書(來)學語言學
 Wo gei ni mai-le yi-ben shu (lai)
 1SG GEI 2SG buy-PERF 1 -CL book in:order:to
 xue yuyanxue
 learn linguistics

I have bought a book for you to learn linguistics.

Similarly, in S'1+4 we can choose among several verbs for the purposive clause, a situation which is excluded in S'2:

- (84) 我們給他三天時間(來)準備／考慮問題
 Women gei ta san-tian shijian (lai)
 1PL give 3SG 3 -day time in:order:to
 zhunbei/ kaolü wenti / ...
 prepare think problem
 We give him three days to get himself ready/ to think about the problem/ ...

Another difference between S'2 and S'5, on the one hand, and S'1+4, on the other hand, is the fact that verbs of giving Va are felt to be more natural in S'1+4 than in S'2: for in S'1+4 (cf. (86)), a verb of giving can place both its arguments within the matrix clause, which is not possible in S'2 (cf. (85)):

- (85) 我送了一把刀(來)給他切肉
 Wo song-le yi-ba dao (lai) gei ta qie rou
 1SG give-PERF 1 -CL knife in:order:to GEI 3SG cut meat

I have given a knife so that he cuts meat with it.

- (86) 我送(給)他一把刀(來)切肉
 Wo song (gei) ta yi-ba dao (lai) qie rou
 1SG give GEI 3SG 1 -CL knife in:order:to cut meat

I have given him a knife to cut meat with.

The position of the purposive clause marker lai indicates very clearly that gei ta in S'1+4 (86) is part of the matrix clause, whereas gei ta in S'2 (85) belongs clearly to the purposive gei-clause.

In S'2, we can also observe that when the meaning of V2P allows it, the NP preceded by gei is not understood as the

subject of a purposive gěi-clause, but as a benefactive gěi-phrase preceding V2P in a purposive clause with an empty subject. In such a case, the sentence in question is not interpreted as S'2, but is analysed as follows:

[_S NP [_V [_V V NP] [_S ∅ [_V [_{pp} gěi NP] [_V V (NP)]]]]]

(87a) 我從媽媽的口袋裏偷了一點錢(來)給你買東西

Wo cong mama -de koudaili tou -le yi-dian qian
1SG from mother-SUB pocket steal-PERF 1 -little money

(lai) gei ni mai dongxi
in:order:to GEI 2SG buy thing

I have stolen some money out of mother's pocket to buy something for you.

(87b) 我從媽媽的口袋裏偷了一點錢(來)給你花

Wo cong mama -de koudaili tou -le yi-dian qian
1SG from mother-SUB pocket steal-PERF 1 -little money

(lai) gei ni hua
in:order:to GEI 2SG spend

I have stolen some money out of mother's pocket so that you can spend it.

(88a) 我買了一隻筆(來)給你寫字

Wo mai-le yi-zhi bi (lai) gei ni xie zi
1SG buy-PERF 1 -CL pen in:o.:to GEI 2SG write character

I have bought a pen to write with for you.

(88b) 我買了一隻筆(來)給你使

wo mai-le yi-zhi bi (lai) gei ni shi
1SG buy-PERF 1 -CL pen in:order:to GEI 2SG use

I have bought a pen so that you use it.

(The fact that I have given only one translation for (87b) and (88b) is not intended to mean that they are unambiguous, but only that the interpretation of nǐ 'you' as the subject of the purposive gěi-clause is the dominant one.)

As shown in (87a) and (88a), gěi NP is especially liable to be interpreted as a benefactive when the relationship between V2P and the object in the matrix clause is not one of verb - object;

this again confirms our observation of the importance of a tight relationship between V2P and the matrix object for the acceptability of the purposive gěi-clause. The purposive clause marker lái cannot serve to distinguish between these two interpretations of gěi NP, for lái introduces a purposive clause with an overt subject marked by gěi (the purposive gěi-clause) as well as a purposive clause with an empty subject.

To summarize, the purposive gěi-clause is only acceptable if a tight relationship exists between V2P and the object in the matrix clause. If this relationship does not prove to be tight enough and there is another interpretation available, this latter will be chosen. For verbs which are not verbs of giving, a postverbal gěi-phrase (in S'2) is to be derived from a purposive gěi-clause (in S'2); that a postverbal gěi-phrase is not always acceptable can therefore be explained by the non-intentionality of the verb excluding a purposive clause in general, or by the fact that sometimes the appropriate verb required for the purposive gěi-clause cannot be found.

5.2.4.3. S'2 with Vd verbs (S'5)

The problem which still awaits an explanation is why the deletion of V2P is possible in the case of verbs with a concrete object NP (Vb+c), but is excluded for verbs with an abstract object NP (Vd), a phenomenon which had been the main reason for Zhu's distinction between S'2 and S'5.

When we look at a sentence which results from the deletion of V2P in S'2 (cf. (89b)), we see that it shows the same structure as a sentence with a verb of giving (cf. (90)), although the verb in the former sentence type (89b) only subcategorises for one argument. I would therefore like to claim that the structure

'NP V NP gěi NP' (S2) with verbs other than verbs of giving represents an instance of restructuring modeled on the structure originally reserved for verbs of giving, i.e. 'NP V NP gěi NP'. In the case of Vd verbs with an abstract object, then, this kind of restructuring is not possible, because the direct object of a verb of giving is normally a concrete NP:¹⁸

- (89a) 我打了一件毛衣給你穿
 Wo da -le yi-jian maoyi gei ni chuan (S'2)
 1SG knit-PERF 1 -CL sweater GEI 2SG wear

I have knitted a sweater so that you wear it.

- (89b) 我打了一件毛衣給你
 Wo da -le yi-jian maoyi gei ni (S2)
 1SG knit-PERF 1 -CL sweater GEI 2SG

I have knitted a sweater for you.

- (90) 我送了一件毛衣給你
 Wo song-le yi-jian maoyi gei ni
 1SG give-PERF 1 -CL sweater GEI 2SG

I have given you a sweater (as a present).

¹⁸ There are, however, examples of Va verbs with an abstract object NP:

- (i) 他給了我一個好的印象
 Ta gei -le wo yi-ge hao -de yinxiang
 3SG give-PERF 1SG 1 -CL good-SUB impression

He made a good impression on me.

(Peyraube 1988: 28)

- (ii) 他打電話給我
 Ta da dianhua gei wo
 3SG dispatch phone:call GEI 1SG

He rings me up.

But as Peyraube(1988: 28) points out, an abstract direct object is only acceptable in the form of a complex NP; as there is no such constraint on concrete direct objects, the latter can be considered to be the regular case. Sentence (ii) with diànhua 'phone call' seems to contradict Peyraube's observation that an abstract direct object cannot be a simple NP. Dǎ in dǎ diànhuà, however, is special in so far as it functions as a verb of giving Va/c only when in combination with diànhuà.

(91a) *我唱了一首歌給你

Wo chang-le yi-shou ge gei ni
1SG sing -PERF 1 -CL song GEI 2SG

(I have sung a song for you.)

(91b) 我唱了一首歌給你聽

Wo chang-le yi-shou ge gei ni ting (S'5)
1SG sing -PERF 1 -CL song GEI 2SG listen

I sing a song so that you listen to it.

Although S'2 (after the deletion of V2P) can be restructured according to the structure of verbs of giving, this does not mean that the gěi in the purposive gěi-clause has the same function as the gěi with verbs of giving, i.e. indicating the recipient in an exchange where the subject of V1 is the giver. The common interpretation of the subject in the purposive gěi-clause as the recipient is not concluded on the basis of the syntactic structure, but it is the result of a semantic-pragmatic inference: the subject NP being in the possession of the direct object is also responsible for its transfer to the person who is intended to use it (i.e. the subject of the purposive gěi-clause) (cf. above, 5.2.4.1.). In fact, if gěi in the purposive gěi-clause really indicated a transfer, Vd verbs should not be acceptable in S'5, because an abstract object cannot be exchanged. Consequently, S'2 is compatible with a context which explicitly excludes a transfer (from the matrix subject to the subject of the purposive gěi-clause):

(92) 我打了一件毛衣給張三穿，你下一次去他家請把他帶去

Wo da -le yi-jian maoyi gei Zhangsan chuan.
1SG knit-PERF 1 -CL sweater GEI Zhangsan wear

Ni xia yi-ci qu ta jia qing ba ta daiqu
2SG next 1 -time go 3SG home please BA 3SG take

I have knitted a sweater so that Zhangsan wears it. Next time you go to his home, take it with you, please.

That the gěi in the purposive gěi-clause does not express a transfer from the subject of the matrix clause to the subject of

the purposive gěi-clause constitutes further evidence for the distinction between the postverbal gěi indicating the recipient and the gěi introducing the subject of a purposive gěi-clause (Cf. also Huang 1982: 106, footnote 36).

The Vd verbs are useful to show the difference between the gěi in the purposive gěi-clause (S'2) and the gěi in preverbal position (S'3). For in the case of Vd verbs with an abstract object NP, the deletion of V2P in S'2 (leading to a postverbal gěi-phrase) is not allowed; only the purposive gěi-clause is acceptable. S'3, on the contrary, is grammatical without the purposive clause (whose subject is empty):

- (93) 我唱個歌給你 *(聽)
 Wo chang ge ge gei ni *(ting) (S'2)
 1SG sing CL song GEI 2SG listen

I sing a song so that you listen to it.

- (94) 我給你唱個歌(聽)
 Wo gei ni chang ge ge (ting) (S'3)
 1SG GEI 2SG sing CL song listen

I sing a song for you (to listen to)

Thus, the two gěi have different functions: the preverbal gěi (as in (94)) is a preposition indicating the benefactive, whereas the other gěi introduces the subject of the purposive gěi-clause. S'2 and S'3 therefore represent different structures; their quasi-synonymy can be explained by the possibility of using the same lexical items in S'2 and S'3 without there being any relationship on the structural level (cf. above, 5.2.2.).

Once the idea has been abandoned that the gěi in the purposive gěi-clause implies a transfer (as does the gěi with verbs of giving), it is now possible to analyse S'2 and S'5 in the same way, except for the deletability of V2P. For the requirement of a tight relationship between V2P and the object of V1 holds for

both structures, i.e. chuān 'to wear' is the appropriate verb for máoyī 'sweater' (cf. (95)) in the same way as tīng 'to listen' is the appropriate verb for zhōngwén 'Chinese' (cf. (96)):

(95) 我買了一件毛衣給他穿

Wo mai-le yi-jian maoyi gei ta chuan
1SG buy-PERF 1 -CL sweater GEI 3SG wear

I have bought a sweater so that he wears it.

(96) 他說中文給我們聽

Ta shuo zhongwen gei women ting
3SG speak Chinese GEI 1PL listen

He speaks Chinese so that we can listen to it.

Thus, S'2 and S'5 represent, in fact, the same structure. The nature of the object (concrete vs. abstract) becomes relevant only in restructuring of the purposive gěi-clause as a gěi-phrase. With a matrix verb taking a concrete object (Vb+c), the structure resulting from deletion of V2P in S'2 looks like the structure of verbs of giving and, accordingly, is acceptable. However, the deletion of V2P in S'2 where the matrix verb subcategorises for an abstract object, is not allowed. For the resulting structure with an abstract direct object: 'NP V (abstract)NP gěi NP' does not really imitate the typical structure of verbs of giving where the direct object is concrete.

5.2.5. Partial summary

We have seen that there are two different types of purposive clauses following the matrix clause, i.e. the purposive clause with an overt subject marked by gěi (the purposive gěi-clause) and the purposive clause with an empty subject which is normally coreferential with the matrix subject. The VP in the purposive gěi-clause must be appropriate, i.e. predictable on the basis of the matrix object, whereas the relation between the VP in the purposive clause with an empty subject and the matrix clause can

be less tight, provided it is semantically or pragmatically plausible. These two types of purposive clauses can optionally be introduced by the purposive clause marker lái, and they are only acceptable when the verb in the matrix clause is intentional.

The structures discussed by Zhu can be divided into two groups one comprising the structures with an empty subject in the purposive clause, the other those with an overt subject (marked by gěi) in the purposive clause. S'1+4, S'3 and S'6 belong to the first group, whereas S'2 (including Zhu's S'5) represents the second group.

We obtain the same distribution, i.e. S'1+4, S'3 and S'6 on the one hand, and S'2 on the other hand, in the structural analysis. In S'1+4, S'3 and S'6, V2P constitutes the purposive clause (with an empty subject):

S'1+4: [_S NP [_V [_V V (gěi) NP NP] [_S Ø VP]]]

S'3: [_S NP [_V gěi NP [_V V NP] [_S Ø VP]]]

S'6: [_S NP [_V [_V V (NP)] [_S Ø gěi NP kàn]]]

S'2: [_S NP [_V [_V V NP] [_S gěi NP V (NP)]]]

Contrary to the common interpretation, the empty subject of the purposive clause in S'1+4 and S'3 is not coreferential with the matrix subject. The fact that the VP of the purposive clause is related to the direct object in the matrix clause leads to a semantic-pragmatic inference: the person who obtains the direct object (the recipient) or profits from it (the benefactive) must be interpreted as the subject of the purposive clause.

S'6 has to be distinguished from S'1+4 and S'3, for the VP in the purposive clause, gěi NP kàn, is related to the entire action in the matrix clause (V1P), and not only to the matrix object.

This also accounts for the lack of any constraint on the VP in the matrix clause. As the empty subject is understood to be coreferential with the matrix subject, S'6 has approximately the following meaning: 'X does something to show Y the contrary of what Y considered to be true'.

The structural analysis can give indications as to the nature of the relationship between S'n and Sn. S1+4 and S3 can be considered as the basic structures for S'1+4 and S'3, respectively, because the VP in the matrix clause of S'1+4 and S'3 has the same structure as the VP in S1+4 and S3. For S'6, it does not make any sense to inquire about a basic structure it is derived from.

The situation is totally different in the case of S'2, because for verbs other than verbs of giving, it is S'2 which is the basic structure for S2. The postverbal position being reserved for the arguments subcategorised for by the verb, only verbs of giving can appear in S2, because only they take a recipient (marked by gěi) as second argument. For the other verbs, which equally figure in S2 (without subcategorising for a patient and a recipient), S2 must be regarded as the result of restructuring. Restructuring of the purposive gěi-clause as a gěi-phrase (via the deletion of V2P) is only possible when the matrix object is a concrete NP. For the model of restructuring is the structure of the verbs of giving, 'NP V NP gěi NP', where the direct object is normally a concrete NP. This explains why Vd verbs with an abstract object NP do not allow restructuring and, consequently, do not allow a postverbal gěi-phrase, either.

It is the impossibility of restructuring in the case of Vd verbs which made it necessary to modify James C.-T. Huang's analysis which served as the basis of this chapter. According to Huang(p.c.), restructuring of the purposive gěi-clause as gěi-

phrase is only possible if the (deleted) VP of the purposive gěi-clause can be recovered due to its tight relationship with the object in the matrix clause. But such an explanation would falsely predict the acceptability of restructuring for verbs Vd with an abstract object, where the link between V2P and the object of V1 is in fact particularly tight because it is always the verb - object relationship, and furthermore, practically only tīng 'to listen' and kàn 'to look' appear as verbs in the purposive gěi-clause. In my analysis, however, the predictability of V2P is not relevant for restructuring, but for the acceptability of the purposive gěi-clause itself (which then can or cannot undergo restructuring).

As far as the status of gěi in the purposive clause is concerned (which is not discussed by Zhu), Huang (1982: 106, footnote 36) considers it as a complementiser like the for in I bought a book for you to read. Another possible analysis is that gei forms one phrase with the subject of the purposive gěi-clause (Alain Rouveret, p.c.). This idea is interesting in so far as the overt subject in a purposive clause is obligatorily introduced by gěi: 'NP VP, *(gěi) NP VP'. But in order to decide between these analyses, there is still a need to study the purposive gěi-clause in Chinese in more detail than I have been able to do here. But it should be clear by now that whatever the nature of gěi in the purposive gěi-clause may be, it will be different from that of the gěi with verbs of giving.

5.3. The VOPs and the problem of the postverbal gěi-phrase

Last, but not least I will now explain why the outer object of a VOP marked by gěi is acceptable only in the preverbal position (cf. (97a)) and never in the postverbal position (cf. (97b)):

(97a) 醫生給病人開刀

Yisheng gei bingren kai dao
 doctor GEI patient open knife

The doctor operates on the patient.

(97b) *醫生開刀給病人

Yisheng kai dao gei bingren
 doctor open knife GEI patient

Given the preceding discussion of the postverbal gei-phrase in a structure like (97b), i.e. in S2, the first step in its analysis is to determine the origin of this gei-phrase, namely whether it is base-generated or whether it has to be derived from a purposive gei-clause via deletion of V2P. Since the outer object is the argument subcategorised for by the VOP as a whole, and not by the verb alone, the gei-phrase occurring with VOPs must be obtained from a purposive gei-clause. For only in the case of verbs of giving, which subcategorise for a patient and a recipient (in the form of a gei-phrase) does the postverbal gei-phrase not result from restructuring of the purposive gei-clause.

The VOPs admit a purposive clause with an empty subject. This indicates that the unacceptability of the outer object marked by gei in postverbal position cannot be due to the interdiction of a second postverbal constituent by the PSC:

(98) 醫生開刀給學生看

Yisheng kai dao gei xuesheng kan
 doctor open knife GEI student look

The doctor operates to demonstrate to the students.

Like simple intentional verbs, intentional VOPs admit a purposive clause with an empty subject (which is coreferential with the matrix subject); the VP gei NP kàn 'to show sb.' takes the whole action expressed by the VOP as its object. Consequently, (98) can be paraphrased as (99) (cf. above, example (38)):

(99) 醫生給學生看怎麼樣開刀

Yisheng gei xuesheng kan zenmeyang kai dao
 doctor GEI student look how open knife

The doctor shows the students how to operate.

It is evidently possible in (99) to mention the outer object gěi bǐngrén as well which clearly demonstrates that the purposive clause gěi NP kàn has nothing to do with the outer object:

(100) 醫生給病人開刀給學生看

Yisheng gei bingren kai dao gei xuesheng kan
 doctor GEI patient open knife GEI student look

The doctor operates on the patient to show the students.

If there is no general restriction ruling out a constituent following the VOP, then the unacceptability of the outer object marked by gěi can perhaps be explained by the fact that VOPs simply do not allow a purposive gěi-clause (with the outer object as its subject) from which the gěi-phrase could be derived.

It is perhaps useful here to recapitulate the possible relations between the VP of the purposive gěi-clause and the object in the matrix clause. The relation can either be one of verb - object where the object of the matrix verb is also understood to be the object of the verb in the purposive gěi-clause, or it can be a circumstantial relationship. In the second case, the matrix object functions as a circumstantial complement of the VP in the purposive gěi-clause, most often as an instrumental. The tight relation between the VP in the purposive gěi-clause and the object of the matrix verb (the prerequisite of the acceptability of a purposive gěi-clause) becomes evident in the possibility of integrating the matrix object into the purposive gěi-clause when the latter appears separately from the matrix clause (i.e. without gěi). This may seem trivial in the case of a verb - object relationship. In the case of a circumstantial relation-

ship, however, this is more revealing because circumstantial complements show different degrees of cohesion in their relation with the verb concerned.

(101a) 我買了一把刀給他用

Wo mai-le yi-ba dao gei ta yong
1SG buy-PERF 1 -CL knife GEI 3SG use

I bought a knife so that he will use it.

(101b) 他用這把刀

Ta yong zhei-ba dao
3SG use this-CL knife

He uses that knife.

(102a) 我買了一把刀給張三切肉

Wo mai-le yi-ba dao gei Zhangsan qie rou
1SG buy-PERF 1 -CL knife GEI Zhangsan cut meat

I have bought a knife so that Zhangsan will cut meat with it.

(102b) 張三用這把刀切肉

Zhangsan yong zhei-ba dao qie rou
Zhangsan use this-CL knife cut meat

Zhangsan uses this knife to cut meat.

(103a) 我租了一間房間給他堆東西

Wo zu -le yi-jian fangjian gei ta dui dongxi
1SG rent-PERF 1 -CL room GEI 3SG pile:up thing

I have rented a room so that he will put his things there.

(103b) 他用這間房間堆東西

Ta yong zhei-jian fangjian dui dongxi
3SG use this-CL room pile:up things

He uses this room to put his things.

(103c) 他在這間房間堆東西

Ta zai zhei-jian fangjian dui dongxi
3SG in this-CL room pile:up thing

He puts his things in this room.

Let us now apply these observations to the purposive g_{ei}-clause co-occurring with VOPs. First, a verb - object relation between the verb of the purposive g_{ei}-clause and the object of

the matrix verb, i.e. the inner object, is excluded on the ground that the inner object is a possible object (with the meaning it has in a given VOP) only for the verb with which it constitutes the VOP. Thus, there only remains the circumstantial relation to establish the necessary link between the purposive gěi-clause and the matrix clause. Accordingly, in a sentence like (104), the inner object (dāo) would have to serve as a circumstantial complement of the VP in the purposive gěi-clause where the outer object (bìngrén) would play the role of the subject:

(104) *醫生開刀給病人恢復健康

Yisheng kai dao gei bingren huifu jiankang
 doctor open knife GEI patient recover health

(The doctor has operated so that the patient recovers his health.)

Such a relation, however, cannot be established because the VOP functions as a whole and, consequently, the inner object alone cannot entertain relations with elements outside the VOP. It therefore cannot serve as the instrumental for the VP in the purposive gěi-clause (cf. (104)), a relation which would give rise to the unacceptable sentence (105):

(105) *病人用刀恢復健康

Bingren yong dao huifu jiankang
 patient use knife recover health

(cf. (101) -(103))

(In the case of idiomatic VOPs like kāi dāo 'to operate on', the inner object in a sentence like (105) can only have a literal meaning, i.e. it does not, in fact, function as an inner object any more.)

There is, however, no problem for a VOP being followed by a purposive clause with an empty subject like gěi NP kàn 'to show sb.' which refers to the entire VOP (cf. (98)).

Thus, the reason why VOPs do not accept their outer object marked by gěi in postverbal position is that they do not admit a purposive gěi-clause from which the gěi-phrase containing the outer object could be derived. This situation is entirely different from that of Vd verbs with an abstract object for which a purposive gěi-clause is acceptable, and only restructuring of the purposive gěi-clause as a gěi-phrase is excluded due to the non-concrete nature of the object (cf. above, (91)).

I have not yet excluded the possibility that the VOPs are incompatible with a purposive gěi-clause because appropriate verbs for them cannot be found (cf. above, (80) and (81)). Though I consider such an explanation rather improbable, given the inaccessibility of the inner object to relations with elements outside the VOP, there are examples of the type (106) and (107) where the same noun functions as an inner object as well as a "normal" object. If a relationship with the inner object were really possible, (107a) would have to be acceptable, because dài 'to wear, to put on' proves to be the appropriate verb for màozi 'hat' in (106):

(106) 我買了一頂帽子給他戴

Wo mai-le yi-ding maozi gei ta dai
1SG buy-PERF 1 -CL hat GEI 3SG wear

I have bought a hat so that he will wear it.

(107a) *我扣了帽子給他戴

Wo kou-le maozi gei ta dai
1SG put-PERF hat GEI 3SG wear

(107b) 我給他扣了帽子

Wo gei ta kou-le maozi
1SG GEI 3SG put-PERF hat

I have denounced him.

Thus, VOPs do not admit a postverbal gěi-phrase (containing the outer object) because they do not allow a purposive gěi-

clause from which the $\underline{g\check{e}i}$ -phrase could be derived (via deletion of V2P). This is not due to the difficulty of finding an appropriate verb, but can be explained by the inaccessibility of the inner object to relations with elements outside the VOP, more precisely with the VP of a purposive $\underline{g\check{e}i}$ -clause. Such a relation, however, is exactly the prerequisite of the acceptability of the purposive $\underline{g\check{e}i}$ -clause.

After the detailed description of the purposive $\underline{g\check{e}i}$ -clause in Chinese, the question remains to be discussed whether the purposive $\underline{g\check{e}i}$ -clause satisfies the PSC.

The structural analysis I have proposed, namely $[_S NP [_{\bar{V}} [_{\bar{V}} V NP] [_S \underline{g\check{e}i} NP V (NP)]]]$, in fact violates the PSC, for $\bar{V}$ shows a head-initial structure which is only allowed on the lowest level of expansion (here $\bar{V}$). As we have already observed other structures which at first sight seemed to represent a violation of the PSC but which in the end could be explained as cases of restructuring modeled on structures admitted by the PSC, I will try to apply the same procedure here as well.

One of the model structures for restructuring which has already been mentioned is the structure reserved for verbs of giving, namely 'NP V NP $\underline{g\check{e}i}$ NP'. Though I do consider a post-verbal $\underline{g\check{e}i}$ -phrase occurring with verbs different from verbs of giving as the result of restructuring modeled on precisely this structure, such an analysis can, however, not be extended to include a complete purposive $\underline{g\check{e}i}$ -clause whose VP has not been deleted. Even if one accepts the analysis of the $\underline{g\check{e}i}$ in the purposive $\underline{g\check{e}i}$ -clause as a complementiser (cf. Huang 1982) as well as the assumption that a complementiser is in fact very much like a preposition (except that the former takes a sentence as its object, and not a NP), it nevertheless remains impossible to

postulate restructuring of a complex sentence with a purposive gěi-clause according to the model structure 'NP V NP gěi NP'. For after what we have seen so far, it is the surface identity of the two structures which makes restructuring possible; the analogy between [_{pp} Prep NP] and [_s Comp [_s]] is a case of identity which is much too abstract to be relevant for restructuring.

The general idea contained in the PSC which says that the postverbal position is subject to many more constraints than is the preverbal position, however, seems still to be correct and is furthermore confirmed by all the data discussed so far. Thus, the tight relationship between the direct object in the matrix clause and the verb in the purposive gěi-clause (which is necessary for the acceptability of the purposive gěi-clause) is an example of the constraints governing the postverbal position. But the PSC is unable to elucidate the reasons for these constraints; on the contrary, it falsely predicts the unacceptability of complex sentences with a purposive gěi-clause. A reconsideration of the PSC therefore seems necessary and will be the subject of chapter 7.

6. THE POSSESSIVE OBJECT AND THE PHENOMENON OF REANALYSIS

We have already discussed the possessive object as one of the possible forms for the outer object (cf. above, ch. 4.5.). The outer object is said to take on the form of the possessive object when it occupies the modifier position of the inner object and is subordinated by de:

(1) 他丟我的臉了

Ta diu wo -de lian le
3SG lose 1SG-SUB face PART

He made me lose face.

(2) 法院治了他的罪

Fayuan zhi -le ta -de zui
law:court punish-PERF 3SG-SUB guilt

The law court sentenced him.

This reanalysis of the outer object as the modifier of the inner object is motivated by the PSC and results in a structure with only one constituent in the postverbal position, which in fact looks like the following structure:

(3) 他看了三個鐘頭的書

Ta kan -le san-ge zhongtou-de shu
3SG look-PERF 3 -CL hour -SUB book

He read for three hours.

In (3), it is the complement of duration, sān-ge zhōngtóu, which appears as the modifier of the direct object, shū. Restructuring¹ of the complement of duration or of frequency as an apparent

¹ The reason why I make a distinction between "reanalysis" and "restructuring" will become clear later in the discussion (cf. 6.1.). I am indebted to Alain Rouveret who drew my attention to the importance of distinguishing between restructuring and reanalysis. However, the way I developed his suggestion is mine, possible misinterpretations included.

modifier (cf. (3)) is thus another way, besides verb copying (cf. (4)), to make a sentence like (5) acceptable:

- (4) 他看書看了三個鐘頭
 Ta kan shu kan -le san-ge zhongtou
 3SG look book look-PERF 3 -CL hour

He read for three hours.

- (5) *他看了書三個鐘頭
 Ta kan -le shu san-ge zhongtou
 3SG look-PERF book 3 -CL hour

As a first step in my analysis of the possessive object, I will compare it with restructuring of the complement of duration or of frequency as (apparent) modifier of the direct object. Special attention will be paid here to how the NP resulting from the reanalysis of the outer object and the NP resulting from restructuring of the complement behave when the head noun is deleted, taking the behaviour of an ordinary NP with a real modifier (like wǒ-de shū '1SG-SUB book' = 'my book') as the standard of comparison. Furthermore, I will look at the distribution of the reanalysed NP and the restructured NP and explain why the positional restrictions are not the same.

I will then examine whether the reanalysed NP [o.O. de i.O.] is accessible to syntactic transformations, like passivisation and topicalisation, in the same way as an ordinary NP with a real modifier. In this context, I will also deal with VOPs which have a compositional as well as a non-compositional meaning and I will show that they behave like non-ambiguous VOPs. As the last point in the comparison between the reanalysed NP and the ordinary NP, I will examine the VOPs whose inner objects denote a body part. These VOPs are interesting because their comparison with nouns denoting body parts (but not being part of a VOP) can indicate to us whether there are any restrictions for a noun when functioning as the inner object of a VOP.

The discussion of the possessive object is concluded by the description of a special case: the outer object reanalysed as the possessive object does not play the role of patient with respect to the action expressed by the VOP, but rather the role of agent.

Finally, I will present the "possessive subject", i.e. the reanalysis of the subject as modifier of the object (preposed to the verb) and look at the differences between the possessive subject and the possessive object.

6.1. The thematic reanalysis vs. syntactic restructuring

Though superficially the NP resulting from the reanalysis (e.g. tā-de zuì, cf. (2)) and the NP resulting from restructuring (e.g. sān-ge zhōngtóu-de shū, cf. (3)) have the same structure, namely 'NP de N', they behave differently with respect to the deletion of the head noun in coordinated structures ("conjoining deletion").² Whereas the reanalysed NP retains the subordination marker de (cf. (7) - (9)) and thereby shows the same behaviour as a NP with a real modifier (cf. (6)), the restructured NP does not keep de once the head noun has been deleted (cf. (10) and (11)):

(6) 我可以借張三的書，你可以借李四*(的)

Wo keyi jie Zhangsan-de shu ,
1SG can borrow Zhangsan-SUB book

ni keyi jie Lisi *(-de)
2SG can borrow Lisi -SUB

I can borrow Zhangsan's book, you can borrow Lisi's.

² I am thankful to Marie-Claude Paris for having pointed out to me the importance of the conjoining deletion for an analysis of the possessive object.

- (7) 他開別人的玩笑沒關係，別人開他 *(的)就不行
 Ta kai bieren -de wanxiao mei guanxi
 3SG open other:people-SUB joke NEG importance

 bieren kai ta *(-de) jiu bu xing
 other:people open 3SG -SUB then NEG possible

 If he makes fun of other people, it doesn't matter; but
 if other people make fun of him, he cannot stand it.
- (8) 要是你敢說張三的壞話，我也敢說你 *(的)
 Yaoshi ni gan shuo Zhangsan-de huaihua
 if 2SG dare talk Zhangsan-SUB vicious:talk

 wo ye gan shuo ni *(-de)
 1SG also dare talk 2SG -SUB

 If you dare speak ill of Zhangsan, I will dare
 speak ill of you, too.
- (9) 要是你停張三的職，我就停李四 *(的)
 Yaoshi ni ting Zhangsan-de zhi ,
 if 2SG stop Zhangsan-SUB work

 wo jiu ting Lisi *(-de)
 1SG then stop Lisi -SUB

 If you fire Zhangsan, I will fire Lisi.
- (10) 我看了三個鐘頭的書，你看了五個鐘頭 *(的)
 Wo kan -le san-ge zhongtou -de shu ,
 1SG look-PERF 3 -CL hour -SUB book

 ni kan -le wu -ge zhongtou (*-de)
 2SG look-PERF 5 -CL hour -SUB

 I read for three hours, he read for five hours.
- (11) 他學了一年的中文，我學了四年 *(的)
 Ta xue -le yi-nian -de zhongwen,
 3SG learn-PERF 1 -year -SUB Chinese

 wo xue -le si-nian (*-de)
 1SG learn-PERF 4 -year -SUB

 He learnt Chinese for one year, I learnt it for four
 years.

Sentences (10) and (11) show that when the verb is accompanied by only one constituent (the complement of duration), there is no restructuring; the complement of duration follows the verb and does not allow de, because there is no NP it could be part of. The possessive subject, however, keeps de as does a real modifier

when the head noun is deleted. This is, in fact, the expected behaviour because without de the outer object is understood to be the direct object of the verb itself. But as the outer object is an argument subcategorised for by the VOP as a whole, and not by the verb, such a sentence without de is unacceptable:

- (12) 他開別人的玩笑沒關係，*別人開他就不行
 Ta kai bieren -de wanxiao mei guanxi ,
 3SG open other:people-SUB joke NEG importance

 *bieren kai ta jiu bu xing
 other:people open 3SG then NEG possible (cf. (7))

In the elliptical answer, we observe the same situation as in the conjoining deletion: the possessive object (cf. (14), (15)), like a real modifier (cf. (13)), keeps de after the deletion of the head noun, whereas in the case of the restructured complement of duration (cf. (16), (17)), de is deleted as well as the head noun:³

³ However, the reanalysed NP [inner object de outer object] differs from an ordinary NP with a real modifier in two respects. First, in the conjoining deletion, the inner object (the head noun) cannot be deleted if the coordinated structures have identical subjects (cf. (i)), whereas this does not influence the deletability of the head noun in an ordinary NP (cf. (ii)):

- (i) 他開了張三的玩笑，也開了李四的*(玩笑)
 Ta kai -le Zhangsan-de wanxiao
 3SG open-PERF Zhangsan-SUB joke

 ye kai -le Lisi-de *(wanxiao)
 also open-PERF Lisi-SUB joke

 He made fun of Zhangsan, and he also made fun of Lisi.

 (ii) 他借了張三的書，也借了李四的(書)
 Ta jie -le Zhangsan-de shu
 3SG borrow-PERF Zhangsan-SUB book

 ye jie -le Lisi-de (shu)
 also borrow-PERF Lisi-SUB book

 He borrowed Zhangsan's book, and he also borrowed Lisi's (book).

Secondly, in order for the deletion of the inner object (in head noun position) to be acceptable in an answer, the answer

- (13) 你借了誰的書？李四 *(的)

Ni jie -le shei-de shu ? Lisi *(-de)
 2SG borrow-PERF who -SUB book Lisi -SUB

Whose book did you borrow? Lisi's.

- (14) 他開了誰的玩笑？李四 *(的)

Ta kai -le shei-de wanxiao ? Lisi *(-de)
 3SG open-PERF who -SUB joke Lisi -SUB

Whom did he make fun of? Lisi.

- (15) 他停了誰的職？李四 *(的)

Ta ting-le shei-de zhi ? Lisi *(-de)
 3SG stop-PERF who -SUB work Lisi -SUB

Whom did he fire? Lisi.

- (16) 你看了幾個鐘頭的書？（我看了）三個鐘頭(*(的)

Ni kan -le ji -ge zhongtou-de shu ?
 2SG look-PERF how:many-CL hour -SUB book

(wo kan -le) san-ge zhongtou (*-de)
 1SG look-PERF 3 -CL hour -SUB

For how many hours did you read ? (I read) for three hours.

- (17) 他學了幾年的中文？（他學了）五年(*(的)

Ta xue -le ji -nian-de zhongwen ?
 3SG learn-PERF how:many-year-SUB Chinese

(ta xue -le) wu-nian (*-de)
 3SG learn-PERF 5 -year -SUB

For how many years did he learn Chinese? (He learnt it) for five years.

 (footnote 3 continued)

must be elliptical:

- (iii) 他停了誰的職？

Ta ting-le shei-de zhi ?
 3SG stop-PERF who -SUB work

- a) *他停了李四的

Ta ting-le Lisi-de
 3SG stop-PERF Lisi-SUB

- b) 李四的

Lisi-de
 Lisi-SUB

Whom did he fire ? Lisi.

I do not have an explanation for this phenomenon. It is perhaps interesting to note that there are no such restrictions for the restructuring of the complement of duration (cf. (16) and (17)).

On the basis of these data it is now possible to determine the status of de in both the reanalysis and restructuring.

Following Teng's(1974: 460) terminology, the de of the possessive object, which has to be retained when the head noun, the inner object, is deleted, is a "genuine possessive marker", i.e. it behaves like the de in an ordinary NP with a real modifier (e.g. Zhāngsān-de shū 'Zhangsan's book').

The de occurring with the restructured complement of duration (or of frequency), however, is a "pseudo-possessive marker" (ibid.).⁴ But what is the nature of this pseudo-possessive marker? According to James C.-T. Huang (p.c.), it represents the same de as the one which optionally appears between the quantifier phrase 'Q-CL' and the head noun. To be more precise (cf. Paris 1981: 91), de is only admitted in the case of nouns functioning like classifiers and denoting a volume or a surface, e.g. kuāng 'basket', bāo 'packet' and zhuōzi 'table' (cf. (20) - (22)). The "real" classifiers, i.e. those which exclusively function as classifiers, however, never allow de:

- (18) 一本(*的)
 yi-ben (*-de) shu
 1 -CL -SUB book
 one book
- (19) 兩個(*的)人
 liang-ge (*-de) ren
 2 -CL -SUB person
 two persons

⁴ Though the deletion test as well as the terminology are those used by Teng(1974), my result differs from his. For Teng(1974: 459) regards the de of the possessive object as a "pseudo-possessive marker", i.e. he treats it on a par with the de in sān-ge zhōngtóu-de shū. I have shown that in these two cases de's behaviour differs.

That the de in restructuring is the same as the de with nouns functioning like classifiers is confirmed by the fact that the latter de must not be retained in the conjoining deletion:⁵

- (20) 他昨天買了三筐的梨，我只買了兩筐(*的)

Ta zuotian mai-le san-kuang -de li
3SG yesterday buy-PERF 3 -basket-SUB pear

wo zhi mai-le liang-kuang (*-de)
1SG only buy-PERF 2 -basket -SUB

He bought three baskets of pears yesterday, I only bought two baskets.

- (21) 張三買了四包的香烟，李四買了五包(*的)

Zhangsan mai-le si-bao -de xiangyan
Zhangsan buy-PERF 4 -packet-SUB cigarette

Lisi mai-le wu-bao (*-de)
Lisi buy-PERF 5 -packet -SUB

Zhangsan bought four packets of cigarettes, Lisi bought five packets.

- (22) 我家昨天剩了一桌子的菜，今天又剩了一桌子(*的)

Wo jia zuotian sheng -le yi-zhuozi-de cai
1SG home yesterday leave:over-PERF 1 -table -SUB dish

jintian you sheng -le yi-zhuozi (*-de)
today again leave:over-PERF 1 -table -SUB

Yesterday, I had a whole table full of leftovers, and today, it's the same again.

The de in restructuring thus behaves like the de which can follow nouns functioning as classifiers. This seems a plausible result in so far as de only appears in the restructured NP when the complement of duration or of frequency itself has the structure 'Q-CL N' (like sān-ge zhōngtóu 'three hours' in (3)) and, consequently, does not fit in the position of the quantifier phrase 'Q-CL', but must assume the form of a modifier

⁵ It is true that the judgments of the well-formedness of sentences (20) - (22)) vary. But for my purpose it suffices to show that those speakers who do accept de between the quantifier phrase and the head noun do not accept it once the head noun has been deleted.

subordinated by de. If, however, the complement of duration or of frequency consists of a quantifier and a classifier, de is not allowed in the restructured NP, as is the case in an ordinary NP with a real classifier (cf. (18) and (19)):

- (23) 他只看過一次(*的)電視
 Ta zhi kan -guo yi-ci (*-de) dianshi
 3SG only look-PERF 1 -time -SUB television
 He has watched television only once.

- (24) 他進了一趟城
 Ta jin -le yi-tang cheng
 3SG enter-PERF 1 -time town
 He has been downtown once.

The data from the conjoining deletion have shown that, syntactically speaking, the possessive object behaves in the same way as any other NP in the modifier position; what distinguishes the possessive object from a real modifier is the fact that the former represents the outer object of a VOP and can therefore in principle be passivised (cf. above, ch. 4.5.). We thus have a reinterpretation on the level of thematic relations or thematic roles like 'agent' and 'patient', which is the reason why I have chosen the term "thematic reanalysis" for this phenomenon. In the case of the possessive object, the NP in the modifier position, which is not linked to any particular thematic role, plays in fact the role of the patient of the VOP; accordingly, biérén-de wánxiào 'other people-SUB joke' in (7) is understood as 'the fun made of other people'. Interestingly enough, a NP like biérén-de wánxiào is also acceptable when biérén 'other people' is interpreted as a real modifier, the whole NP then having the meaning of 'other people's fun'. This is possible because biérén is compatible with the head noun wánxiào.

Now if a reanalysed NP as such (i.e. under a literal interpretation) is well-formed (the reanalysis taking place on the

level of thematic roles), one would expect it to be able to occur in other positions than in only the object position. Sentences (25b), (25c) and (25d) where the reanalysed NP is the subject in a passive sentence, precedes the verb (being marked by bǎ) or occupies the topic position, respectively, show that this prediction is borne out:

(25a) 上級停了張三的職

Shangji ting-le Zhangsan-de zhi
boss stop-PERF Zhangsan-SUB work

The boss fired Zhangsan.

(25b) 張三的職被上級停了

Zhangsan-de zhi bei shangji ting-le
Zhangsan-SUB work PASS boss stop-PERF

Zhangsan was fired by the boss.

(25c) 上級把張三的職停了

Shangji ba Zhangsan-de zhi ting-le
boss BA Zhangsan-SUB work stop-PERF

The boss fired Zhangsan.

(25d) 張三的職上級不敢停

Zhangsan-de zhi, shangji bu gan ting
Zhangsan-SUB work boss NEG dare stop

The boss does not dare fire Zhangsan.

The situation is totally different in the case of restructuring, which provides further evidence for the distinction between these two phenomena. A restructured NP like sān-ge zhōngtóu-de shū '3-CL hour SUB book' (cf. (3)) is restricted to the object position, because elsewhere (or in isolation) it is not acceptable, shū 'book' not being compatible with a temporal expression as its modifier. When the restructured NP sān-ge zhōngtóu-de shū occupies the object position, however, sān-ge zhōngtóu 'three hours' can be interpreted as the (restructured) complement of duration, which as an outer complement of the verb has to be compatible with the verb, and

not with the direct object.⁶

This relation between the verb and the complement of duration or of frequency becomes very clear in sentences (26) and (27), which have the same direct object, xìn 'letter', but different verbs; the non-durative verb shōudào 'to receive' not being compatible with a complement of duration, (27) is unacceptable:

⁶ A temporal expression in the modifier position of the direct object can, but need not be interpreted as a complement of duration or of frequency, because it is possible for the temporal expression to function as a real modifier, provided that it is compatible with the head noun. Therefore, the following example by Teng has two interpretations: in (a), sì-tiān 'four days' is a real modifier, whereas sì-tiān 'four days' in (b) is the restructured complement of duration:

(i) 我買了四天的糧食

Wo mai-le si-tian-de liangshi
1SG buy-PERF 4 -day -SUB provisions

a) I have bought four days' provisions.

b) For four days, I have been buying provisions.

(Teng 1974: 460)

But when liángshi 'provisions' occupies the topic position, there is no ambiguity. If sì-tiān 'four days' is a real modifier, it is followed by de, which shows that it is part of a NP whose head noun has been deleted (cf. (ii)). If sì-tiān 'four days' functions as the complement of duration, de does not appear, because with the direct object being topicalised, there is no need for restructuring and sì-tiān 'four days' simply follows the verb (cf. (iii)):

(ii) 糧食我買了四天的，蠟燭我買了六天的

Liangshi, wo mai-le si-tian-de
provisions 1SG buy-PERF 4 -day -SUB

lazhu, wo mai-le liu-tian-de
candle 1SG buy-PERF 6 -day -SUB

Provisions, I have bought them (to last) for four days;
candles, I have bought them (to last) for six days.

(iii) 糧食我買了四天，蠟燭我買了六天

Liangshi, wo mai-le si-tian
provisions 1SG buy-PERF 4 -day

lazhu, wo mai-le liu-tian
candle 1SG buy-PERF 6 -day

Provisions, I have been buying them for four days;
candles, I have been buying them for six days.

- 26) 他寫了三個鐘頭的信
 Ta xie -le san-ge zhongtou-de xin
 3SG write-PERF 3 -CL hour -SUB letter
 He wrote a letter for three hours.

- (27) *他收到了三個鐘頭的信
 Ta shoudao-le san-ge zhongtou-de xin
 3SG receive-PERF 3 -CL hour -SUB letter

The contrast of acceptability between (26) and (27) can only be explained by referring to the phenomenon of restructuring; for a real modifier is only related to the head noun and does not depend on the characteristics of the verb.

The following sentences illustrate my claim that the restructured NP can only appear in the object position, i.e. in the postverbal position:

- (28a) 他等了半天的信
 Ta deng-le bantian -de xin
 3SG wait-PERF long:time-SUB letter
 He long waited for a letter.

- (28b) *半天的信終於到了
 bantian -de xin zhongyu dao -le
 long:time-SUB letter finally arrive-PERF

- (29a) 他寫了三個鐘頭的信
 Ta xie -le san-ge zhongtou-de xin
 3SG write-PERF 3 -CL hour -SUB letter
 He wrote a letter for three hours.

- (29b) *我把三個鐘頭的信看得很仔細
 wo ba san-ge zhongtou-de xin kan de hen zixi
 1SG BA 3 -CL hour -SUB letter look DE very careful

The ungrammaticality of (28b) shows that the NP containing the restructured complement of duration is restricted to the object position because elsewhere the temporal expression cannot be interpreted as a complement, but must be understood as a real modifier. Bàntiān 'a long time', however, is not a possible modifier for xìn 'letter', and even though the preceding sentence

makes clear that bàntiān is not a real modifier, but plays the role of the complement of duration, sentence (28b) is not acceptable. Example (29b) is particularly interesting because it clearly shows that it is the object position, and not the object role which is relevant for restructuring; though xìn 'letter' in (29b) functions as the object, it does not allow a restructured complement of duration in its modifier position, because instead of occupying the object position, i.e. the postverbal position, it precedes the verb (being marked by bǎ).

There are, however, sentences which seem to invalidate my claim that the restructured NP is confined to the object position:

- (30) 連一分鐘的書我都不能看

Lian yi-fenzhong-de shu wo dou bu neng kan
even 1 -minute -SUB book 1SG DOU NEG can look

I cannot even read for a single minute.

- (31) 連半分鐘的電視他都不讓我看

Lian ban -fenzhong-de dianshi ta dou bu rang wo kan
even half-minute -SUB telev. 3SG DOU NEG let 1SG look

He doesn't even let me watch television for a single moment.

In both (30) and (31), the NP in the topic position must be interpreted as a restructured NP because neither yī-fēnzhōng 'one minute' nor bàn-fēnzhōng '30 seconds' are possible modifiers for shū 'book' and diànshì 'television', respectively. But it is important to notice that (30) and (31) both involve the (lián)... dōu construction in negation.⁷ Now it is well-known that the constituent the (lián) ... dōu construction bears on cannot follow the verb, but must occupy a preverbal position:

⁷ What I say in the following equally applies to (lián) ...yě.

- (32a) 連孩子他都不喜歡
 Lian haizi ta dou bu xihuan
 even child 3SG DOU NEG like
 He does not even like children.

- (32b) *他都不喜歡連孩子
 Ta dou bu xihuan lian haizi
 3SG DOU NEG like even child

As example (32) shows, the object NP háizi 'child' has to appear in preverbal position, i.e. it has to precede the universal quantifier dōu in order to fall into the scope of dōu (cf. Huang 1982: 310, footnote 12; Li/Thompson 1981: 336; Paris 1979: 62). Consequently, in the case of the (lián) ... dōu construction, the position of the object is preverbal (i.e. before dōu), and not postverbal. If we apply this to the apparent counterexamples (30) and (31), we see that the restructured NP occurs in precisely the position reserved for the object NP in such a construction.

That the preverbal position is allowed for the restructured NP only in the (lián) ... dōu construction (involving negation) is confirmed by the unacceptability of (33b) which shows the restructured NP in the topic position (i.e. preverbally), but without (lián) ... dōu:

- (33a) 他讓我看四個鐘頭的電視
 Ta rang wo kan si-ge zhongtou-de dianshi
 3SG let 1SG look 4 -CL hour -SUB television
 He let me watch television for four hours.

- (33b) *四個鐘頭的電視他讓我看
 Si-ge zhongtou-de dianshi , ta rang wo kan
 4 -CL hour -SUB television 3SG let 1SG look

Let us summarize the results obtained so far from the comparison between the thematic reanalysis and syntactic restructuring.

Both the thematic reanalysis and syntactic restructuring are

motivated by the PSC and consist in merging two postverbal constituents into one NP, with the object of the verb as its head and the other constituent as modifier. This "new" NP displays different properties depending on whether it is the complement of duration or of frequency which occupies the modifier position, or whether it is the outer object.

In the case of restructuring the complement of duration or of frequency, the subordinator de appears only before the head noun if the structure of the complement is 'Q-CL N' (like sān-ge zhōngtóu 'three hours'). If, however, the complement has the structure 'Q-CL' (like wǔ-cì 'five times') it can occupy the position of the quantifier phrase (in the restructured NP) and there is no de.

The complement of duration or of frequency is restructured only if necessitated by the PSC, i.e. if the direct object and the complement both follow the verb.⁸ If the direct object is deleted in coordinated structures or in an elliptical answer, the complement of duration or of frequency simply follows the verb, restructuring does not take place and there is no de, either. This shows that restructuring depends on a particular syntactic configuration (the modification of which renders restructuring impossible) which has motivated the choice of the term syntactic restructuring in contrast to thematic reanalysis.

As an outer complement of the verb, the (restructured) complement of duration or of frequency must be compatible with the verb, and not with the head noun, i.e. the direct object. This dependence on the verb clearly distinguishes it from a real

⁸ Another way of avoiding a violation of the PSC in this case is verb copying (cf. above, chapter 3.2.).

modifier which is related only to the head noun. It also explains why the restructured NP is restricted to the object position, which normally is equivalent to the position immediately following the verb, except in the (lián) ... dōu construction where it is in front of dōu. Only in the object position can the apparent modifier NP be identified as the restructured complement of duration or of frequency. Elsewhere it must be interpreted as a real modifier which in most cases, though, is impossible due to the incompatibility of the head noun with a temporal expression as its modifier (cf. footnote 6).

The NP resulting from the thematic reanalysis where the outer object in form of the possessive object is subordinated (by de) to the inner object, behaves like an ordinary NP. Consequently, when the head noun, the inner object, is deleted, de is retained as is the de in an ordinary NP with a real modifier under deletion of the head noun. Contrary to the restructured NP, the reanalysed NP can also appear elsewhere other than in the object position, i.e. elsewhere other than in the position where the reanalysis helps to avoid a violation of the PSC.

Thus, it is not the syntactic behaviour which distinguishes the reanalysed NP from an ordinary NP, but the fact that the NP in the modifier position of the restructured NP is the outer object of a VOP. It is related to the inner object (which accounts for the conservation of de under deletion of the head noun) as well as to the verb, which is not visible in the reanalysed NP itself, but which is reflected in the passivisability (excluded for a real modifier) of the outer object.

Contrary to restructuring of the complement of duration or of frequency, the reanalysis of the outer object as the possessive object cannot be predicted. This is not surprising because we

have already seen in chapter 4 that the form of the outer object is in general not predictable and must therefore be included in the lexical entry of each VOP.

6.2. The reanalysed NP vs. the ordinary NP

We have seen in the preceding section that the reanalysed NP can appear elsewhere other than in the object position, e.g. in the subject position of a passive sentence (cf. (34b)):

- (34a) 上級停了張三的職
 Shangji ting-le Zhangsan-de zhi (= (25a))
 boss stop-PERF Zhangsan-SUB work

The boss has fired Zhangsan.

- (34b) 張三的職被上級停了
 Zhangsan-de zhi bei shangji ting-le (= (25b))
 Zhangsan-SUB work PASS boss stop-PERF

Zhangsan has been fired by the boss.

Sentence (34b) is interesting in so far as it illustrates a case of optional reanalysis (in contrast to (34a) where the reanalysis is necessary for the acceptability of the sentence). For as shown in (35), if the outer object Zhāngsān alone is passivised, the resulting sentence is perfectly grammatical, the verb being followed by only one constituent, the inner object:

- (35) 張三被上級停了職
 Zhangsan bei shangji ting-le zhi
 Zhangsan PASS boss stop-PERF work

Zhangsan has been fired by the boss.

In fact, (34b) shows the same behaviour as an ordinary NP with a real modifier, where only the entire NP can be passivised:

- (36a) 我打破了李四的杯子
 Wo dapo -le Lisi-de beizi
 1SG break-PERF Lisi-SUB cup

I have broken Lisi's cup.

(36b) *李四被我打破了杯子

Lisi bei wo dapo -le beizi
 Lisi PASS 1SG break-PERF cup

(36c) 李四的杯子被我打破了

Lisi-de beizi bei wo dapo -le
 Lisi-SUB cup PASS 1SG break-PERF

Lisi's cup has been broken by me.

If one examines whether other VOPs whose outer object appears in the form of the possessive object and which allow the passivisation of the outer object alone also admit the passivisation of the entire reanalysed NP, one sees that this is not systematically the case. At the same time acceptability judgments vary among native speakers; this is probably due to the optionality of reanalysis here. In the case of chāi tái 'to take apart platform' = 'to let sb. down', jiē dǐ 'to take off bottom' = 'to reveal', for example, the outer object as well as the reanalysed NP can be the subject of a passive sentence, whereas for kāi wánxiào 'to open joke' = 'to make fun of' and zhàn piányi 'to seize cheap' = 'to profit at sb.'s expense' the passivisation of the reanalysed NP seems to be not entirely acceptable to some native speakers, though the passivisation of the outer object does:

(37a) 你被他拆台了

Ni bei ta chai tai le
 2SG PASS 3SG take:apart platform PART

(37b) 你的台被他拆了

Ni -de tai bei ta chai -le
 2SG-SUB platform PASS 3SG take:apart-PERF

He let you down.

(38a) 這件事情被他揭底了

Zhei-jian shiqing bei ta jie di le
 this-CL matter PASS 3SG take:off bottom PART

- (38b) 這件事情的底被他揭了
 Zhei-jian shiqing-de di bei ta jie le
 this-CL matter -SUB bottom PASS 3SG take:off PART
 This affair has been revealed by him.

- (39a) 他被人家開了個大玩笑
 Ta bei renjia kai -le ge da wanxiao
 3SG PASS people open-PERF CL big joke

- (39b) 他的玩笑被人家開了
 Ta -de wanxiao bei renjia kai ge mei wan
 3SG-SUB joke PASS people open CL NEG finish
 He was continuously made fun of.

- (40a) 你被他佔便宜
 Ni bei ta zhan -le pianyi
 2SG PASS 3SG seize-PERF cheap

- (40b) 你的便宜被他佔
 Ni -de pianyi bei ta zhan -le
 2SG-SUB cheap PASS 3SG seize-PERF
 He profits at your expense.

The reanalysed NP can also behave like an ordinary NP with respect to topicalisation. Thus, the reanalysed NP Zhāngsān-de zhí is as acceptable in the topic position as an ordinary NP with a real modifier like Zhāngsān-de gǒu 'Zhangsan's dog:

- (41) 張三的職上級不敢停
 Zhangsan-de zhi, shangji bu gan ting
 Zhangsan-SUB work boss NEG dare stop
 The boss does not dare fire Zhangsan.

- (42) 張三's狗我已經看過了
 Zhangsan-de gou, wo yijing kan -guo le
 Zhangsan-SUB dog 1SG already look-EXP PART
 Zhangsan's dog, I have already seen it.

But as the following sentences show, not every VOP whose outer object assumes the form of the possessive object allows the reanalysed NP in the topic position:

- (43) 你的便宜他佔個沒完
 Ni -de pianyi, ta zhan ge mei wan
 2SG-SUB cheap 3SG seize CL NEG finish
 He does not stop profiting at your expense.
- (44) ?你的台張三敢拆嗎?
 Ni -de tai, Zhangsan gan chai ma ?
 2SG-SUB stage Zhangsan dare pull:down PART
 Does Zhangsan dare let you down?
- (45) ?他的玩笑誰敢開?
 Ta -de wanxiao, shei gan kai ?
 3SG-SUB joke who dare open
 Who would dare making fun of him?

Like the passivisation of the reanalysed NP, the topicalisation of the reanalysed NP is not necessary to satisfy the PSC, for with the outer object topicalised, there is only one postverbal constituent, i.e. the inner object (cf. (46) and (47)). This probably explains why the acceptability of the reanalysed NP in the topic position is not predictable.

- (46) 這件事情我們終於揭底了
 Zhei-jian shiqing, women zhongyu jie di le
 this-CL matter 1PL finally take:off bottom PART
 This affair, we have finally revealed it.
- (47) 張三上級要停職，李四上級要調職
 Zhangsan, shangji yao ting zhi
 Zhangsan boss want stop work
 Lisi, shangji yao diao zhi⁹
 Lisi boss want move work
 Zhangsan, the boss wants to fire him; Lisi, the boss wants to transfer him to a new post.

⁹ Contrary to the topicalisation of an inanimate outer object like zhèi-jian shìqíng 'this matter' in (46), the topicalisation of an animate outer object is fully acceptable only in a contrastive context like that in (47).

A comparison between the sentences illustrating the passivisation of the reanalysed NP (cf. (37) - (40)) with those illustrating the topicalisation of the reanalysed NP (cf. (43) - (45)) further shows that there is no relation between these two phenomena. Zhàn piányi 'to profit at sb.'s expense', for example, allows the reanalysed NP to appear in the topic position (cf. (43)), whereas its acceptability in the subject position of a passive sentence is limited (cf. (40b)); and for chāi tái 'to let sb. down', we observe the opposite situation (cf. (37b) and (44)).

With respect to passivisation and topicalisation, the reanalysed NP (consisting of the inner object as the head and the outer object in the modifier position) shows the same behaviour as an ordinary NP with a real modifier. The acceptability of these two processes, however, is not predictable for a particular VOP, probably because neither the passivisation nor the topicalisation of the reanalysed NP is necessary to satisfy the PSC. Furthermore, there is no relation between the passivisation and the topicalisation of the reanalysed NP in the sense that the acceptability of the one would imply the acceptability of the other, and vice-versa.

6.2.1. VOPs with a literal and an idiomatic meaning

For the VOPs discussed so far, no difference has been observed between the passivisation of the outer object and the passivisation of the reanalysed NP. But in the case of VOPs having a literal as well as an idiomatic meaning, passivisation of the entire reanalysed NP is preferred for the literal meaning, whereas passivisation of the outer object alone is primarily associated with the idiomatic meaning. This has been pointed out by Chappell(1983):

Idioms are striking evidence of the difference between the two kinds of bèi-passive. When it is possible to split idioms of the V-O form apart which involve a body part term, the latter become part of the new subject of a regular passive [i.e. the passive with the entire reanalysed NP in the subject position; W.P.] and the meaning changes to a literal interpretation with the consequent loss of the idiomatic meaning.

(Chappell 1983: 123 - 124)

- (48a) 他被人揪住了辮子

Ta bei renjia jiu -zhu -le bianzi
3SG PASS people seize-hold-PERF pigtail

She had her shortcomings seized upon by people.

- (48b) 他的辮子被人揪住了

Ta -de bianzi bei renjia jiu -zhu -le
3SG-SUB pigtail PASS people seize-hold-PERF

Her pigtail was grabbed hold of by people.

- (49a) 他被朋友扣上了帽子

Ta bei pengyou kou-shang-le maozi
3SG PASS friend put-DIR -PERF hat

He was denounced by friends.

- (49b) *他的帽子被朋友扣上了

Ta -de maozi bei pengyou kou-shang-le
3SG-SUB hat PASS friend put-DIR -PERF

(Chappell 1983: 115)

The association of a particular type of passive with a particular meaning is, however, not as clear-cut as presented by Chappell. It is true that (48a) only allows for the idiomatic meaning of jiū biànzi, i.e. 'to seize sb.'s shortcomings', but (48b) can also have this idiomatic meaning, even though the literal interpretation of jiū biànzi, 'to grab sb.'s pigtail', is clearly dominant here. Example (49a) with màozi 'hat' as an inner object shows that Chappell's observation can also be applied to VOPs whose inner object does not denote a body part. That kòu màozi in (49b) does not have a literal meaning of 'to put a hat on sb.' according to Chappell(1983: 115) is due to the fact that other than in the VOP the appropriate verb for màozi is dài 'to wear,

to put on', and not kòu. Since the idiomatic meaning of kòu màozi, 'to denounce', is excluded as well because of the passivisation of the entire reanalysed NP, (49b) is unacceptable.

There are additional data which weaken the claim that on the one hand the passivisation of the outer object alone is associated with the idiomatic meaning of a VOP, and that on the other hand the passivisation of the entire reanalysed NP is linked to the literal meaning of the VOP. The VOPs zhuā (xiǎo) biànzi 'to seize (small) pigtail' = 'to seize sb.'s shortcomings' (which my informants preferred to jiū biànzi) and zhāi màozi 'to take off hat' = 'to remove a dishonourable label, to rehabilitate', for example, have their idiomatic meaning in the case of the passivisation of the reanalysed NP (cf. (50a) and (51a)) as well as in the case of the passivisation of the outer object (cf. (50b) and (51b)):

(50a) 他的辮子終於被我抓住了

Ta -de bianzi zhongyu bei wo zhua -zhu -le
3SG-SUB pigtail finally PASS 1SG seize-hold-PERF

(50b) 他終於被我抓住了辮子

Ta zhongyu bei wo zhua -zhu -le bianzi
3SG finally PASS 1SG seize-hold-PERF pigtail

I finally succeeded in seizing her shortcomings.

(51a) 他的帽子被摘掉了

Ta -de maozi bei zhai -diao-le
3SG-SUB hat PASS take:off-fall-PERF

(51b) 他被摘掉了帽子

Ta bei zhai -diao-le maozi
3SG PASS take:off-fall-PERF hat

He was rehabilitated.

(52a) 他的反革命的帽子被摘掉了

Ta -de fangeming -de maozi bei zhai -diao-le
3SG-SUB antirevolution.-SUB hat PASS take:off-fall-PERF

(52b) 他被摘掉了反革命的帽子

Ta bei zhai -diao-le fangeming -de maozi
 3SG PASS take:off-fall-PERF antirevolutionary-SUB hat

He was freed from the false reputation of an anti-revolutionary.

In order for (50) and (51) to be interpreted literally (i.e. as 'I finally succeeded in grabbing her pigtail' and 'His hat was taken off', respectively), a special context is needed, for example, that we speak of a children's game where one has to grab the other children by their pigtails or where one has to try to take off somebody's hat.

In another structure where the verb and the inner object are separated, namely in the structure where the reanalysed NP occupies the topic position, the VOPs do not automatically lose their idiomatic meaning, either, but remain ambiguous:

(53) 他的辮子我終於抓住了

Ta -de bianzi, wo zhongyu zhua -zhu-le
 3SG-SUB pigtail 1SG finally seizehold-PERF

(i) I finally succeeded in seizing her shortcomings.

(ii) Her pigtail, I finally succeeded in grabbing it.

(54) 他的帽子他們終於摘掉了

Ta -de maozi, tamen zhongyu zhai -diao-le
 3SG-SUB hat 3PL finally take:off-fall-PERF

(i) They have finally rehabilitated him.

(ii) His hat, they have finally taken it off.

The VOPs with a literal and an idiomatic meaning have been shown to behave like non-ambiguous VOPs because they in principle allow the passivisation and the topicalisation of the entire reanalysed NP. They differ from non-ambiguous VOPs only in so far as we can still obtain an acceptable sentence (with a literal interpretation) even if the passivisation or topicalisation of the reanalysed NP is excluded because the NP 'N de NP' can also be considered as an ordinary NP whose passivisation and topicalisation are much less restricted than that of the reanalysed NP.

6.2.2. The VOPs whose inner object denotes a body part

The phenomenon that a complex NP 'NP de N' can be either analysed as an ordinary NP or as a reanalysed NP is not restricted to ambiguous VOPs, but can also be observed in the case of VOPs whose inner object denotes a body part. This possibility of two analyses makes it difficult to determine the relation between (quasi-synonymous) sentences like Tā bèi rénjiā jiū-zhù-le biànzi (48a) and Tā-de biànzi bèi rénjiā jiū-zhù-le (48b).¹⁰ Sentence (48a) clearly illustrates the passivisation of the outer object tā of the VOP jiū biànzi 'to grab the pigtail' and has the underlying representation [_V[_V jiū biànzi] ta]. (48b), however, can be analysed in two different ways: either tā-de biànzi is the reanalysed NP with tā-de as the possessive object (and accordingly, (48b) is a variant of (48a) and has the same underlying representation), or tā-de biànzi is an ordinary NP with tā as a real modifier, and there is no VOP involved. In the latter case, the link between sentences (48a) and (48b) cannot be provided by the derivation from the same underlying representation. They are instead related by their quasi-identical meaning due to the part-whole relation existing between biànzi 'pigtail' and tā 'she'. Like Tā-de biànzi bèi rénjiā jiū-zhù-le, the sentence Rénjiā jiū-zhù-le tā-de biànzi 'People have grabbed hold of her pigtail' can also be derived from two different underlying representations. In one, tā is the outer object of the VOP jiū biànzi and accordingly tā-de biànzi results from the reanalysis of the outer object as the possessive object; in the other, tā-de biànzi, being an ordinary NP with tā as a real modifier, is the direct object of

¹⁰ In the case of ambiguous VOPs, I will only take into consideration the literal meaning here.

jiū (cf. above, 4.6.). Sentences like Tā-de biànzi bèi rénjiā jiū-zhù-le as well as Rénjiā jiū-zhù-le tā-de biànzi are thus always acceptable because tā-de biànzi can be interpreted as an ordinary NP and need not be regarded as a reanalysed NP.

The acceptability of a sentence like Tā bèi rénjiā jiū-zhù-le biànzi (cf. (48a)), however, depends on the possibility of the body part term (here biànzi 'pigtail') to function as the inner object of a VOP. Whereas this is possible for biànzi 'pigtail', huái 'ankle', for example, cannot play the role of the inner object of a VOP (cf. (55)), though there is evidently no problem in its functioning as the head noun of the NP tā-de huái 'his ankle' (cf. (56)):¹¹

- (55) *他被子彈打碎了踝
Ta bei zidan dasui-le huai
3SG PASS bullet smash-PERF ankle

- (56) 他的踝被子彈打碎了
Ta -de huai bei zidan dasui-le
3SG-SUB ankle PASS bullet smash-PERF

His ankle was smashed when a bullet hit it.

(Chappell 1983: 166)

Judging by the list where Chappell(1983: 160, 165) mentions the body part terms which are liable to function as the inner object of a VOP and those which are not, a VOP of the type 'verb body part' apparently must have a high degree of Transitivity in order to be acceptable.¹² This is confirmed by Chappell's

¹¹ Some of my informants, though, did accept (55) as well as (59) (cf. below) which somewhat weakens Chappell's claim of a distinction between body part terms liable to function as the inner object of a VOP and those which are not.

¹² "Only parts of the body which are well-defined in form and easily individualizable can be encoded as the postverbal NP in this construction [i.e. as the inner object of a VOP in a passive sentence; W.P.]. These are: gēbo 'arm', yǎnjing 'eye', ěrdùo

observation (p. 121) that e.g. in (57), shāng 'to hurt' is obligatory whereas it is optional in (58):

(57) 他被敵人打*(傷)了腿

Ta bei diren da *(-shang)-le tui
3SG PASS enemy beat -hurt -PERF leg

He was leg-wounded [sic] through the enemy's firing.

(58) 他的腿被敵人打(傷)了

Ta -de tui bei diren da (-shang)-le
3SG-SUB leg PASS enemy beat -hurt -PERF

His leg was wounded by the enemy's firing.

(Chappell 1983: 120)

A complex verb like dǎ-shāng 'to beat-to hurt' where the second verb indicates the result of the action of the first verb is called a "resultative compound" (cf. Li/Thompson 1981: 54 ff). The need for such a resultative compound (instead of a simple verb) can also be observed in the case of the "normal" passive

(footnote 12 continued)

'ear', yá 'tooth', bízi 'nose', tóufa 'hair', nǎodai 'head', shǒu 'hand', zhítou 'finger', zhǐjiā 'fingernail', tuǐ 'leg', jiǎo 'foot' and jiǎozhǐ 'toe'. ... As a consequence of the requirement of form, places on the body or areas of the body are not permissible. Thus, huái 'ankle', xīgàigǔ 'kneecap', jiānbǎng 'shoulder', liǎn 'face', étou 'forehead', xiōngtáng 'chest', dùzi 'belly', dàtuǐ 'thigh' and yāo 'waist' are all excluded as they designate either joints of the body or surface areas."

(Chappell 1983: 160)

"Body part terms denoting internal organs are mainly excluded ... from the passive of bodily effect [i.e. from the passive having the form 'NP bei V body part'; W.P.]. This list includes gǎnzàng 'liver', pí 'spleen', yaōzi 'kidney', hóulóng 'throat', sǎngzi 'voice' or 'throat', xīnzàng 'heart', dùzi 'guts' or 'belly' and nǎozi 'brain'. Gǔtōu 'bones' when used in the sense of most of the skeletal framework (not just a few insignificant bones) may be felicitously employed in this construction:

(i) 他被汽車壓碎了骨頭

Ta bei qiche ya -sui -le gutou
she BEI car crush-smash-COM bone

She was bones crushed [sic] by a car.

(ibid., p. 164 - 165)

(i.e. with verbs and not with VOPs)¹³ and is in general interpreted as a means to heighten the degree of Transitivity of the verb phrase. Thus, in order for a VOP 'verb body part' to be acceptable, the "right" body part term has to be chosen and the verb phrase must express the impact of the action on the outer object. Consequently, if the verb in question denotes an action exclusively affecting the body part, and not the entire person, this verb and the body part term cannot form a VOP, and the passivisation reserved for VOPs is not acceptable, either (but see footnote 11):

- (59) *他被機器震壞了耳朵
 Ta bei jiqi zhen -huai -le erduo
 3SG PASS machine vibrate-destroy-PERF ear

- (60) 他的耳被機器震壞了
 Ta -de erduo bei jiqi zhen -huai -le
 3SG-SUB ear PASS machine vibrate-destroy-PERF

His ears were hurt through the vibrating of the machines.

(Chappell 1983: 123)

Those body part terms which can play the role of the inner object of a VOP do not admit modifiers in this function (cf. Chappell 1983: 154 ff):

- (61a) 張三被李四打歪了(*鷹鉤)鼻子
 Zhangsan bei Lisi da -wai -le (*yinggou) bizi
 Zhangsan PASS Lisi beat-crooked-PERF aquiline nose

Zhangsan was nose knocked [sic] out of shape by him.

¹³ Sentence (i) is not acceptable with mài 'to sell' alone, but only if mài 'to sell' forms a resultative verb compound with guāng 'used up' because then the degree of Transitivity of the verb phrase is high enough for passivisation to be acceptable:

- (i) 書被賣光了
 Shu bei mai -guang -le
 book PASS sell-used:up-PERF

The books are sold out.

- (61b) 他的(鷹鉤)鼻子被李四打歪了

Ta -de (yinggou) bizi bei Lisi da -wai -le
3SG-SUB aquiline nose PASS Lisi beat-crooked-PERF

His (roman) nose was knocked out of shape by Lisi.

(ibid., p. 154)

- (62a) 張三被捕奴人割掉了(*他的)耳朵

Zhangsan bei bunuren ge -diao-le (*ta -de) erduo
Zhangsan PASS slave:trader cut-fall-PERF 3SG-SUB ear

Zhangsan had his ears cut off by the slave traders.¹⁴

- (62b) 他的耳朵被捕奴人割掉了

Ta -de erduo bei bunuren ge -diao-le
3SG-SUB ear PASS slave:trader cut-fall-PERF

His ears were cut off by the slave traders.

(ibid., p. 158)

This observation by Chappell is interesting in so far as in the case of VOPs with a body part term as the inner object, there exists corresponding sentences with a quasi-identical meaning where the body part term does not belong to a VOP and therefore can be modified (cf. (61b), (62b)).¹⁵ The inacceptability of yīnggōu and tā-de in (61a) and (62a), respectively, can therefore not be explained by a general incompatibility of body part terms with modifiers, but is due to the special function of the body part terms here, i.e. to be the inner object of a VOP.

¹⁴ Since Zhāngsān (in the subject position) is the (passivised) outer object, tā-de in tā-de ǎrduo cannot represent the possessive object, but must be a real modifier.

¹⁵ There remains, however, the problem that theoretically, tā-de ǎrduo 'his ears' in (62b) could also be considered as a reanalysed NP which has been passivised, i.e. like tā-de zhí in (i):

- (i) 他的職被上級停了

Ta -de zhi bei shangji ting-le (cf. (25b))
3SG-SUB work PASS boss stop-PERF

He was fired by the boss.

It is in fact the group of examples (63) - (65) (see below) which confirms our analysis of the data presented in (61) and (62). The situation in sentences (63) - (65) is clearer because here the modifier, being an adjective, cannot be confused with the possessive object.

The incompatibility of the inner object with modifiers can be observed in active sentences as well and also holds for an inner object which is in a part-whole relation with the outer object (cf. (65)). As shown in examples (66) and (68), a noun playing the role of the inner object which does not enter into a part-whole relation with the outer object can be modified in the same way as a noun not belonging to a VOP (cf. (67)):

- (63a) 李四把張三打歪了(*鷹鉤)鼻子
 Lisi ba Zhangsan da -wai -le (*yinggou) bizi
 Lisi BA Zhangsan beat-crooked-PERF aquiline nose
 Lisi has knocked out of shape Zhangsan's nose.
- (63b) 李四打歪了張三的(鷹鉤)鼻子
 Lisi da -wai -le Zhangsan-de (yinggou) bizi
 Lisi beat-crooked-PERF Zhangsan-SUB aquiline nose
 Lisi has knocked out of shape Zhangsan's (roman) nose.
- (64a) 李四給張三拔了(*壞)牙
 Lisi gei Zhangsan ba -le (*huai) ya
 Lisi GEI Zhangsan pull:out-PERF bad tooth
 Lisi has pulled out Zhangsan's tooth.
- (64b) 李四拔了張三的(壞)牙
 Lisi ba -le Zhangsan-de (huai) ya
 Lisi pull:out-PERF Zhangsan-SUB bad tooth
 Lisi has pulled out Zhangsan's (bad) tooth.
- (65a) 他把橘子剝了(*厚)皮
 Ta ba juzi bo -le (*hou) pi
 3SG BA orange remove-PERF thick skin
 He has peeled the orange.
- (65b) 他剝了橘子的(厚)皮
 Ta bo -le juzi -de (hou) pi
 3SG remove-PERF orange-SUB thick skin
 He has peeled the thick skin of the orange.
- (66) 他把門按上(監牢的)鎖
 Ta ba men an -shang (jianlao-de) suo
 3SG BA door install-DIR solid -SUB lock
 He has fitted a (solid) lock on the door.

- (67) 他買了一把監牢的鎖
 Ta mai-le yi-ba jianlao-de suo
 3SG buy-PERF 1 -CL solid -SUB lock
 He has bought a solid lock.
- (68) 你先把這篇文章起一個(清楚的)稿
 Ni xian ba zhei-pian wenzhang qi
 2SG first BA this-CL article work:out
 yi-ge (qingchu-de) gao
 1 -CL clear -SUB draft
 Make a clear draft of this article first.

There is thus a general constraint prohibiting the modification of an inner object which is in a part-whole relation with the outer object.

The discussion of VOPs with a literal and an idiomatic meaning has shown that there is no systematic difference in syntactic behaviour between these VOPs and non-ambiguous VOPs. Like non-ambiguous VOPs, they allow passivisation and topicalisation of the entire reanalysed NP, none of these two processes being exclusively associated with one of the two meanings (literal or idiomatic). Furthermore, no correlation exists between the passivisation of the outer object and the idiomatic meaning, on the one hand, and between the passivisation of the entire reanalysed NP and the literal meaning, on the other hand.

A large part of the ambiguous VOPs are VOPs whose inner object denotes a body part. In order for a verb and a body part term to form a VOP, this VOP must express an action whose impact is not confined to the body part, but applies to the person as a whole (i.e. the outer object). A VOP of the form 'verb body part' thus must show a high degree of Transitivity.

6.3. A special case of the possessive object

So far we have only discussed the possessive object representing the patient of the action expressed in the VOP. But there is also a small number of VOPs where the outer object, reanalysed as the possessive object, plays the role of agent instead of that of patient:¹⁶

- (69) 張三吃李四的醋
 Zhangsan chi Lisi-de cu
 Zhangsan eat Lisi-SUB vinegar

Zhangsan is jealous of Lisi.

- (70) 張三吃了李四的虧
 Zhangsan chi-le Lisi-de kui
 Zhangsan eat-PERF Lisi-SUB loss

Zhangsan was exploited by Lisi.

- (71) 張三上了李四的當
 Zhangsan shang-le Lisi-de dang
 Zhangsan go:up-PERF Lisi-SUB trickery

Zhangsan was fooled by Lisi.

In all these examples, it is not the subject noun, Zhangsan, but the possessive object, Lisi, who is responsible for the action

¹⁶ I have not included shòu piàn 'to suffer swindle' = 'to be deceived, cheated' and shòu qì 'to suffer anger' = 'to suffer wrong, to be bullied' because the NP preceding piàn and qì in (i) and (ii) could eventually be interpreted as a real modifier. In this case, a more adequate translation of (i) and (ii) would be 'I have suffered his swindle' and 'I have suffered his anger', respectively:

- (i) 我受了他的騙
 Wo shou -le ta -de pian
 1SG suffer-PERF 3SG-SUB swindle

I was cheated by him.

- (ii) 我受了他的氣
 Wo shou -le ta -de qi
 1SG suffer-PERF 3SG-SUB anger

I was bullied by him.

whose victim is Zhāngsān. We thus observe the exact opposite of the situation with other VOPs where the subject plays the role of agent and the outer object (irrespective of its form) the role of patient (cf. above, chapter 4).

Since this new type of possessive object is different from the possessive object already discussed, how can we be sure that in (69) - (71) the NP in the modifier position really represents the outer object of the VOP, i.e. that it depends on the VOP as a whole and not only on the head noun?¹⁷ As none of the VOPs in (69) - (71) allows passivisation, we cannot use passivisation as a test here to distinguish between the possessive object and a real modifier (cf. above, 4.5.). The impossibility of passivisation, though, is not surprising because these VOPs already show the "perspective" of passive where the patient is encoded as the subject of the sentence.

Passivisation being excluded as a possible test, we have to content ourselves with semantic considerations in order to determine the status of the NP in the modifier position. If Lǐsì in (70) Zhāngsān chī-le Lǐsì-de kuī were a real modifier, (70) would have the meaning of 'Zhangsan has suffered Lisi's losses', i.e. Zhangsan would have suffered the same losses as Lisi. Accordingly, Lisi would not be the person responsible for Zhangsan's (bad) situation. To express this meaning, however, the following sentence must be used:

¹⁷ In his discussion of the possessive object, Lu Shuxiang (1955: 186) also cites an example with chī kuī 'to suffer loss' = 'to suffer losses, to be exploited', but he does not comment on it.

- (72) 張三吃了李四那樣的虧
 Zhangsan chi-le Lisi na -yang-de kui
 Zhangsan eat-PERF Lisi that-sort-SUB loss
 Zhangsan has suffered the same losses as Lisi.
 Zhangsan was exploited in the same way as Lisi.

The possibility to paraphrase sentences (69) - (71) (repeated here in (73a) - (75a)) by sentences (73b) - (75b) constitutes further evidence for my analysis of the modifier NP in (69) - (71) as the agent of the VOP:

- (73a) 張三吃李四的醋
 Zhangsan chi Lisi-de cu
 Zhangsan eat Lisi-SUB vinegar
 Zhangsan is jealous of Lisi.
- (73b) 李四使張三吃醋
 Lisi shi Zhangsan chi cu
 Lisi cause Zhangsan eat vinegar
 Lisi makes Zhangsan jealous.
- (74a) 張三吃了李四的虧
 Zhangsan chi-le Lisi-de kui
 Zhangsan eat-PERF Lisi-SUB loss
 Zhangsan was exploited by Lisi.
- (74b) 李四使張三吃虧
 Lisi shi Zhangsan chi kui
 Lisi cause Zhangsan eat loss
 Lisi causes Zhangsan to be exploited.
- (75a) 張三上了李四的當
 Zhangsan shang-le Lisi-de dang
 Zhangsan go:up-PERF Lisi-SUB trickery
 Zhangsan was fooled by Lisi.
- (75b) 李四使張三上當
 Lisi shi Zhangsan shang dang
 Lisi cause Zhangsan go:up trickery
 Lisi fools Zhangsan.

The paraphrases (73b) - (75b) demonstrate that Lisi^Y is understood to be the agent in (73a) - (75a), for in the causative sentences

(73b) - (75b) it functions as the subject. Zhāngsān here plays the same role as in the sentences (73a) - (75a), namely the "causee", the person affected. The relation between the (a) and (b) sentences of (73) - (75) is such that (a) entails (b). This relation of implication does not exist between sentences which are superficially identical with (73) - (75), but which contain a real modifier:

(76a) 張三吃了李四的飯
 Zhangsan chi Lisi-de fan
 Zhangsan eat Lisi-SUB food
 Zhangsan eats Lisi's meal.

(76b) 李四使張三吃飯
 Lisi shi Zhangsan chi fan
 Lisi cause Zhangsan eat food
 Lisi makes Zhangsan eat.

Let us now examine whether the fact that the possessive object indicates the agent (and not the patient as in all the other cases) is reflected in syntax, i.e. whether the reanalysed NP with such a special possessive object shows the same behaviour as the reanalysed NP discussed above (cf. 6.1.):

(77) 我已經吃過他的虧了再也不要吃別人 *(的)了
 Wo yijing chi-guo ta -de kui le
 1SG already eat-EXP 3SG-SUB loss PART
 zai ye bu yao chi bieren *(-de) le
 again also NEG want eat other:people -SUB PART
 I already was exploited by him, I don't want to be exploited again by other people.

(78) 你上了張三的當，我上了李四 *(的)
 Ni shang-le Zhangsan-de dang,
 2SG go:up-PERF Zhangsan-SUB trickery
 wo shang-le Lisi *(-de)
 1SG go:up-PERF Lisi -SUB
 You were fooled by Zhangsan, I was fooled by Lisi.

(79) 他吃誰的醋？李四*(的)

Ta chi shei-de cu ? Lisi *(-de)
 3SG eat who -SUB vinegar Lisi -SUB

Whom is he jealous of? Lisi.

(80) 他上了誰的當？李四*(的)

Ta shang-le shei-de dang ? Lisi *(-de)
 3SG go:up-PERF who -SUB trickery Lisi -SUB

By whom was he fooled? By Lisi.

In the conjoining deletion and the deletion in elliptical answers, the possessive object indicating the agent of the VOP behaves exactly like the possessive object representing the patient: if the head of the reanalysed NP, the inner object, is deleted, de is retained as when following a real modifier.

The reanalysed NP of the type Lisì-de kuī not only occupies the postverbal position, but also the topic position. (The other position where the reanalysed NP can in principle occur, namely the subject position in a passive sentence (cf. above, 6.2.), is evidently excluded here, passivisation not being possible for the VOPs in question.)

(81) 他的當我上過好幾次了

Ta -de dang , wo shang-guo haoji -ci le.
 3SG-SUB trickery 1SG go:up-EXP how:many-time PART

I have been fooled by him so many times.

(82) 張三的虧我吃過好幾次了

Zhangsan-de kui , wo chi-guo haoji -ci le
 Zhangsan-SUB loss 1SG eat-EXP how:many-time PART

I have been exploited by Zhangsan so many times.

The possessive object representing the agent thus shows the same syntactic behaviour as the possessive object representing the patient, except that the outer object of VOPs like chī kuī 'to be exploited', being the agent, cannot be passivised.

6.4. The possessive subject

After the discussion of the possessive object, it is interesting to compare it with another case of reanalysis which, however, does not involve elements in the postverbal position, but elements in the preverbal position. Here, the subject is reanalysed as the modifier of the object which has been preposed to the verb, i.e. the subject and the object, being adjacent, form one NP in the subject position; accordingly, I propose to call this type of reanalysis the "possessive subject". If there is no reanalysis as the possessive subject (cf. (83a) - (87a)) nor object preposing (cf. (83b) - (87b)), verb copying must apply, the verb being followed by its direct object as well as by a complement of manner (cf. (83c) - (87c)):

(83a) 他的法文講得很好

Ta -de fawen jiang de hen hao
3SG-SUB French speak DE very good

(83b) 他法文講得很好

Ta fawen jiang de hen hao
3SG French speak DE very good

(83c) 他講法文講得很好

Ta jiang fawen jiang de hen hao
3SG speak French speak DE very good

He speaks French very well.

(84a) 他的數學教得很好

Ta -de shuxue jiao de hen hao
3SG-SUB mathematics teach DE very good

(84b) 他數學教得很好

Ta shuxue jiao de hen hao
3SG mathematics teach DE very good

(84c) 他教數學教得很好

Ta jiao shuxue jiao de hen hao
3SG teach mathematics teach DE very good

He teaches mathematics very well.

- (85a) 他的舞跳得很好
 Ta -de wu tiao de hen hao
 3SG-SUB dance dance DE very good
- (85b) 他舞跳得很好
 Ta wu tiao de hen hao
 3SG dance dance DE very good
- (85c) 他跳舞跳得很好
 Ta tiao wu tiao de hen hao
 3SG dance dance dance DE very good
 He dances very well.
- (86a) 媽媽的毛衣打得很好
 Mama -de maoyi da de hen hao
 mother-SUB sweater knit DE very good
- (86b) 媽媽毛衣打得很好
 Mama maoyi da de hen hao
 mother sweater knit DE very good
- (86c) 媽媽打毛衣打得很好
 Mama da maoyi da de hen hao
 mother knit sweater knit DE very good
 Mother knits sweaters very well.
- (87a) 他的頭髮理得很好
 Ta -de toufa li de hen hao
 3SG-SUB hair cut DE very good
- (87b) 他頭髮理得很好
 Ta toufa li de hen hao
 3SG hair cut DE very good
- (87c) 他理頭髮理得很好
 Ta li toufa li de hen hao
 3SG cut hair cut DE very good
 He cuts hair very well.

Examples (83b) – (87b) show that if the object is preposed and no reanalysis takes place, the sentence is nevertheless acceptable, contrary to the reanalysis of the outer object as the possessive object which is obligatory when there are no other forms for the outer object. The optionality of the reanalysis as the possessive subject is due to the lack of constraints limiting the number of

constituents in the preverbal position¹⁸ which shows the preverbal position to be fundamentally different from the postverbal position where such constraints exist.

There is, however, one condition to be satisfied in the reanalysis as the possessive subject: only inanimate NPs can be moved from the object position to the position between the subject and the verb, whereas this is not possible for animate NPs (cf. Hou 1979: 62):

(88a) 我批評了那篇文章
 Wo piping -le nei -pian wenzhang
 1SG criticize-PERF that-CL article

(88b) 我那篇文章批評了
 Wo nei -pian wenzhang piping -le
 1SG that-CL article criticize-PERF
 I have criticized that article.

(89a) 我看見過那個女孩兒
 Wo kanjian-guo nei -ge nühair
 1SG see -EXP that-CL girl
 I have already seen that girl.

(89b) *我那個女孩兒看見過
 Wo nei -ge nühair kanjian-guo
 1SG that-CL girl see -EXP

Consequently, the unacceptability of (90a) can be explained by the unacceptability of object preposing for an animate NP like gǒu 'dog' (cf. (90b)):

(90a) *他的狗愛得要命
 Ta -de gou ai de yaoming
 3SG-SUB dog love DE extremely

¹⁸ With the exception of the subject and the (preposed) object, a NP in preverbal position must be marked by a preposition (cf. Huang 1982: 169).

(90b) *他狗愛得要命

Ta gou ai de yaoming
3SG dog love DE extremely

(90c) 他愛狗愛得要命

Ta ai gou ai de yaoming
3SG love dog love DE extremely

He loves dogs very much.

That the possessive subject really represents an instance of reanalysis, and not one of restructuring is demonstrated by the fact that reanalysed NPs like tā-de fàwén (83a) and tā-de wǔ (85a) are perfectly acceptable as such, i.e. outside the context of reanalysis, having the meaning of 'his French' and 'his dance', respectively.

Thus, sentences (86a) and (87a) (repeated below in (91) and (92)) are in fact ambiguous and the interpretation of māma 'mother' and tā 'he' as real modifiers is the predominant one. For the other sentences (83a) - (85a), the interpretation of the complex NP in the subject position as an ordinary NP with a real modifier is excluded for semantic reasons and we can only obtain the in fact secondary meaning (cf. Huang 1982: 56 - 57, 1982': footnote 20; Mei 1978: 526 - 527).

(91) 媽媽的毛衣打得很好

Mama -de maoyi da de hen hao
mother-SUB sweater knit DE very good

Mother's sweater is knit very well.

(92) 他的頭髮理得很好

Ta -de toufa li de hen hao
3SG-SUB hair cut DE very good

His hair is cut very well.

Like the possessive object, the possessive subject represents a case of reanalysis, the NP in the modifier position being, in principle, compatible with the head noun. This NP, however, does not modify the head noun, but is the subject of the sentence.

Furthermore, the noun in the position of the head noun is in fact the object of the verb, and not the subject as is implied by its being the head of a NP in the subject position. In the case of the possessive object, however, the head noun of the reanalysed NP in the object position really is the object, namely the inner object, and only the NP in the modifier position turns out to have a different role than that expected from its position.

Let us now examine the behaviour of the NP reanalysed as the possessive subject in the conjoining deletion and in the elliptical answer. In order to obtain a clear result, we will not take ambiguous sentences like (86a) and (87a) into consideration here:¹⁹

- (93) 張老師的數學教得很好，李老師(*的)也教得很好
 Zhang laoshi -de shuxue jiao de hen hao
 Zhang teacher-SUB mathematics teach DE very good
- Li laoshi (*-de) ye jiao de hen hao
 Li teacher -SUB also teach DE very good
- Teacher Zhang teaches mathematics very well, and so does teacher Li.

¹⁹ If māma 'mother' in (86a) functions as a real modifier, the subordinator de must evidently be retained when the head noun is deleted:

- (i) 媽媽的毛衣打得很好，你的也打得很好
 Mama -de maoyi da de hen hao
 mother-SUB sweater knit DE very good
- ni -de ye da de hen hao
 2SG-SUB also knit DE very good
- Mother's sweater is knit very well, and yours is, too.
- (ii)a) 誰的毛衣打得很好？
 Shei-de maoyi da de hen hao ?
 who -SUB sweater knit DE very good
- b) 媽媽的
 Mama -de
 mother-SUB
- Whose sweater is knit very well? Mother's.

In fact, (iia) is ambiguous because it can also be understood as 'Who knits sweaters very well?'. It is the conservation of de in the answer which makes clear the function of māma 'mother' as a real modifier.

- (94) 張三的舞跳得很好，李四(*的)也跳得很好
 Zhangsan-de wu tiao de hen hao
 Zhangsan-SUB dance dance DE very good

Lisi (*-de) ye tiao de hen hao
 Lisi -SUB also dance DE very good

Zhangsan dances very well, and so does Lisi.

- (95) 張三的法文講得很好，李四(*的)也講得很好
 Zhangsan-de fawen jiang de hen hao
 Zhangsan-SUB French, speak DE very good

Lisi (*-de) ye jiang de hen hao
 Lisi -SUB also speak DE very good

Zhangsan speaks French very well, and so does Lisi.

The deletion in coordinated structures shows that when the head noun, the preposed object, is deleted, de does not appear, either. The possessive subject does not behave like a real modifier which keeps de under the deletion of the head noun. This is confirmed by the behaviour of the possessive subject in the elliptical answer:

- (96) 誰的數學教得最好，李老師(*的)還是張老師(*的)? 李老師(*的)
 Shei-de shuxue jiao de zui hao, Li laoshi (*-de)
 who -SUB mathematics teach DE most good Li teacher -SUB

haishi Zhang laoshi (*-de) ? Li laoshi (*-de)²⁰
 or Zhang teacher -SUB Li teacher -SUB

Who teaches mathematics the best, teacher Li or teacher Zhang? Teacher Li.

²⁰ If Lǐ lǎoshī and Zhāng lǎoshī were real modifiers, de would be retained in such a disjunctive question:

- (i) 誰的太太最好看? 李老師的還是張老師的? 李老師的
 Shei-de taitai zui haokan ? Li laoshi -de
 who -SUB wife most pretty Li teacher-SUB

haishi Zhang laoshi -de ? Li laoshi -de
 or Zhang teacher-SUB Li teacher-SUB

Whose wife is the prettiest? Teacher Li's or teacher Zhang's? Teacher Li's.

- (97) 誰的舞跳得最好？李四(*的)
 Shei-de wu tiao de zui hao ? Lisi (*-de)
 who -SUB dance dance DE most good Lisi -SUB
 Who dances best ? Lisi.

The NP reanalysed as the possessive subject does not behave like a real modifier, for when the head noun of the reanalysed NP, the preposed object, is deleted, there is no de, either. We thus find the same situation as in a sentence with a (non-reanalysed) preposed object (cf. (83b) - (87b)), i.e. the deletion of the preposed object will not affect the subject.

The possessive subject illustrates a type of reanalysis which is different from the possessive object in several respects.

First, this reanalysis involves elements in the preverbal position, and not in the postverbal position. It can therefore not be motivated by the PSC which imposes restrictions on the postverbal position only. According to Mei(1978: 527), the reanalysis of the two preverbal NPs into one NP occupying the subject position is modeled on the structure 'NP V NP' which is the canonical structure for Chinese sentences (where in the ideal case the postverbal NP is the object of the verb). This is in fact the same structure as that proposed as the model structure for the thematic reanalysis and syntactic restructuring (cf. above, 6.1.), except that the NP involved in that case was in the postverbal, and not in the preverbal position.

Secondly, this reanalysis involves two NPs; because for the subject to be able to be reanalysed as the possessive subject, the object must be preposed and be reanalysed as the head of the NP in the subject position with the subject as its apparent modifier. The reanalysis of the outer object as the possessive object and restructuring of the complement of duration or of frequency as the modifier of the direct object, however, only

involve one element, i.e. the element in the modifier position; whereas the head noun of the reanalysed or the restructured NP occupies a position corresponding to its actual role (i.e. being the inner object or the direct object, respectively).

Thirdly, when the head noun (the preposed object) is deleted, de does not appear. The possessive subject therefore does not show the same behaviour as the possessive object which retains de in the deletion (as does a real modifier NP). The lack of de in the deletion is in fact expected because the NP reanalysed as the possessive subject is the subject of the verb and therefore not related to the object. This distinguishes it clearly from the possessive object; the outer object (being subcategorised for by the VOP as a whole) depends on the verb as well as on the inner object (which accounts for the conservation of de in the deletion of the head noun).

6.5. Some properties of the reanalysis and of restructuring

As shown in the preceding section (6.4.), the reanalysis of the subject as the possessive subject differs in several respects from the reanalysis of the outer object as the possessive object as well as from restructuring of the complement of duration or of frequency as the modifier of the direct object. The differences between these two types of reanalysis, on the one hand, and between the reanalysis and restructuring, on the other hand, can give us some indications as to the nature of the reanalysis and restructuring. This is the reason why I have included the possessive subject in the present discussion, even though it has no direct bearing on the analysis of VOPs. (When I want to refer to both phenomena, reanalysis and restructuring, I will use the term "Reanalysis" with a capital R in order to distinguish it from the thematic reanalysis.)

Since in the structures resulting from the Reanalysis, the Reanalysed NPs all appear in the modifier position of a NP, it is impossible to separate the NPs having different functions by looking at the surface structure only. When the Reanalysed structure is modified, however, the differences between these NPs become visible. Accordingly, when the head noun of the Reanalysed NP is deleted, only the possessive object retains the subordinator de which shows that it depends not only on the verb, but also on the inner object. The possessive subject and the restructured complement of duration or of frequency, on the other hand, are not related to the head noun (i.e. the direct object) and consequently, there is no de when the head noun is deleted. The possessive subject is especially interesting here because the NP in the modifier position can also be interpreted as a real modifier. In this case, it is the absence or presence of de (under deletion) which disambiguates the sentence in question. Thus, the Reanalysed structures reflect, to a certain extent, the underlying relations between the elements involved.

It is more difficult to account for the positional restrictions existing in some cases of Reanalysis. We have seen that the restructured NP 'Complement de direct object' (e.g. sān-ge zhōngtóu-de shū '3-CL hour-book') is confined to the object position (cf. above, 6.1.), whereas the reanalysed NP 'outer object de inner object' can in principle appear elsewhere other than in the object position. Now, one important difference between restructuring of the complement of duration or of frequency and the reanalysis as the possessive object is the fact that the elements involved in the reanalysis are both arguments (subcategorised for by the verb and the VOP, respectively) in contrast to restructuring where only one of the elements is an argument (the direct object), the second being an

outer complement. One could speculate that the relation of subcategorisation holding between the verb (or the VOP) and its argument(s) guarantees that the argument can always be identified as such (even if it does not occupy the appropriate position) thus accounting for the rather free distribution of the reanalysed NP. The identification of the complement of duration or of frequency, however, depends upon its position, the relation between the verb and its outer complement being less tight than that between the verb and its inner complement.

It is difficult to decide whether a reanalysed NP can appear elsewhere other than in the original position of the reanalysis only if the NPs involved are all subcategorised arguments or whether it suffices that they all be arguments (subcategorised or not). Contrary to the reanalysis as the possessive object which exclusively affects subcategorised arguments (of the verb and the VOP, respectively), the reanalysis as the possessive subject applies to one subcategorised argument (the preposed object) and to one non-subcategorised argument (the subject). But the latter type of reanalysis cannot serve as a test for our problem here, for the very nature of this reanalysis excludes the possibility of the reanalysed NP appearing other than in the subject position, the reanalysis crucially depending on the preposing of the direct object.

The fact that under the deletion of the head noun, the preposed object, de does not appear suggests that the reanalysed NP containing the possessive subject would probably not be acceptable in another position than in the subject position. This speculation seems to indicate that the relation between the two nouns composing the reanalysed NP is also relevant for the distribution of the reanalysed NP. If there exists a relation (outside of the context of reanalysis) between the NP in the modifier

position and the head noun, de is retained under deletion of the head noun, and the reanalysed NP can appear elsewhere other than in the original position of the reanalysis. If, however, the NP in the modifier position bears no relation to the head noun, there is no de under deletion, and the reanalysed NP is confined to the position where the reanalysis applies.

As to the predictability of the Reanalysis, the status of argument vs. the status of non-argument of the elements involved seems to be the decisive factor. Thus, restructuring of the complement of duration or of frequency will be acceptable in the case of a verb which is followed by its direct object and a complement.²¹ The same holds for restructuring modeled on the double object construction which can always take place if there is a referential and animate/definite object and a complement of duration or of frequency (cf. above, 3.2.1.):

- (98) 我等了三個鐘頭
 Wo deng-le ta san-ge zhongtou
 1SG wait-PERF 3SG 3 -CL hour
 I waited for him for three hours.

Restructuring of the outer object as the indirect object which also takes the double object construction as a model is, however, not predictable and must be considered an idiosyncratic property of the VOP in question (cf. above, 4.7.):

²¹ Provided that the direct object is not a pronoun or a proper noun, for in this case restructuring of the complement as a modifier of the direct object is excluded:

- (i) *我拜訪了三個鐘頭的他／張三
 Wo baifang-le san-ge zhongtou-de ta / Zhangsan
 1SG call:on-PERF 3 -CL hour -SUB 3SG Zhangsan

Huang(p.c.) proposes to explain this constraint by the general incompatibility of pronouns and proper nouns with a modifier.

- (99) 他佔你便宜
 Ta zhan ni pianyi
 3SG seize 3SG cheap
 He profits at your expense.

- (100a) *他拍你馬屁
 Ta pai ni mapi
 3SG pat 2SG horse:buttocks

- (100b) 他給你拍馬屁
 Ta gei ni pai mapi
 3SG GEI 2SG pat horse:buttocks
 He flatters you.

The comparison between (98) and (99), (100) clearly demonstrates the importance of the status (argument vs. non-argument) of the elements involved in the Reanalysis. Though modeled on the same structure, the double object construction, restructuring which applies to the inner complement (the direct object) and to the outer complement (the complement of duration or of frequency) can be predicted, whereas restructuring affecting two arguments, the inner object and the outer object, takes place in an arbitrary fashion. Consequently, the acceptability of this kind of restructuring must be included in the lexical entry of the VOP in question together with the "normal" form(s) of the outer object (cf. above, ch. 4.5.).

The predictability of the possessive subject represents a rather complex problem, for it differs in too many respects from the possessive object (i.e. by the nature of the elements involved and by the position where the reanalysis applies) for a comparison between these two types to give a clear result. Furthermore, there seems to exist another constraint on the preposing of the direct object besides that of inanimateness:

- (101a) *他的書看得很累
 Ta -de shu kan de hen lei
 3SG-SUB book look DE very tired

- (101b) 他看書看得很累
 Ta kan shu kan de hen lei
 3SG look book look DE very tired
 He read until getting very tired.

- (102) 他的書看得很仔細
 Ta -de shu kan de hen zixi
 3SG-SUB book look DE very careful
 He reads very carefully.

Example (101) shows that a complement referring to the state of the subject (resulting from the action) makes reanalysis impossible, contrary to a complement indicating the manner of the action. (A more detailed analysis of the possessive subject which is beyond the scope of the present study would profitably take the contrast displayed in (101) - (102) as a starting point.)

Since only very few studies have dealt with the phenomenon of Reanalysis in Chinese, the analysis proposed here is necessarily incomplete and rather speculative. Nevertheless, I hope to have shown that the Reanalysis is extremely productive in Chinese and therefore plays an important role in the syntax of Chinese.

In this chapter, I have introduced the distinction between "syntactic restructuring", i.e. restructuring of the complement of duration or of frequency as the modifier of the direct object, and the "thematic reanalysis", i.e. the reanalysis as the possessive object or as the possessive subject. I have provided evidence for this distinction using data concerning the behaviour of the restructured NP and the reanalysed NP in the conjoining deletion and the elliptical answer, as well as their distribution.

In the case of syntactic restructuring, the complement of duration or of frequency appears in the modifier position of the direct object and is subordinated by de if the structure of the

complement is 'Q-CL N'; if, however, the complement has the structure 'Q-CL', it can occupy the position of the quantifier phrase (in the restructured NP) and there is no de. The de in restructuring does not behave like the de following a real modifier, but like the de after nouns which, designating a volume or a surface, can function as classifiers, i.e. de is not retained in the conjoining deletion nor in the elliptical answer. This behaviour of the pseudo-possessive de reflects the lack of a relation between the complement of duration or of frequency and the direct object. Instead, this complement, as an outer complement of the verb, is related to the verb which explains why the restructured NP is confined to the object position, i.e. the postverbal position (except in the case of the (lián) ... dōu construction where it is left to the verb). For only in the object position can the temporal expression in the modifier position of the object NP be identified as the (restructured) complement of duration or of frequency. Elsewhere it must be interpreted as a real modifier which, in most cases, leads to unacceptable sentences, the head noun not being compatible with a temporal expression as its modifier.

In the thematic reanalysis, the outer object of a VOP occupies the modifier position of the inner object and is subordinated by de, i.e. it appears in the form of the possessive object. (This applies to the outer object representing the patient of the VOP as well as to the outer object indicating the agent of the VOP.) The NP resulting from the reanalysis, 'outer object de inner object', shows the same behaviour as an ordinary NP with a real modifier. Consequently, when the head noun, the inner object, is deleted, de is retained as is the case under the deletion of the head noun with a real modifier. Contrary to the restructured NP, the reanalysed NP is not

confined to the object position, but can also occupy the topic position or the subject position in a passive sentence. The acceptability of the reanalysed NP in these positions can however, not be predicted, probably because the topicalisation and the passivisation of the entire reanalysed NP is not required to satisfy the PSC. Furthermore, there is no relation between the topicalisation and the passivisation of the reanalysed NP in the sense that the acceptability of the one would imply the acceptability of the other, and vice-versa. Neither can a correlation be observed in the case of ambiguous VOPs between the topicalisation or passivisation of the entire reanalysed NP and the literal meaning of the VOP, on the one hand, and between the topicalisation or passivisation of the outer object alone and the idiomatic meaning of the VOP, on the other hand. What distinguishes the reanalysed NP from an ordinary NP with a real modifier then is not the syntactic behaviour (which is the same), but the fact that the NP in the modifier position does not modify the head noun and plays the role of the outer object of a VOP. Accordingly, this NP is not only related to the head noun, the inner object, but also to the verb, the outer object being subcategorised for by the VOP as a whole.

The reanalysis of the subject as the modifier of the preposed object (as the possessive subject) differs from the reanalysis as the possessive object in several respects. First, applying to elements in the preverbal position, this type of reanalysis is not motivated by the PSC, but by the tendency to conform to the canonical structure 'NP V NP'. Secondly, two NPs are affected here: neither does the NP in the modifier position function as a modifier, nor is the head noun really the subject of the sentence as implied by its position. In the reanalysis of the outer object as the possessive object as well as in restructuring of the

complement of duration or of frequency as the modifier of the direct object, however, only one element, the NP in the modifier position, is affected (i.e. plays a role different from that implied by its position). The head noun of the reanalysed or the restructured NP, on the contrary, occupies a position which corresponds to its actual function of being the object of the verb. Thirdly, when the head noun, the preposed object is deleted, there is no de. The possessive subject therefore does not behave like the possessive object which retains de under deletion of the head noun (as does a real modifier). The possessive subject has in common with the possessive object that it can also be interpreted as a real modifier, in principle being compatible with the head noun. The absence of de under deletion of the head noun is in fact expected, for the NP reanalysed as the possessive subject is in reality the subject and therefore not related to the object. This makes it clearly different from the possessive object because the outer object, being subcategorised for by the VOP as a whole, depends on the verb as well as on the inner object.

Our discussion of the different types of Reanalysis has shown that the Reanalysed structures, to a certain extent, reflect the underlying relations between the elements involved in the Reanalysis.

7. A RECONSIDERATION OF THE "PHRASE STRUCTURE CONDITION"

As shown in chapter 5, the PSC proposed by Huang(1982') incorrectly predicts the unacceptability of the purposive gěi-clause. In the present chapter, we will review the data contradicting the PSC in detail and will finally replace it with two conditions which together account for those phenomena left unexplained by the PSC.¹

7.1. The Bi-uniqueness Condition

First, let us briefly reconsider the conclusions from chapter 5 and their consequences for the PSC.

Chinese has two different types of purposive clauses following the matrix clause, one with an overt subject marked by gěi (the purposive gěi-clause) and one with an empty subject (which is normally coreferential with the matrix subject); both can optionally be introduced by the purposive clause marker lái. Consequently, there are the following two structures: $[_S NP [_{\bar{V}} [_{\bar{V}} V NP] [_S \underline{g\check{e}i} N V (NP)]]]$ for the purposive gěi-clause, and $[_S NP [_{\bar{V}} [_{\bar{V}} V (NP)] [_S \emptyset V (NP)]]]$ for the purposive clause with an empty subject. These two structures clearly violate the PSC, because the head of $\bar{V}$ is in the phrase-initial position which is only allowed for the lowest level of expansion, here $\bar{V}$. That the (first) $\bar{V}$ is really the head is confirmed by the compatibility of the first verb with aspect markers and the incompatibility of the second verb with aspect markers:

¹ This chapter represents, for a large part, the results of several discussions with Alain Rouveret.

- (1a) 我買了一把刀給你切肉

Wo mai-le yi-ba dao gei ni qie rou
 1SG buy-PERF 1 -CL knife GEI 2SG cut meat

I have bought a knife so that you cut meat with it.

- (1b) *我買一把刀給你切了肉

Wo mai yi-ba dao gei ni qie-le rou
 1SG buy 1 -CL knife GEI 2SG cut-PERF meat

- (2a) 我買了一把刀(來)送他。

Wo mai-le yi-ba dao (lai) song ta
 1SG buy-PERF 1 -CL knife in:order:to give 3SG

I have bought a knife to give it to him (as a present).

- (2b) *我買一把刀(來)送了他

Wo mai yi-ba dao (lai) song-le ta
 1SG buy 1 -CL knife in:order:to give-PERF 3SG

As shown at the end of chapter 5, it is not possible to "save" the structure with a purposive gěi-clause with respect to the PSC by interpreting it as an instance of restructuring modeled on a structure in harmony with the PSC. For there is no structure which could have served as the model structure for restructuring of a complex sentence with a complete purposive gěi-clause. Only in the case of a purposive gěi-clause whose VP has been deleted can restructuring (modeled on the structure reserved for verbs of giving: NP V NP gěi NP) apply.

We thus have the situation where verb copying and the different types of Reanalysis can be accounted for by the PSC, whereas complex sentences with a purposive gěi-clause are incorrectly ruled out by the PSC. When comparing these two groups of phenomena, one observes that in the first group (confirming the PSC) it is the position immediately following the verb which is accounted for by the PSC, while in the second group (contradicting the PSC) the position in question is the position after the first $\bar{V}$. Apparently, the PSC makes correct predictions for the position adjacent to the verb, but is too restrictive in

its prediction as to what is allowed in the position after $\bar{V}$.

The PSC implies that the constraints governing the postverbal position can be captured in the form of well-formedness conditions on possible phrase structures, formulated in terms of "head-initial" vs. "head-final". Thus, Huang associates the $\bar{V}$ -level with subcategorised NPs, and the $\bar{\bar{V}}$ -level with non-subcategorised NPs. This approach, however, runs counter to the original motivation of the $\bar{X}$ -Theory which had introduced intermediate levels in order to avoid such an association and in order to allow subcategorised NPs to appear on the $\bar{\bar{V}}$ -level as well.

Let us now try to formulate the constraints governing the position immediately following the verb without taking into consideration the bar-level in question.

The first postulate to make is that the position immediately following the verb is reserved for the argument(s) subcategorised for by the verb. But it also has to be taken into account that a NP need to be identified by a predicator, i.e. by a verb or a preposition, thus entailing a 1:1 relationship between a verb and a NP.² In the case of the double object verbs, the verb is evidently followed by two NPs, because each NP is subcategorised for by this same verb and must therefore appear in the position reserved for arguments, i.e. the postverbal position.

The requirement of a 1:1 relation between a verb and a NP, which also holds for non-subcategorised NPs, leads to the following situation: when the inner complement is not in the

² This does not hold for a NP in the subject position or in the topic position.

postverbal position, the outer complement can occupy this position (cf. (3)); when the inner complement and the outer complement both follow the verb, verb copying must apply (cf. above, ch. 3.2.):

- (3) 這本書我看了三個鐘頭
 Zhei-ben shu, wo kan -le san-ge zhongtou
 this-CL book 1SG look-PERF 3 -CL hour
 This book, I read it for three hours.

- (4) 我看書看了三個鐘頭
 Wo kan shu kan -le san-ge zhongtou
 1SG look book look-PERF 3 -CL hour
 I read for three hours.

As shown in sentence (4), the outer complement sān-ge zhōngtóu 'three hours' must be identified by a verb in the same way as the inner complement, shū 'book', which explains the unacceptability of the corresponding structure without verb copying:
 *Verb inner Complement outer Complement.

It is important to point out that not just any NP can occupy the postverbal position. Temporal NP-adverbs like zuótiān 'yesterday' and xīngqīyī 'Monday' are unacceptable in the postverbal position and must precede the verb:

- (5a) *這本書我看了星期一
 Zhei-ben shu, wo kan -le xingqiyi
 this-CL book 1SG look-PERF Monday
- (5b) 這本書我星期一看了
 Zhei-ben shu, wo xingqiyi kan -le
 this-CL book 1SG Monday look-PERF
 This book, I read it on Monday.

- (6) *我看書看了昨天
 Wo kan shu kan -le zuotian
 1SG look book look-PERF yesterday

These examples show that the non-subcategorised NPs, acceptable in the postverbal position, are limited to NPs functioning as the

outer complement. A complement of duration or of frequency differs from a temporal NP-adverb by the fact that it is chosen in terms of the characteristics of the verb (like [+durative]). A temporal NP-adverb like zuótiān 'yesterday', however, must be compatible with the entire verb phrase in order not to give rise to ungrammatical sentences like (7):

- (7) *他明天已經洗了澡
 Ta mingtian yijing xi -le zao
 3SG tomorrow already wash-PERF bath

That in (3) the outer complement sān-ge zhōngtóu 'three hours' can follow the verb when the inner complement zhèi-běn shū 'this book' is topicalised, is thus not as self-evident as it seems at first sight, for the outer complement is the only element besides the inner complement which is allowed to appear in the postverbal position. It can therefore be considered as a "quasi-argument" in so far as it imitates the behaviour of a real subcategorised argument. Adverbs, however, cannot function as quasi-arguments and are confined to the preverbal position.

Examples (5a) and (6a) not only show that the postverbal position is reserved for the inner complement(s) and the outer complement, respectively, but also that the PSC is insufficient here. The PSC incorrectly predicts the acceptability of (5a) and (6a) where the verb is followed by only one constituent (a temporal NP-adverb) and thus proves to be too permissive. Neither can the PSC account for the difference in acceptability between (8) and (9) which should both be acceptable, the head-initial phrase being on the lowest level, $\bar{V}$, as predicted by the PSC:

- (8) 他哭了三天
 Ta ku -le san-tian
 3SG cry-PERF 3 -day
 He cried for three days.

- (9) *他哭了昨天
 Ta ku -le zuotian³
 3SG cry-PERF yesterday

In our analysis, however, the difference of acceptability between (8) and (9) is explained by the different status of sān-tiān 'three days' and zuótiān 'yesterday': zuótiān, being a temporal NP-adverb, is confined to the preverbal position, whereas sān-tiān as the complement of duration occupies the postverbal position.

Thus, the PSC is not sufficiently restrictive because it does not at all take into account the content of the postverbal position. Our generalization, however, correctly predicts which elements can follow the verb. Therefore, we propose to replace the PSC, in a first step, by the "Bi-uniqueness Condition":

The Bi-uniqueness Condition (BC)

- a) A NP need to be identified by a predicator, i.e. by a verb or a preposition.
- b) The postverbal position is reserved for the subcategorised argument(s) of the verb (i.e. the inner complement(s)), and, by extension,⁴ for the quasi-argument (i.e. the outer complement).

³ Sentence (9) can be acceptable under the reading 'He deplored the past', i.e. with kū 'to cry, to deplore' as a transitive verb which takes zuótiān 'yesterday' as its object.

⁴ The following sentences seem to be problematic for the BC because here the postverbal position is occupied by a NP which apparently plays the role of the subject:

- (i) 下雨了
 Xia yu le
 go:down rain PART
 It's raining.
- (ii) 刮風了
 Gua feng le
 blow wind PART
 The wind is blowing.

There are, however, sentences which seem to contradict the claim made by the BC that an outer complement occupies the same position as the subcategorised argument(s), i.e. the postverbal position:⁵

(footnote 4 continued)

(iii) 生了人
Sheng-le ren
bear -PERF person

Somebody was born.

(iv) 來了五個人
Lai -le wu-ge ren
come-PERF 5 -CL person

Five people came.

(v) 昨天死了三棵樹
Zuotian si -le san-ke shu
yesterday die-PERF 3 -CL tree

Three trees died yesterday. (Hou 1979: 21)

(vi) 今天早上逃了一個犯人
Jintian zaoshang tao -le yi-ge fanren
today morning escape-PERF 1 -CL prisoner

One prisoner escaped this morning. (Hou 1979: 30)

But in fact there are good reasons for analysing the alleged subject as the subcategorised argument of the verb, i.e. as the direct object (cf. Audrey Li: 1985: 247ff). This is to say that the verbs appearing in sentences (i) - (vi) are "ergative" verbs, i.e. verbs whose only argument in the D-structure is the direct object (cf. Burzio 1981: 45).

Another way to integrate these data into the BC is to slightly modify the formulation of the BC using Williams' (1984) distinction between "internal" argument and "external" argument: the external argument is "the argument that is assigned to a NP external to the maximal projection of the verb", and the internal argument that assigned to a NP within VP (Williams 1984: 641). Accordingly, ergative verbs can be described as having only an internal argument, but no external argument. Thus, if in the second part of the BC, "internal argument" is substituted for "subcategorised argument", the BC accounts for the position of the direct object of "normal" transitive verbs like kàn 'to look' as well as for the position of the so-called postverbal subject. We will not pursue any further which approach is the more advantageous here.

⁵ Sentences like (i) and (ii) where the temporal expression appears in the preverbal position and is followed by dōu are no counterexamples to the BC because, as observed by James C.T. Huang, "they each take on a definite interpretation and are by definition an expression of time like 'today', not one of

他三個月只看了中文書

- (10) Ta san-ge yue zhi kan -le zhongwen shu
3SG 3 -CL month only look-PERF Chinese book

During three months, he only read Chinese books.

他三年搬了五次家

- (11) Ta san-nian ban -le wu-ci jia
3SG 3 -year move-PERF 5 -time home

He moved five times in three years.

他兩個禮拜收到了三封信

- (12) Ta liang-ge libai shoudao-le san-feng xin
3SG 2 -CL week receive-PERF 3 -CL letter

He received three letters in two weeks.

In sentences (10) - (12), a temporal expression which seems to indicate the duration, appears in the preverbal position, and not in the postverbal position as predicted by the BC. However, at least for (12) it is immediately evident that liǎng-ge lǐbài 'two weeks' cannot express the duration, shōudào 'to receive' being a non-durative verb; instead, liǎng-ge lǐbài refers to the span of time under consideration here. In fact, I would like to argue in general (i.e. also including sentences with durative verbs (cf. (10), (11))) that the temporal expression in the preverbal position indicates the span of time within which the event in question takes place, and does not refer to the duration

(footnote 5 continued)

duration or frequency:

(i) 他三天都沒睡覺

- Ta san-tian dou mei shui jiao
3SG 3 -day all NEG sleep sleep

For all the three days he didn't sleep.

(ii) 他兩次都來了

- Ta liang-ci dou lai -le
3SG 2 -time all come-PERF

He came on both occasions. (Huang 1982': 2.4.)

Also cf. Audrey Li(1987: 29).

of the event itself. Thus, sān-nián 'three years' in (11) is the period within which he moved five times. Syntactic evidence confirming my point of view can be divided into three parts.

First, the presence of the preverbal temporal expression does not exclude the presence of a complement of duration, thus demonstrating that the former cannot function as a complement of duration:

- (13) 他三個月只學了一個禮拜的中文
Ta san-ge yue zhi xue -le yi-ge libai-de zhongwen
3SG 3 -CL month only learn-PERF 1 -CL week -SUB Chinese

In three months he only learnt Chinese for one week.

- (14) 他一個禮拜只能看四個鐘頭的電視
Ta yi-ge libai zhi neng kan si-ge zhongtou-de dianshi
3SG 1 -CL week only can look 4 -CL hour -SUB telev.

He is only allowed to watch TV for four hours a week.

- (15) 我兩個禮拜只跟他說了半個鐘頭的話
Wo liang-ge libai zhi gen ta shuo
1SG 2 -CL week only with 3SG talk

ban -ge zhongtou-de hua⁶
half-CL hour -SUB words

In two weeks I only talked to him for half an hour.

- (16) 他整天只休息了十分鐘
Ta zheng-tian zhi xiuxi-le shi-fenzhong
3SG whole-day only rest -PERF 10 -minute

The whole day, he only rested for ten minutes.

⁶ I have shown in chapter 6.1. that the temporal expression in the modifier position of the direct object (cf. (13) - (15)) is the (restructured) complement of duration. In fact, restructuring of the complement of duration is clearly preferred to verb copying here; thus, Tā sān-ge yuè xué zhōngwén zhǐ xué-le liǎng-ge lǐbài (the corresponding verb copying construction for sentence (13)) was accepted only with hesitation and immediately replaced with Tā sān-ge yuè zhǐ xué-le liǎng-ge lǐbài-de zhōngwén.

Secondly, a preverbal temporal expression can be followed by nèi 'inside, within' which emphasizes its function of indicating the span of time within which the event occurs:

- (17) 他一個月內休息了八天

Ta yi-ge yue nei xiuxi-le ba-tian
3SG 1 -CL month inside rest -PERF 8 -day

In one month, he had 8 days off.

- (18) 他三年內住了幾個月的旅館

Ta san-nian nei zhu -le ji -ge yue de lüguan
3SG 3 -year inside live-PERF several-CL month SUB hotel

During 3 years he lived in hotels for several months.

Thirdly, a preverbal temporal expression can be modified by the demonstrative zhè 'this' which is excluded for a complement of duration (cf. (22)):

- (19) 他這五天休息了一天就上班

Ta zhe wu-tian xiuxi-le yi-tian, jiu shang ban
3SG this 5 -day rest -PERF 1 -day then go:up work

In the last five days he had one day off and then went to work.

- (20) 他這三個月只學了一個禮拜的中文

Ta zhe san-ge yue zhi xue -le yi-ge libai de
3SG this 3 -CL month only learn-PERF 1 -CL week SUB

zhongwen
Chinese

In the last three months he learnt Chinese only for one week.

- (21) 我這兩個禮拜只跟他說了半個鐘頭的話

Wo zhe liang-ge libai zhi gen ta shuo-le
1SG this 2 -CL week only with 3SG talk

ban -ge zhongtou de hua
half-CL hour SUB word

In the last two weeks I talked to him only for half an hour.

- (22) 他休息了(*這)三天

Ta xiuxi-le (*zhe) san-tian
3SG rest -PERF this 3 -day

He rested for three days.

The data presented here all show that a temporal expression in the preverbal position does not function as the complement of duration, but indicates the span of time within which the event in question occurs. Accordingly, the prediction made by the BC that the outer complement occupies the postverbal position is confirmed.

The situation is the same for negative sentences, because like in the positive sentences (13) – (16) cited above a preverbal temporal expression does not exclude the presence of a complement of duration (cf. (25)):

(23) 我三天没做事

Wo san-tian mei zuo shi
1SG 3 -day NEG do matter

I have not done things for three days.

(24) 我三天没來了

Wo san-tian mei lai -le
1SG 3 -day NEG come-PERF

I have not been here for three days. (Li 1987: 29-30)

(25) 他三天没休息一分鐘一直工作

Ta san-tian mei xiuxi yi-fenzhong, yizhi gongzuo
3SG 3 -day NEG rest 1 -minute continuous work

He did not rest for a minute in three days and worked without a break.

I thus disagree with Audrey Li(1987: 30) who claims that a preverbal temporal expression in negative sentences indicates duration. Admittedly, the situation is not very clear-cut in a sentence like (23) where sān-tiān 'three days' can in principle be interpreted as the duration of the action of doing nothing, or as the span of time within which the action of doing nothing occurs. (24), on the other hand, seems a very obvious case to me lái 'to come' not being a durative verb, except if one accepts a second (derived) meaning for lái, i.e. 'to be, to stay', which is durative. Judging from her translation, that is apparently what

Audrey Li does. (For a discussion of non-durative verbs occurring with an apparent complement of duration, cf. above, ch.3, footnote 11.)

Last, but not least, it is interesting to note that a temporal expression in the preverbal position can be preceded by yǒu 'to have' (which sometimes is even preferred) (cf. Audrey Li 1987: 29)):

- (26) 我有三個月只看了幾個鐘頭的書
 Wo you san-ge yue zhi kan -le ji -ge
 1SG have 3 -CL month only look-PERF several-CL
 zhongtou de shu
 hour SUB book
 In three months I only read for several hours.

- (27) 我有兩個禮拜只跟他說了半個鐘頭的話
 Wo you liang-ge libai zhi gen ta shuo-le
 1SG have 2 -CL week only with 3SG talk-PERF
 ban -ge zhongtou de hua
 half-CL hour SUB word
 In two weeks I only talked to him for half an hour.

- (28) 我有三天沒做事 (cf. (23))
 wo you san-tian mei zuo shi
 1SG have 3 -day NEG do matter
 I have not done anything in three days.

- (29) 我有三天沒來了 (cf. (24))
 wo you san-tian mei lai le
 1SG have 3 -day NEG come PART
 I have not been here for three days.

Sentences (26) and (27) show that Audrey Li's (1987: 30) observation that yǒu 'to have' preceding a preverbal temporal expression is only allowed in negative sentences, does not hold in general:

- (30a) *我有三天來了
 Wo you san-tian lai le
 1SG have 3 -day come PART

(30b) *我有三天做事了

Wo you san-tian zuo shi le
1SG have 3 -day do matter PART

The comparison between Audrey Li's examples (30a), (30b) and sentences (26), (27) suggests that the difference in acceptability has to do with the lack of a second temporal expression in (30a) and (30b) (i.e. with the absence of a complement of duration); for in (26) and (27) it is the presence of the complement of duration jǐ-ge zhōngtóu 'several hours' and bàn-ge zhōngtóu 'half an hour', respectively, which makes clear the role of the preverbal temporal expression. In other words, (30a) and (30b) are unacceptable because sān-tiān 'three days' is understood to be the complement of duration, but does not occupy the appropriate position, i.e. the postverbal position.⁷

The preceding discussion further confirms the claim made by the BC that NPs in general (with the exception of the subject NP and the topic NP) need to be identified by a predicator. The complementary distribution of yǒu 'to have' preceding a preverbal temporal expression and nèi 'inside' following it might suggest

⁷ Sentence (30a), however, still remains somewhat problematic, because lái 'to come' being a non-durative verb should in any case not allow a complement of duration. But, in fact, the following sentence where lái occurs with a temporal expression occupying the position of the complement of duration is completely acceptable:

(i) 他來了三年了

Ta lai -le san-nian le
3SG come-PERF 3 -year PERF

He has been here for three years. (Li 1987: 26)

As mentioned in our discussion of example (24) above, it is perhaps possible to treat lái occurring in sentences like (i) as a verb different from the (non-durative) verb lái 'to come'. The fact that other non-durative verbs are also allowed in the complement structure illustrated in (i), however, suggests that a more general explanation must be found (cf. above, ch. 3, footnote 11).

including postpositions like nèi among the category of predicators:

- (31a) 他有一個月(*內)休息了八天
 Ta you yi-ge yue (*nei) xiuxi-le ba-tian
 3SG have 1 -CL month within rest -PERF 8 -day
- (31b) 他(*有)一個月內休息了八天
 Ta (*you) yi-ge yue nei xiuxi-le ba-tian
 3SG have 1 -CL month within rest -PERF 8 -day

In one month, he rested eight days.

7.1.1. The Bi-uniqueness Condition and verb copying

In the light of the BC, let us now briefly review the phenomenon of verb copying:

- (32) 我看書看了三個鐘頭
 wo kan shu kan -le san-ge zhongtou
 1SG look book look-PERF 3 -CL hour

I read for three hours.

The first $\bar{V}$ contains the subcategorised argument of kàn 'to look, to read', i.e. shū 'book', whereas the outer complement, the complement of duration sān-ge zhōngtóu 'three hours' is in the second $\bar{V}$. As a quasi-argument, san-ge zhongtou appears in the postverbal position and the second $\bar{V}$ is thus modeled upon the relation between a verb and its real argument existing in the first $\bar{V}$. That only the second verb can be marked for aspect (e.g. for perfectivity as in (32)) can be explained as follows. The aspect marker -le, for example, is only compatible with bounded events; events can be bounded by e.g. an indication of their duration or their frequency (cf. Li/Thompson 1981: 185ff). Since the first verb in the verb copying construction takes the direct object, the complement of duration or of frequency has to be identified by the second verb which, being bounded, carries the aspect marker. Consequently, the incompatibility of the first

verb with aspect markers does not so much indicate its lack of verbal properties, but is rather due to the necessity of a 1:1 relation between a NP and a verb.

A similar analysis can be applied to de introducing the complement of manner. De, excluding aspect markers, can in fact be considered as a kind of aspect marker itself. Accordingly, de needs a verb to be attached to and this must be the second verb, the first verb already being related to the argument of the verb:

- (33) 他看書看(了)得很仔細
 Ta kan shu kan (*-le) de hen zixi
 3SG look book look -PERF DE very careful
 He reads very carefully.

Verb copying in the case of double object verbs also shows that the first verb behaves in fact like a full-fledged verb, at least with respect to its function as a predicator:

- (34) 我教他英文教了三年了
 Wo jiao ta yingwen jiao -le san-nian le
 1SG teach 3SG English teach-PERF 3 -year PART
 I taught him English for three years.

Jiào 'to teach' is followed by its two subcategorised arguments, and it is the presence of the outer complement sān-nián 'three years' which makes verb copying necessary. (34) is in harmony with the BC, but is problematic for the PSC which, in fact, incorrectly predicts its unacceptability. For with the second $\bar{V}$ being the head of $\bar{V}$, the first $\bar{V}$, [jiào tā yīngwén], must be interpreted as a kind of prepositional phrase, which, however, in Chinese can only have one form: [_{pp} Prep NP].

As for the VOPs and verb copying, the analysis given in chapter 3.2.4. is still valid, except that verb copying has been shown to be motivated by the BC, and not by the PSC. Thus, the second verb in the verb copying construction cannot mark the

outer object of a VOP because the latter is not an outer complement of the verb (as its position following the second verb would imply), but the argument subcategorised for by the VOP as a whole.

We still have to explain why the structure $[_{\bar{V}}[_{\bar{V}} \text{ verb in.Obj.}] \text{ out.Obj.}]$ is not appropriate for the VOPs, either, for we have seen that the association of the $\bar{V}$ -level with non-subcategorised NPs is not correct. This structure is excluded for VOPs because the BC would interpret each of the postverbal NPs in 'V NP NP' as an argument subcategorised for by the verb. Though this is correct for double object verbs like sòng 'to give (as a present)', it is incorrect for VOPs because the second NP, the outer object, is the argument of the entire VOP.

7.1.2. The Bi-uniqueness Condition and the Reanalysis

As in the case of verb copying, the analysis proposed for the reanalysis and restructuring remains essentially the same, except that the condition underlying these two phenomena is now of a different nature.

The model structures for the Reanalysis are the same as those under the explanation in terms of the PSC, namely 'NP V NP' and 'NP V NP (gěi) NP'. Their choice, however, is not motivated by the fact that the VP shows a flat structure here in accordance with the PSC, but by the fact that in both structures the verb is followed by its subcategorised argument(s) only.

The difference between the analysis in terms of the BC and in terms of the PSC is more visible in the case of the Reanalysis modeled on the double object construction. Since the Reanalysis imitates a structure allowed by the BC, it is evident

that the Reanalysed structure immediately "benefits" from the "privilege" the two arguments of a double object verb have, i.e. to both appear in the postverbal position. The "detour" observed in Huang's(1982') analysis can therefore be dispensed with, i.e. that the aim of the Reanalysis is to arrive at a flat structure $\bar{V}$ which, on its part, is the level reserved for the subcategorised arguments of the verb.

The claim that all the subcategorised arguments have to be on the $\bar{V}$ -level is particularly difficult to sustain in the case of the verbs of giving where the indirect object, being marked by gěi, follows the direct object: NP V NP gěi NP. In our analysis, however, nothing prevents the NP marked by gěi from appearing on the $\bar{V}$ -level. Since this structure is in harmony with the BC (both arguments following the verb), it can serve as the model structure for restructuring of the purposive gěi-clause as a gěi-phrase when the VP in the purposive gěi-clause is deleted (cf. above, chapter 5.2.4.2.).

We have shown that the predictions made by the PSC for the position immediately following the verb are not correct. In order to determine the positions available for the inner complement(s) and the outer complement of a verb, we have proposed the Bi-uniqueness Condition which, representing a generalization from the point of view of the verb as well as from that of a NP, comprises two parts. Thus, the necessity for a NP to be identified by a verb together with the fact that the postverbal position is reserved for the subcategorised argument(s) of the verb and, by extension, for the quasi-argument (i.e. the outer complement), entails a 1:1 relation between a verb and a NP. In the case of a verb which subcategorises for two arguments (e.g. a double object verb), the verb is related to both of them: V NP NP.

The comparison of the BC with the PSC shows that the constraints governing the positions available for the inner and the outer complement of a verb as formulated by the BC, in fact, correspond to the constraints Huang(1982') postulates for $\bar{V}$ and $\bar{\bar{V}}$, respectively. Thus, Huang's "error" consists in the association of $\bar{V}$ with the inner complement(s) and the association of $\bar{\bar{V}}$ with everything else (but the inner complement) without any further differentiation.

7.2. The Predication Condition

The Bi-uniqueness Condition can account for the form of a VP composed of a verb and the NPs depending on it, i.e. the inner complement(s) and the outer complement. The VP in a complex sentence with a purposive clause having an overt or an empty subject (cf. above, (1a), (2a)), however, is beyond its scope, and the constraints governing it still have to be formulated. As for the PSC, we have shown in chapter 5 that it incorrectly predicts the unacceptability of the two types of purposive clauses following the matrix clause, because the head of $\bar{\bar{V}}$, i.e. $\bar{V}$, occupies the phrase-initial position, and not the phrase-final position as required by the PSC.

The fact that a tight relationship must hold between the purposive gěi-clause and the matrix clause suggests that the constraints for a complex VP be formulated in terms of the relationship holding between the matrix clause and the purposive clause. We will therefore examine the two types of purposive clauses from this point of view.

For the purposive gěi-clause, the necessary tight relationship exists if the verb phrase of the purposive gěi-clause can be predicted on the basis of the direct object in the

matrix clause. As shown in chapter 5, the following two relations count as sufficiently tight: a) the direct object of the matrix clause is also understood as the object of the verb in the purposive gěi-clause (cf. (35)); b) the object of the matrix clause functions as the circumstantial complement (instrumental or location: cf. (36)) of the VP in the purposive clause (cf. above 5.2.4.2.):

- (35) 我買了一本書給他看
 Wo mai-le yi-ben shu gei ta kan
 1SG buy-PERF 1 -CL book GEI 3SG look
 I have bought a book so that he reads it.

- (36) 我租了一間房間給他堆東西
 Wo zu -le yi-jian fangjian gei ta dui dongxi
 1SG rent-PERF 1 -CL room GEI 3SG pile:up thing
 I have rented a room so that he can put his things there.

It is important to point out that in a complex sentence with a purposive gěi-clause, the object of the matrix clause remains always implicit in the purposive gěi-clause, i.e. there are no sentences like (37):

- (37) *我買了一本書給他看書
 Wo mai-le yi-ben shu gei ta kan shu
 1SG buy-PERF 1 -CL book GEI 3SG look book

In the case of a purposive clause with an empty subject which (in general) is interpreted as being coreferential with the matrix subject, the relationship between the purposive clause and the matrix clause can be less tight, provided it is semantically or pragmatically plausible:

- (38) 我買了一本書(來)送他
 Wo mai-le yi-ben shu (lai) song ta
 1SG buy-PERF 1 -CL book in:order:to give 3SG
 I have bought a book to give it to him (as a present).

(39) 他搬出去(來)照顧他媽媽

Ta banchuqu (lai) zhaogu ta mama
 3SG move in:order:to take:care 3SG mother

He has moved to take care of his mother.

If the object of the purposive clause is understood to be coreferential with the object of the matrix clause (cf. (38)), it must remain implicit as in the purposive gěi-clause.

The characteristic common to both types of purposive clauses is the fact that a constituent of the matrix clause also plays a role in the purposive clause: in the purposive gěi-clause, it is the object of the matrix clause which is implicitly present in the purposive gěi-clause; in the purposive clause with an empty subject, it is the subject which is coreferential with the matrix subject.

The two types of purposive clauses differ from each other with respect to the relationship holding between the purposive clause and the matrix clause. The purposive gěi-clause must entertain a very close relationship with the matrix clause, whereas the relation between the purposive clause with an empty subject and the matrix clause can be less tight, provided it is semantically or pragmatically plausible. The tight relationship between the purposive gěi-clause and the matrix clause is necessary to avoid the NP marked by gěi (i.e. the subject of the purposive gěi-clause) being understood as an argument of the matrix verb or as a benefactive. For as shown above (cf. chapter 5.2.4.2., examples (87), (88)) the latter interpretation imposes itself when the relation between the VP of the purposive gěi-clause and the matrix object is not maximally tight, i.e. if it is a relation different from the relation verb - object.

Besides the two types of purposive clauses, there is another set of data violating the PSC:

- (40) 我有一個朋友很聰明
 Wo you yi-ge pengyou hen congming
 1SG have 1 -CL friend very intelligent
 I have a very intelligent friend.
- (41) 昨天發生了一件車禍很可怕
 Zuotian fasheng-le yi-jian chehuo hen kepa
 yesterday happen -PERF 1 -CL accident very terrible
 There was a terrible accident yesterday.
- (42) 張三有一本書我看不懂
 Zhangsan you yi-ben shu wo kan bu dong
 Zhangsan have 1 -CL book 1SG look NEG understand
 Zhangsan has a book which I don't understand.
- (43) 他炒了一個菜我很喜歡吃
 Ta chao-le yi-ge cai wo hen xihuan chi
 3SG fry -PERF 1 -CL dish 1SG very like eat
 He has prepared a dish which I like very much.

In (40) - (43), the object NP is followed by a VP or a sentence which from a semantic point of view functions as a non-restrictive relative clause; accordingly, Li/Thompson(1981: 611) call them "descriptive clauses". The phenomenon of the descriptive clause contradicts two predictions made by the PSC: first, that all the nominal modifiers precede the head noun and secondly, that a head-initial structure is allowed only on the lowest level. Concerning the latter point, I have already demonstrated the inadequateness of the PSC in this respect; as to the generalization which says that the NP is always head-final, however, I agree with Huang that it remains valid and that the data presented in (40) - (43) are only apparent counter-examples.

Huang(1982': 2.3) gives three arguments in order to show that the descriptive clause is not part of the object NP and therefore not in a post-head position within that NP.

First, the descriptive clause only occurs in the postverbal position:

- (44) 我打破了一個茶杯很值錢
 Wo dapo -le yi-ge chabei hen zhiqian
 1SG break-PERF 1 -CL teacup very valuable
 I have broken a very valuable teacup.

- (45) *一個茶杯很值錢被我打破了
 Yi-ge chabei hen zhiqian bei wo dapo -le
 1 -CL teacup very valuable PASS 1SG break-PERF

Secondly, only indefinite NPs can be followed by a descriptive clause:

- ((46) 我買了一本/*那本小說很有趣
 Wo mai-le yi-ben/*zhei-ben xiaoshuo hen youqu
 1SG buy-PERF 1 -CL / this-CL novel very interesting
 I have bought a very interesting novel.

Thirdly, the descriptive clause exclusively appears in sentences which are existential (in a broad sense), i.e. in sentences with an existential verb like yǒu 'to have' or with an "existential" aspect marker (like the perfective -le) on the verb. Thus, the verbs in (40) - (42) are existential verbs, namely yǒu 'to have' and fāshēng 'to happen'. The perfective aspect marker -le in (43), implying the existence of the object, contrasts with e.g. the durative aspect marker zài in (47b) which indicates the coming into existence of the object:

- (47a) 他在炒一個我很喜歡吃的菜
 Ta zai chao yi-ge wo hen xihuan chi de cai
 3SG DUR fry 1 -CL 1SG very like eat SUB dish
 He is preparing a dish which I like very much.

- (47b) *他在炒一個菜我很喜歡吃
 Ta zai chao yi-ge cai wo hen xihuan chi
 3SG DUR fry 1 -CL dish 1SG very like eat

- (48a) 我想買一本很有趣的書
 wo xiang mai yi-ben hen youqu -de shu
 1SG want buy 1 -CL very interesting-SUB book
 I want to buy a very interesting book.

(48b) *我想買一本書很有趣

Wo xiang mai yi-ben shu hen youqu
 1SG want buy 1 -CL book very interesting

These three constraints on the descriptive clause are incompatible with an analysis which considers the descriptive clause as part of the object NP. For it would be very unusual for a nominal modifier to be allowed only if the NP in question is indefinite and occupies the postverbal position and moreover to depend on the nature of the verb (existential vs. non-existential).

If it is clear that the descriptive clause does not belong to the object NP, but forms a constituent on its own, then what is its relation with the preceding NP?

According to Huang(1982': 2.3.; 1984b: 568), the descriptive clause can be considered as a predicate (in the sense of Williams 1980) of the preceding NP. The descriptive clause thus involves the same kind of relation as that holding between a subject NP and the VP (cf. (49)) or between a NP in the topic position and the comment (cf. (50)):

(49) 這件車禍很可怕

Zhei-jian chehuo hen kepa (cf. (41))
 this-CL accident very terrible

This accident was terrible.

(50) 這個菜我很喜歡吃

Zhei-ge cai, wo hen xihuan chi (cf. (43))
 this-CL dish 1SG very like eat

This dish, I like very much.

The condition that the object NP must be indefinite in order to be followed by a descriptive clause (according to Huang) represents a characteristic typical of existential sentences. He does, however, not explain why a predication can be made on NPs in the postverbal position only. For the BC, this constraint

makes sense. Thus, the postverbal position is the normal position for a NP because it needs to be identified by a verb. The NPs exempt from this necessity are the NP in the subject position and the NP in the topic position which consequently do not admit a descriptive clause (cf. above, example (25)). (It would also be possible to explain the unacceptability of a descriptive clause in the case of a subject NP or a topic NP by the fact that here the VP and the comment sentence, respectively, already make a predication on the NP in question and thereby exclude a second predicate in the form of a descriptive clause.)

Contrary to the predictions made by the PSC, the structure $[_{\bar{V}}[_{\bar{V}} V NP] [_S \quad]]$ with $\bar{V}$ as the head of $\bar{V}$ is acceptable, provided that the constituent following $\bar{V}$ makes a predication on the object NP in $\bar{V}$.

I propose to apply the same analysis to complex sentences with a purposive gei-clause, i.e. to consider the purposive gei-clause as a predicate of the object in the matrix clause (cf. Huang 1984b: 568 - 569). This is more plausible as the purposive gei-clause is only acceptable when the matrix clause contains an object NP.⁸

For the purposive clause with an empty subject, though, this analysis seems somewhat problematic. Since in this type of purposive clause the matrix clause need not contain an object, it

⁸ This holds irrespective of whether the object actually occupies the postverbal position or has moved into another position, e.g. into the topic position:

- (i) 這把刀我借給你使
 Zhei-ba dao , wo jie gei ni shi (Zhu 1983: 163)
 this-CL knife 1SG lend GEI 2SG use

This knife, I lend it so that you use it.

is evidently impossible to consider the purposive clause with an empty subject as a predicate of the matrix object. Apparently, it is the coreferentiality of the empty subject in the purposive clause and the matrix subject which establishes the relation between the matrix clause and the purposive clause and which makes it possible for the purposive clause to follow the matrix clause. (For if the two subjects are not coreferential nor is it possible to predict the purposive clause on the basis of the matrix object, giving rise to the purposive gěi-clause, a purposive clause must precede the matrix clause and be introduced by wèi(le) 'for the sake of, in order to'.)

But the the fact that the (empty) subject of the purposive clause and the matrix subject are coreferential cannot be the crucial factor alone because coreferentiality of an empty subject and an overt subject can be observed in a coordinated structure as well:

- (51) 他脫了鞋進去
 Ta tuo -le xie jinqu
 3SG take:off-PERF shoe go:in

He took off his shoes and went in.

However, this sentence can also be understood as a complex sentence with a purposive clause (whose subject is empty), i.e. as 'He took off his shoes to go in' (cf. Li/Thompson 1974: 269). The comparison between these two interpretations for (51) shows that by giving the reason for the action expressed in the matrix clause (thereby excluding other possible reasons), the purposive clause has a delimiting function with respect to the matrix clause, i.e. a modifying function (in a broad sense). This is to say that the purposive clause with an empty subject, in fact, functions as a predicate which from a semantic point of view is tantamount to a modification. The predication here is, however,

of the verb phrase, and not of the NP (as in the case of the purposive gěi-clause and the descriptive clause).

In order to account for the data discussed so far which lie beyond the scope of the BC we propose the "Predication Condition":

The Predication Condition (PC)

A constituent (which is neither argument nor quasi-argument) occupies the postverbal position if it makes a predication on (part of) the verb phrase.

In this chapter, we have demonstrated the inadequateness of the PSC and we have replaced it with two conditions which together can also account for the data left unexplained by the PSC.

The first condition is the Bi-uniqueness Condition (BC) according to which the position immediately to the right of a category X is reserved for the argument which X subcategorises for and, by extension, for the quasi-argument.

This also makes the correct predictions for the category N, namely that N does not allow a constituent following it (which would be part of the NP); for N, not being a predicator, does not subcategorise for arguments. The head-final structure the NP shows on all levels is therefore a consequence of the BC and need not be stipulated as under the PSC.

The outer complement of a verb counts as "quasi-argument" because its choice depends on the properties of the verb. When a verb is followed by the inner complement(s) and the outer complement, verb copying must apply in order to maintain a 1:1 relation between a verb and a NP.

The second condition, the Predication Condition (PC), licences a constituent (which is not a complement of the verb) in

the postverbal position if it makes a predication on the preceding verb phrase or the object NP contained in it. Thus, the descriptive clause as well as the two types of purposive clauses are accounted for.

It is evident that the analysis presented here is far from being definitive. Thus, the two conditions proposed here still leave a lot of unanswered questions, e.g. that of the possibility to subsume the two conditions under a single one. What I hope to have shown, however, is that the principle to be looked for cannot be reduced to a well-formedness condition on possible phrase structures (like the PSC), but must be a principle operating on a deeper level where not only the positions, but also their contents are taken into consideration.

8. CONCLUSION

The purpose of this study was two-fold: to give a detailed description of transitive verb-object phrases and to uncover the fundamental principles of Chinese syntax.

While showing the syntactic behaviour of a phrase, from a semantic point of view, a VOP behaves like a unit and is accordingly included in the lexicon. The lexical entry of a VOP indicates the form(s) of its outer object and, in the case of an idiomatic VOP, also its meaning. That the form of the outer object cannot be predicted is due to the impossibility of determining the degree of Transitivity for a given VOP in isolation, i.e. without knowing its behaviour in a sentence. The degree of Transitivity corresponds to the extent of the impact of the action on the outer object and can vary from one VOP to the other, the role of the outer object always being that of the patient.

The fact that the outer object is the argument subcategorised for by the entire VOP can account for the following phenomena:

First, the outer object of a VOP can be passivised like the object of a simple verb (provided the degree of Transitivity is sufficiently high).

Secondly, the second verb in the verb copying construction cannot mark the outer object of a VOP because this position is only appropriate for the outer complement of a verb.

Thirdly, the structure of double object verbs, namely 'NP V NP NP', is unacceptable for the VOPs because the outer object is not a second inner complement of the verb.

Fourthly, VOPs do not allow a postverbal gěi-phrase containing the outer object because the VOPs are not compatible with a purposive gěi-clause from which the postverbal gěi-phrase would have to be derived.

The discussion of the problems encountered in the analysis of the VOPs has shown the inadequateness of the working hypothesis, the PSC. Instead, two conditions were proposed, the "Bi-uniqueness Condition" and the "Predication Condition", in order to account for the constraints governing the postverbal position.

The Bi-uniqueness Condition (BC) determines the distribution of the inner and the outer complement of a verb, demanding a 1:1 relationship between a verb and a NP; if a verb subcategorises for more than one argument, it is related to all of them.

The Predication Condition (PC) is not based on the relation of complementation, but on the relation of predication. The PC predicts that a constituent which is not a complement of the verb is acceptable in the postverbal position only if it makes a predication on the preceding noun phrase or verb phrase.

The structure of the VP in Chinese can thus be regarded as the result of a collaboration between the BC and the PC.

Postulating the BC as one of the principles of Chinese syntax is compatible with the claim that the object position (i.e. the position immediately following the verb) is not limited to the object role, due to the phenomenon of Reanalysis. The Reanalysis consists in modeling structures otherwise excluded by the BC on structures in harmony with it, which explains why the model structures for the Reanalysis only comprise two structures: 'NP V NP' and 'NP V NP (gěi) NP'.

The great extent to which Reanalysis applies as well as its systematic character show clearly that the Reanalysis occupies a central position in the syntax of Chinese and is based on the privileged relation verb - object. Thus, further research into this phenomenon would certainly prove fruitful for a better understanding of Chinese syntax.

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